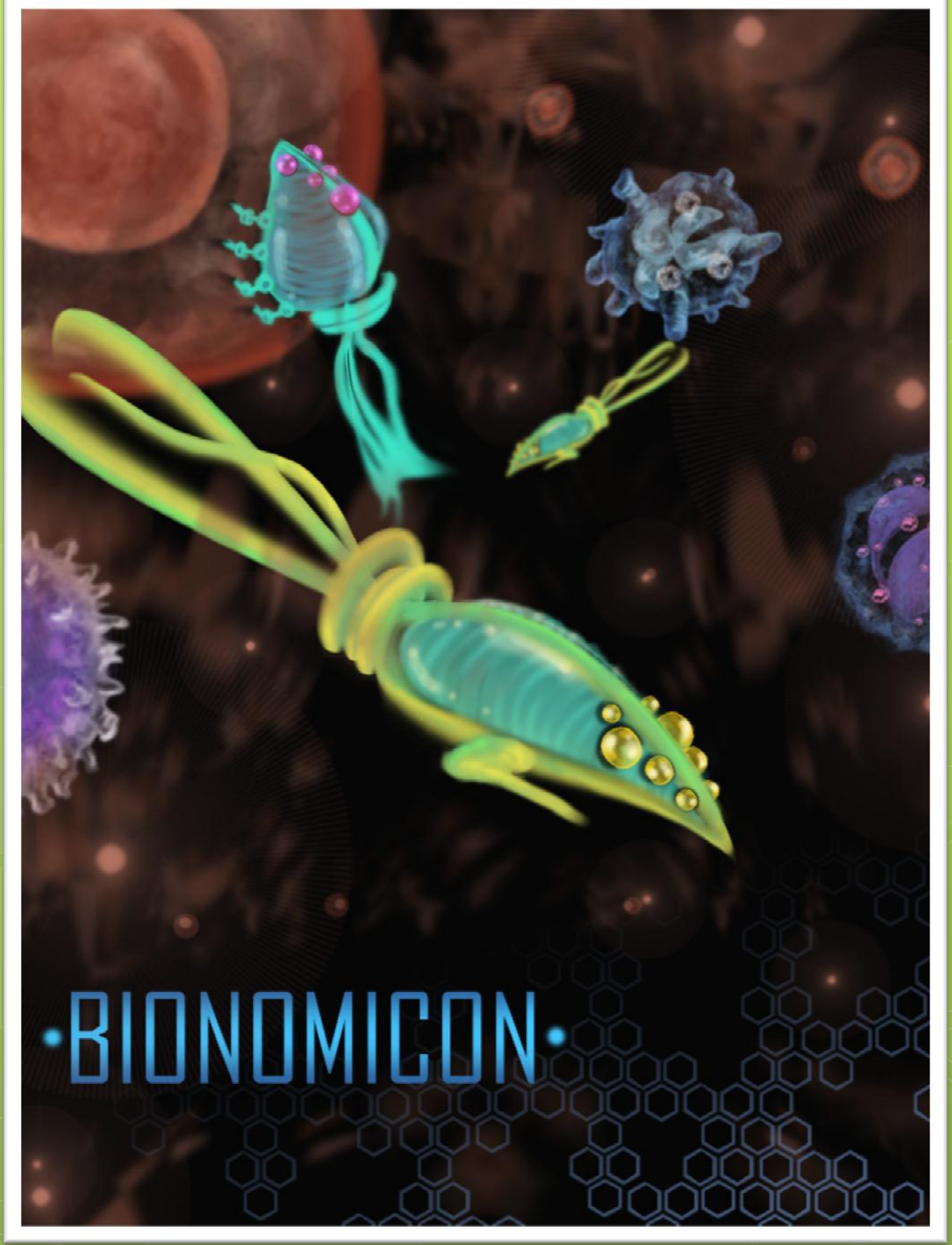

BIONOMICON

OVERVIEW

The apocalypse is coming.

Our immune system can't handle the post apocalyptic biological pathogens that will be thrown at it.

Our only chance of survival is to enhance our immune system and prepare for the unknown.

BIONOMICON

Bionomicon has developed advanced Artificial Life nano-technology capable of learning and adapting.

As a scientist at Bionomicon, your job is to train the nano-bots to fight along side the human immune system against pathogenic invaders.

SIMULATOR

To aid in the training of nano-bots, Bionomicon has developed a state-of-the-art “biological simulator”.

The simulator can recreate any environment inside the human body using holographic technology.

The human immune system and pathogen invaders are also simulated with advanced AI.

Think “Holodeck” from Star Trek TNG
but at a cellular level.

GAMEPLAY

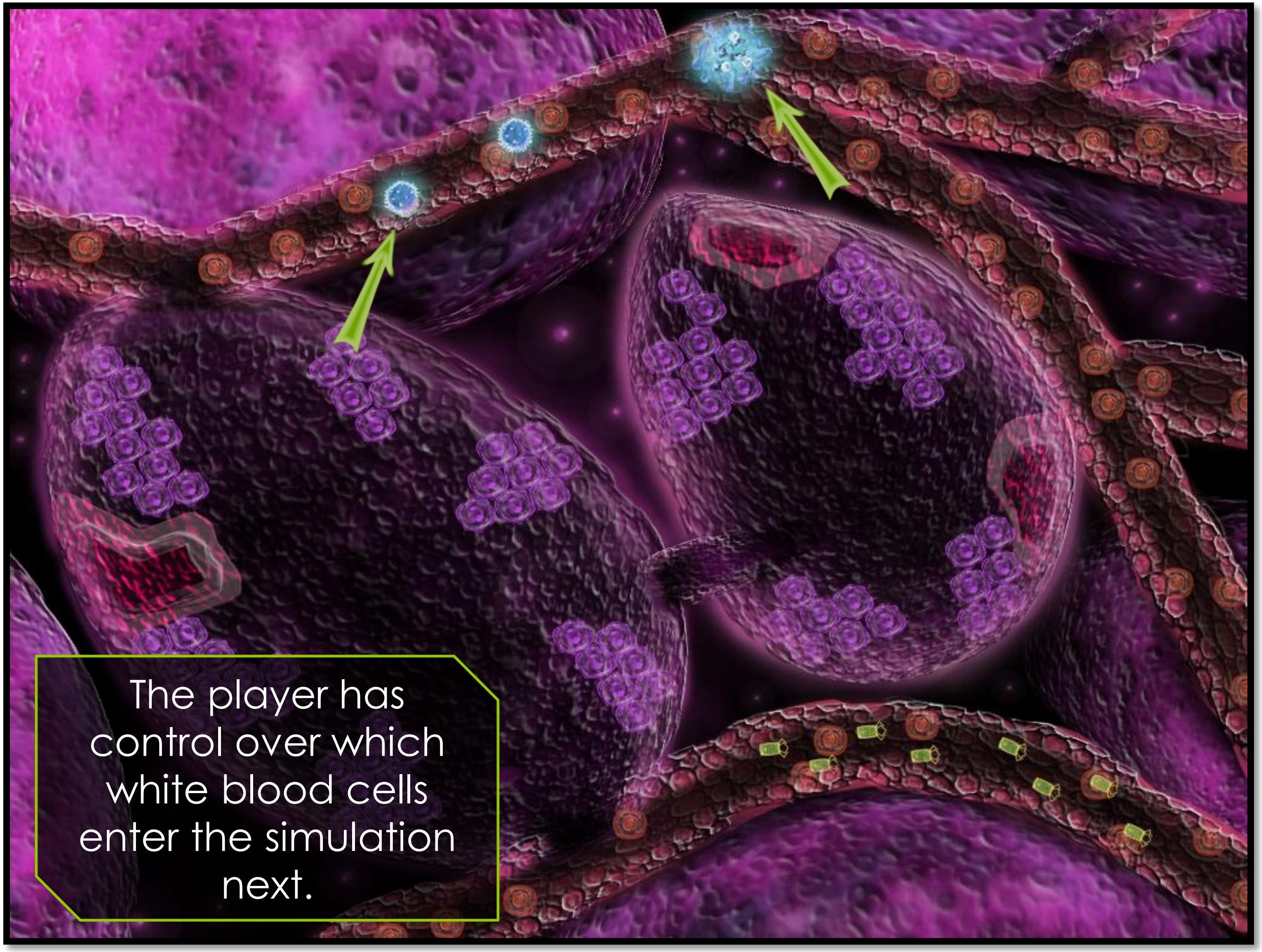
Game play draws inspiration from the Tower Defense and Real-Time Strategy genres.

GAMEPLAY

GAMEPLAY

Cells and Pathogens enter the simulation in waves through the blood vessels.

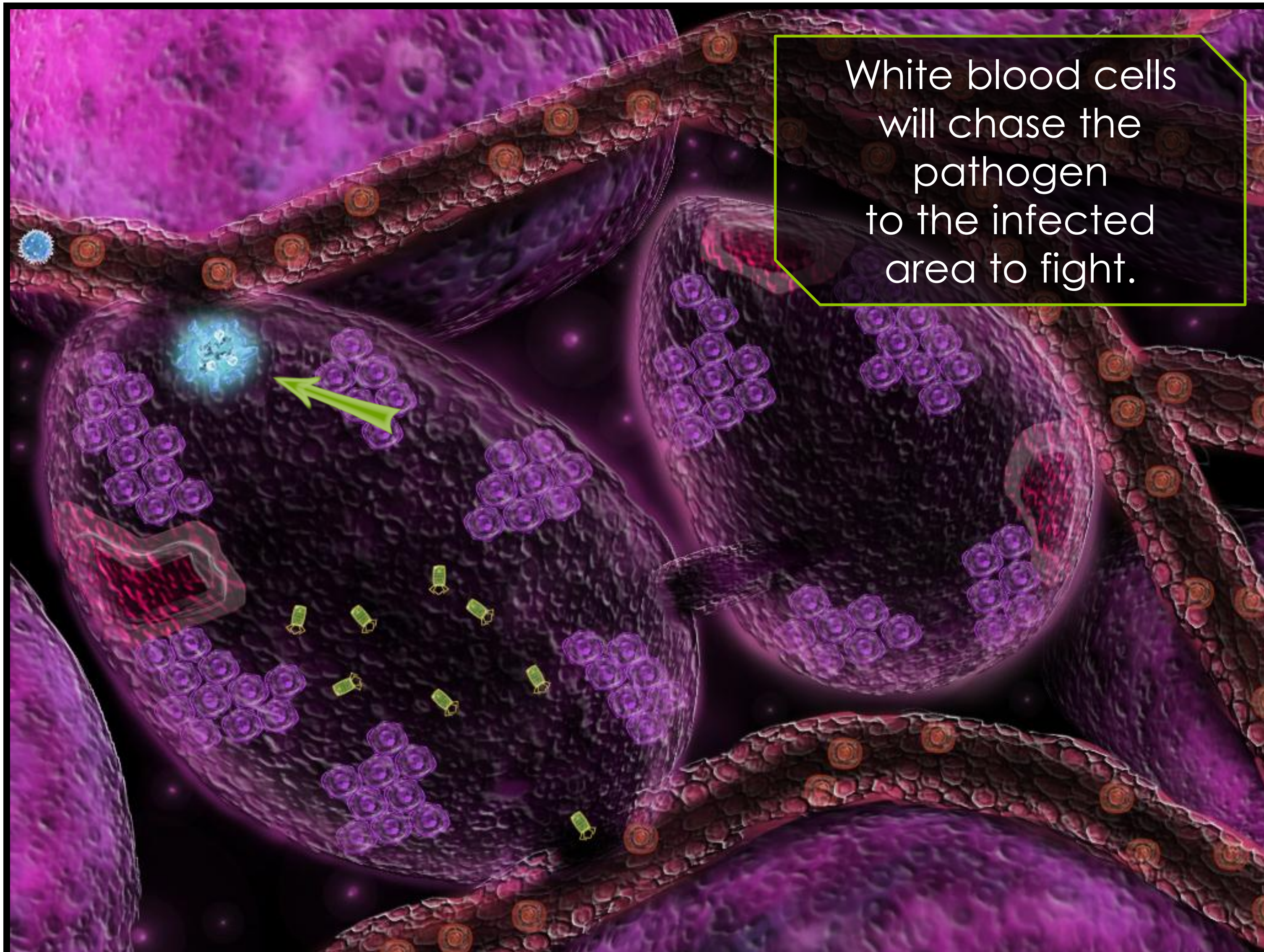
GAMEPLAY



The player has control over which white blood cells enter the simulation next.

GAMEPLAY

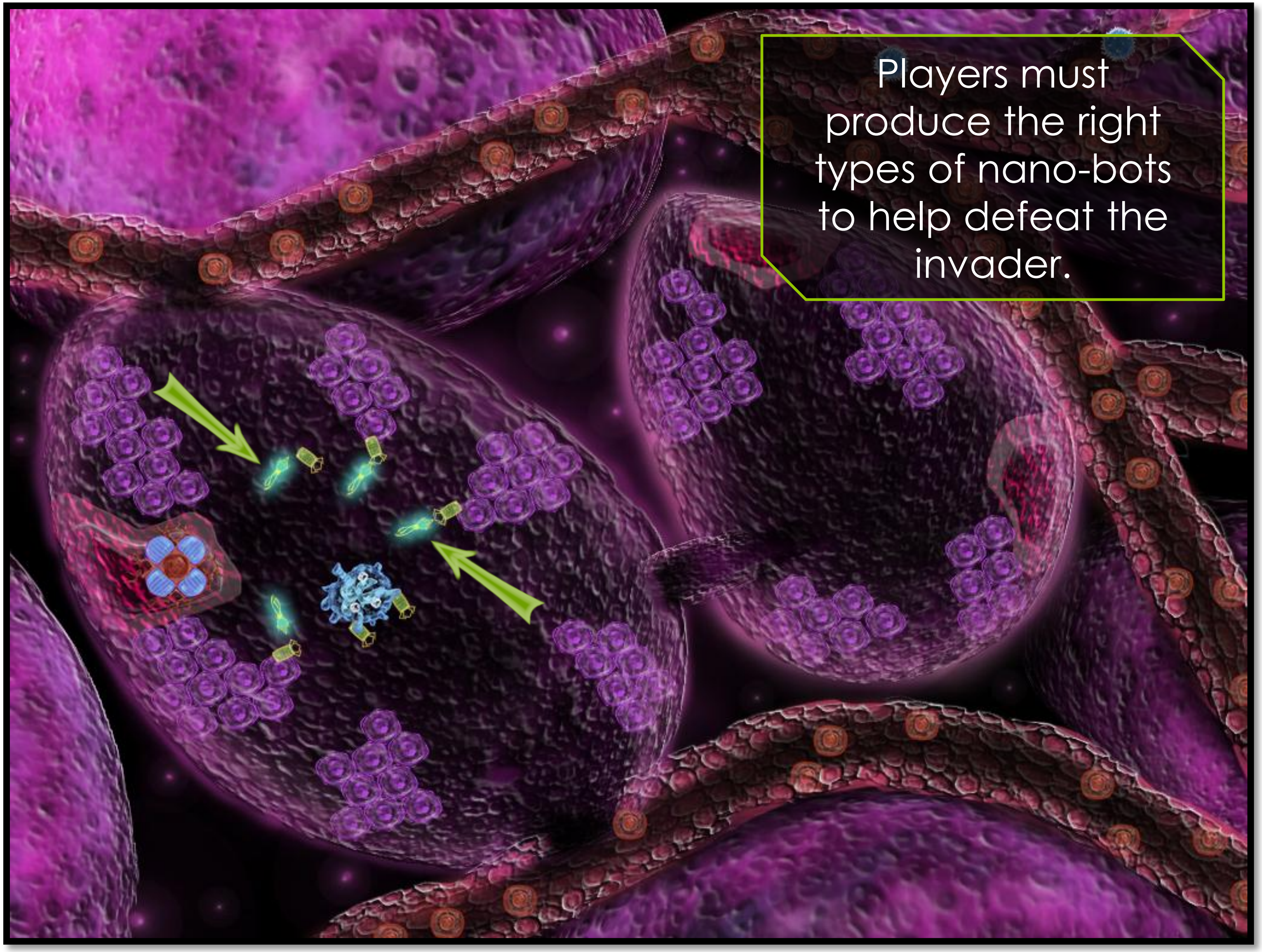
GAMEPLAY


White blood cells
will chase the
pathogen
to the infected
area to fight.

GAMEPLAY


Players build nano-factories where they can queue up various nano-bots for production.

GAMEPLAY



Players must produce the right types of nano-bots to help defeat the invader.

GAMEPLAY

Each scenario will have win conditions like "destroy all the pathogens and clean up the dead cells".

RESOURCES

The player will have to keep track of many resources during game play.

Cytokine levels control how many white blood cells can be called into an area. A player can choose which cells will enter the simulation next.

Immune system awareness. The player uses dendritic cells to inform the lymph node of the pathogen. This brings in more B and T cells coded to kill the pathogen.

Cell reproduction rate. Clumps of cells will slowly regenerate over time if damaged.

RESOURCES

Nano-materials are used for producing bots at a factory. The player will use Scavenger bots to collect dead cell remnants and expired nano-bots in the field.

Nano-rejection levels must be monitored and balanced during the simulation. Too much nano-technology in an area will cause the immune system to attack.

RESOURCES

Picture of Level with UI overlay showing different resources…

INSPIRATION

The mechanics of the immune system and various pathogens will be modeled as accurately as possible to their real-world counterparts.

This gives the player an unprecedented exposure to the core concepts of the actual human immune system through a game.

Our goal is to inspire players to learn more about the human immune system and deadly diseases - planting seeds for **future thinking** on the subjects.

FUTURE THINKING

BIONOMICON

Simulation Actors

HUMAN CELLS

Human cells are ultimately what the player is trying to protect and keep healthy during a simulation.

The simulator can model the full life cycle of a human cell including reproduction. Interactions with pathogens and cancerous states can also be simulated.

Cellular walls like the inside lining of the stomach or skin can be modeled as well as cell “clumps” with confined reproduction.

IMMUNE SYSTEM

The simulator models the behavior of the human immune system down to the cellular level.

Immune system actors include white blood cells like Neutrophils, Macrophages, Dendritic, B-Cells, and T-Cells.

Cytokines are a key communication mechanism used by the immune system to recruit reinforcements.

Players do not have direct control over the immune system units in a simulation. They can influence the type of cell the immune system should respond with.

CYTOKINES

Cytokines are used to attract white blood cells. They are produced by dying white blood cells and T-Helpers.

A certain amount of cytokines are required to attract white blood cells. For example, a small amount of cytokine is required to attract a Neutrophil but a large amount is required to attract the bigger Macrophage.

Cytokines dissipate over time. Therefore a player must keep cytokine levels high in order to attract bigger units throughout the scenario.

ANTIBODIES

Antibodies are released by B-Cells.

They attach to a pathogen and slow down movement.

Attaching to a pathogen can prevent cell penetration.

The attached antibodies also act as a “tagging” system letting other white blood cells in the area to consume the pathogen.

Antibodies do have a life span so the player is encouraged to keep the antibody levels high during a heavy battle.

IMMUNE SYSTEM CELLS

There are two types of immune system cells in the game; Innate and Acquired.

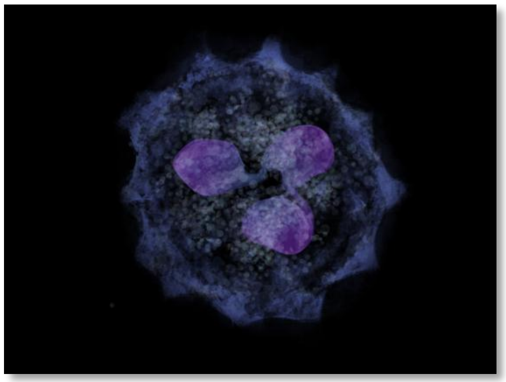
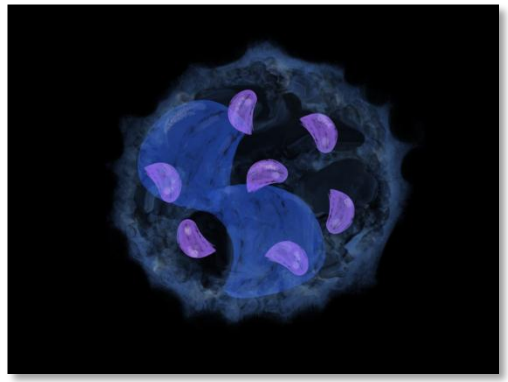
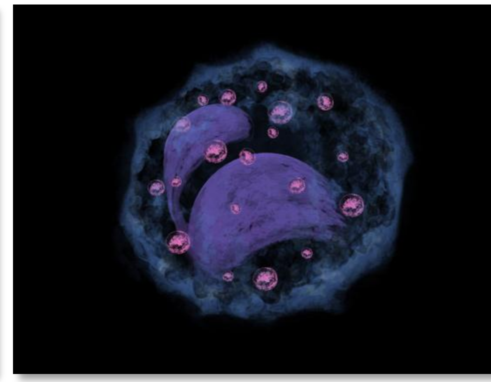
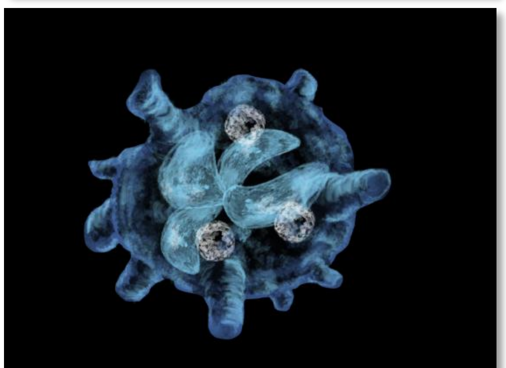
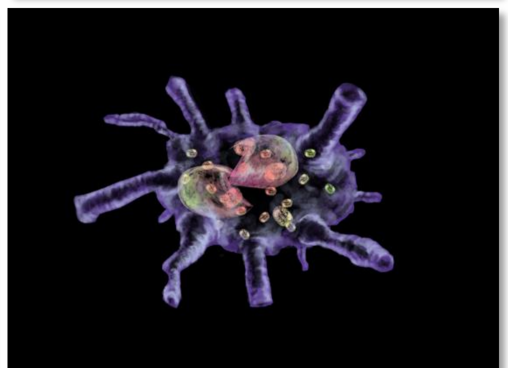
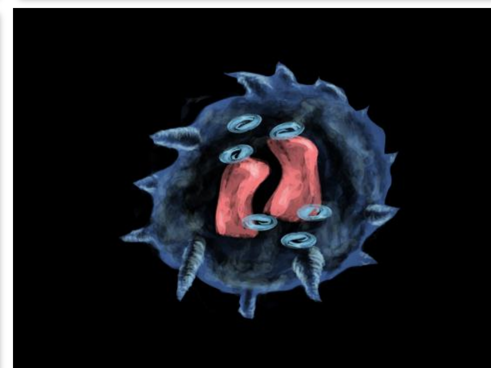
Innate cells will hunt down and kill anything that looks like a bad guy in the body.

Acquired cells are coded to kill a specific pathogen or infected cell.

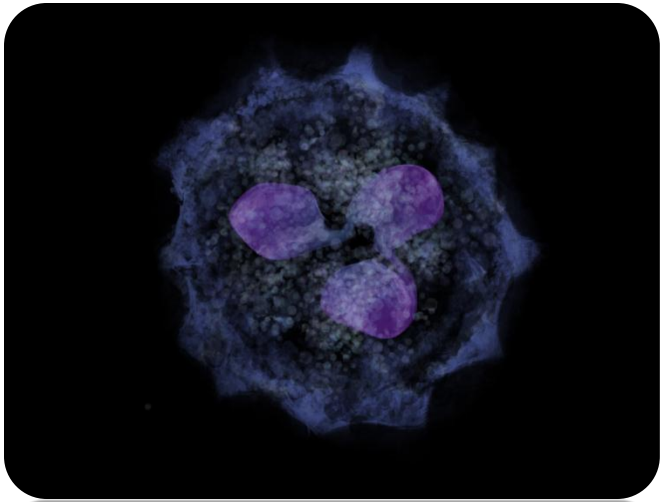
The fight against a pathogen usually starts with Innate units which then have the ability to call in Adaptive units.

INNATE CELLS

There are six types of innate immune system cells, each with a different benefit to the player.

NEUTROPHIL



Name: Neutrophil

Purpose: Consume pathogens

Attraction Level: 5 cytokines

Speed: Medium

Lifespan: 60 seconds or 1 pathogen

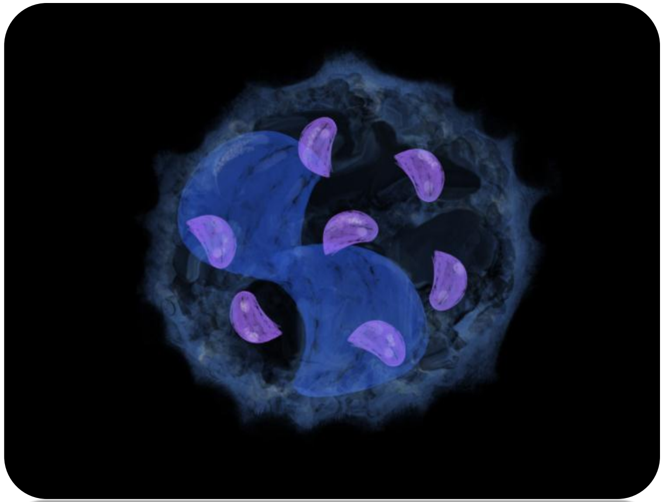
Death: Release 20 cytokines

Overview

Neutrophils are the most common white blood cells in our immune system. Their sole purpose is to consume pathogens.

After they consume a pathogen, they die. Just before death, they release cytokines which call in other white blood cells to the area.

BASOPHIL



Name: Basophil

Purpose: Consume pathogens

Attraction Level: 10 cytokines

Speed: Fast

Lifespan: 90 seconds or 1 pathogen

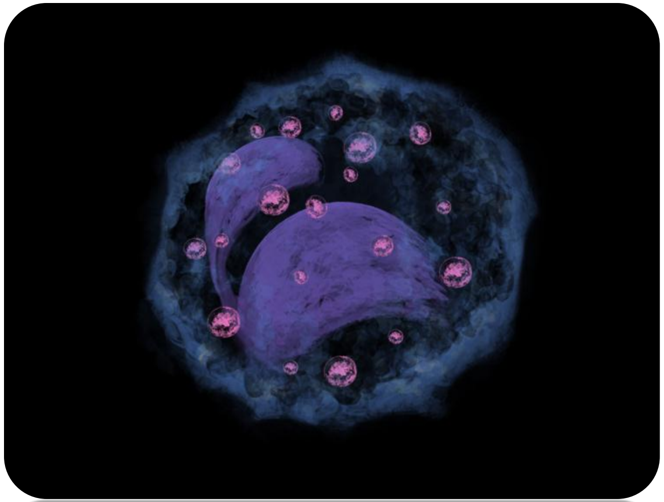
Death: Release histamine that slow nearby pathogens

Overview

Basophils, like neutrophils, are good at engulfing all basic types of pathogens. They are particularly effective against parasites.

Upon death, the Basophil releases histamines that will immobilize nearby parasites completely and slow down the movement of other types of pathogens.

EOSINOPHIL



Name: Eosinophil

Purpose: Consume pathogens

Attraction Level: 10 cytokines

Speed: Medium

Lifespan: 90 seconds or 1 pathogen

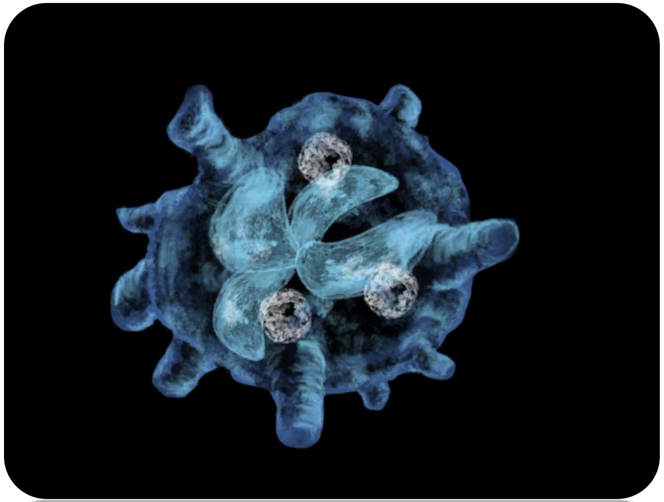
Death: Release toxic cloud that kills pathogens

Overview

Eosinophils release toxic proteins and free radicals that are especially good at killing bacteria and parasites.

Toxins can damage tissue, so these toxins must be tightly regulated by the body. There should be a limit of the number of Eosinophils that can be added to the system at once.

MACROPHAGE



Name: Macrophage

Purpose: Consume pathogens

Attraction Level : 25 cytokines

Speed: Slow

Lifespan: 120 seconds or 10 pathogens

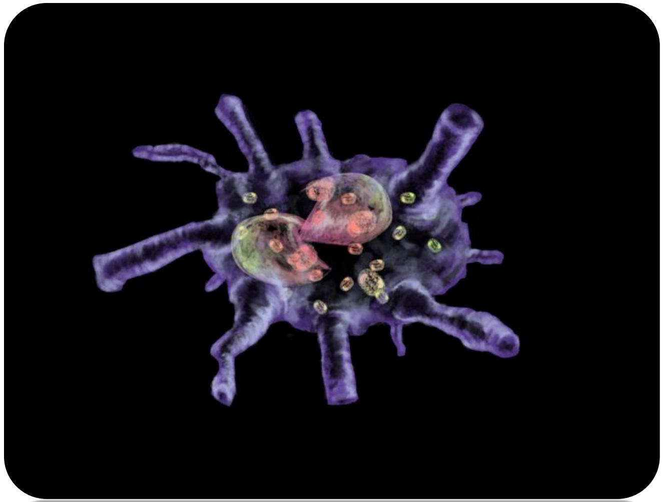
Death: Release 100 cytokines

Overview

Macrophages are the "big eaters" of our immune system. They can consume up to 100x more pathogens than a Neutrophil. They therefore require more cytokine in an area to be attracted and move very slowly.

After they consume 10 pathogens, they die. Just before death, they release cytokines which call in other white blood cells to the area.

DENDRITIC



Name: Dendritic

Purpose: Recruit B-Cells & T-Helpers

Attraction Level : 35 cytokines

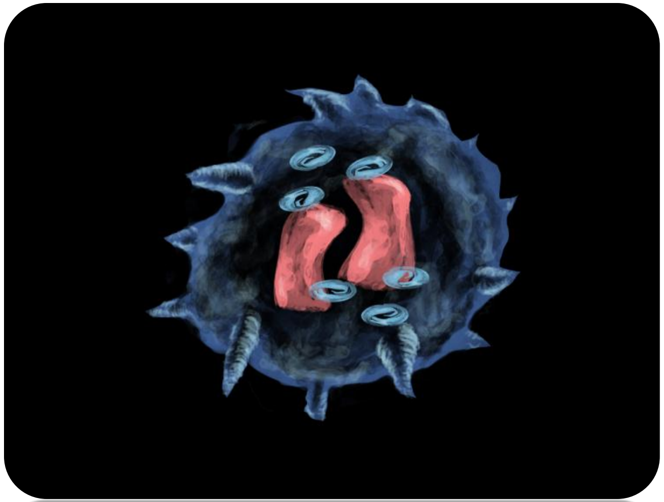
Speed: Fast

Lifespan: Consume 1 pathogen and bring it back to the lymph node. Upon completion of the trip, the player will have 1 additional B or T-Helper cell.

Overview

The Dendritic Cell is responsible for consuming a pathogen and then reporting back detailed information to the lymph node where B-Cells and T-Helpers are produced specifically to take out the specific pathogen.

NATURAL KILLER



Name: Natural Killer

Purpose: Kill infected or cancerous cells.

Attraction Level : 50 cytokines

Speed: Medium

Lifespan: 60 seconds

Activation: Bond with an infected or cancerous cell and release toxins into it.

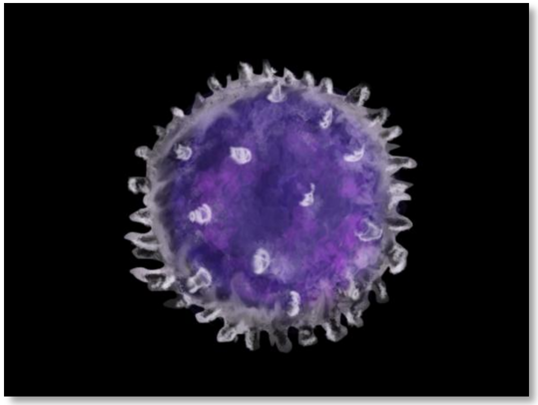
Overview

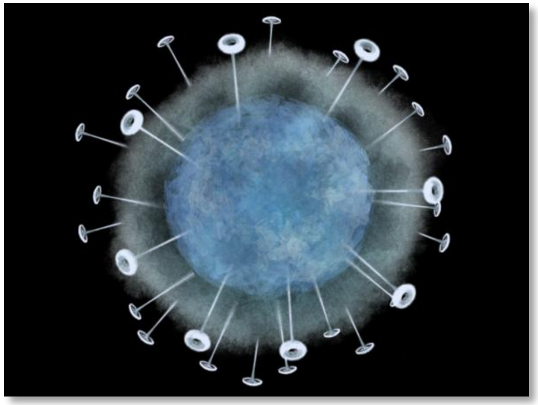
The Natural Killer cell has one job – to kill cells that it finds to be defective. It does this by injecting toxins into the cell and killing it from the inside. These toxins may leak out and damage nearby health cells.

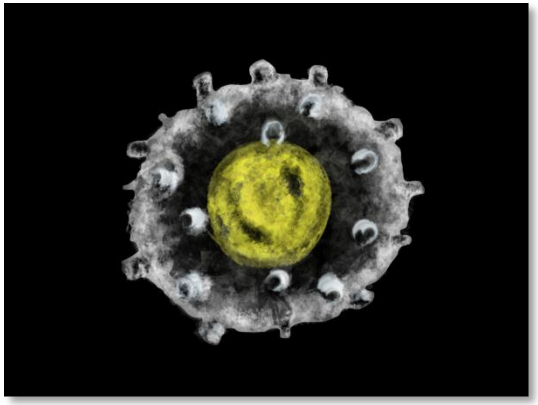
Because the NK cell is not “targeted” like a T-Killer cell, it can sometimes become confused and attack healthy cells. This is what leads to autoimmune disorders.

TARGETED CELLS

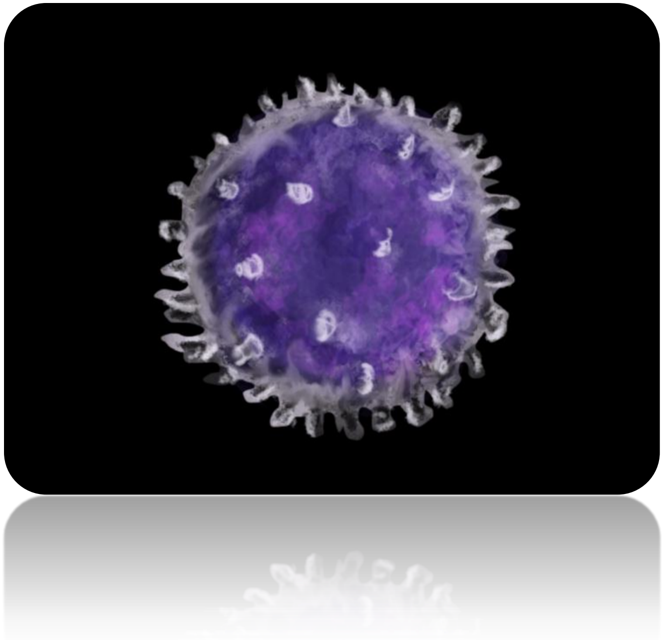
There are three types of adaptive immune system cells programmed into the simulator.







B-CELL



Name: B-Cell

Purpose: Release antibodies to slow down and tag pathogens for death.

Speed: Medium

Lifespan: Until the simulation is complete.

Activation: Consume a pathogen and release antibodies.

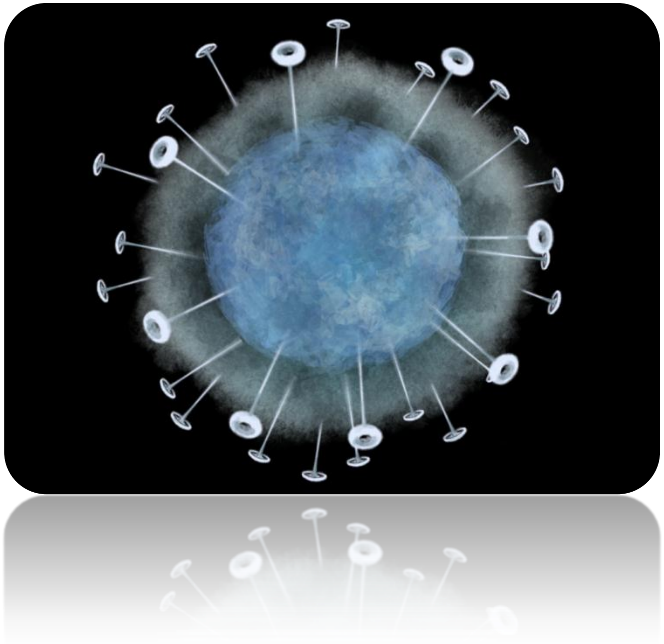
Activation: Replicate a B-Cell when stimulated by a T-Helper.

Overview

B-Cells are coded to target a specific pathogen. When they consume the pathogen, they become "activated" and release antibodies.

The antibodies hunt down and attach themselves to the specific pathogen. This slows down the pathogen and also tags it for termination by Macrophages.

T-HELPER



Name: T-Helper

Purpose: Attract T-Killers and stimulate B-Cells.

Speed: Fast

Lifespan: Until the simulation is complete.

Activation : Bond with a WBC that has consumed a pathogen and release 10 cytokines.

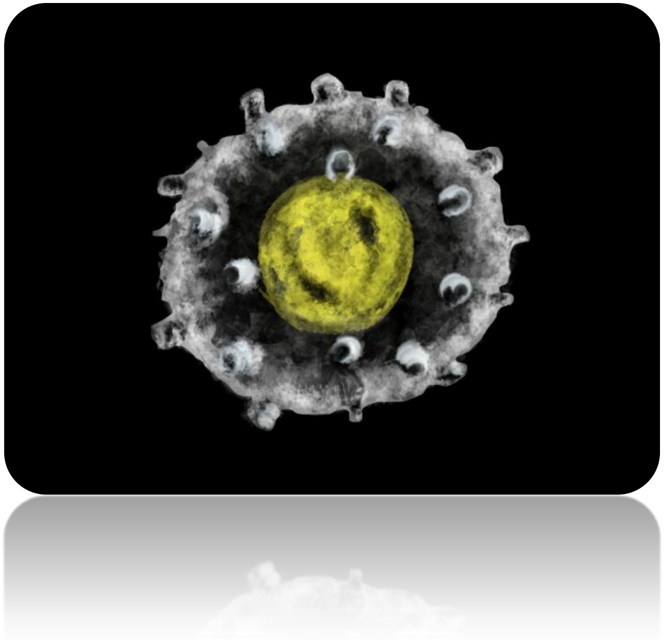
Activation : Bond with an infected or cancerous cell and release 10 T-cytokines.

Activation : Bond with a B-Cell to stimulate replication and increase antibody production.

Overview

T helper cells are a type of T cells that play an important role in the immune system, particularly in the adaptive immune system. They help the activity of other immune cells by releasing T cell cytokines. They are essential in B cell antibody class switching, in the activation and growth of cytotoxic T cells, and in maximizing bactericidal activity of phagocytes such as macrophages.

T-KILLER



Name: T-Killer

Purpose: Kill infected or cancerous cells.

Attraction Level : 100 T-cytokines

Speed: Medium

Lifespan: Until the simulation is complete.

Activation: Bond with an infected or cancerous cell and trigger Apoptosis (programmed cell death).

Overview

A T-Killer cell is a T lymphocyte (a type of white blood cell) that kills cancer cells, cells that are infected (particularly with viruses), or cells that are damaged in other ways.

Most cytotoxic T cells express T-cell receptors (TCRs) that can recognize a specific antigen. An antigen is a molecule capable of stimulating an immune response, and is often produced by cancer cells or viruses.

RECRUITMENT

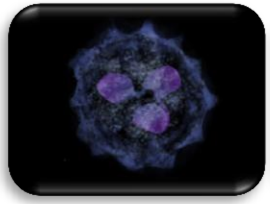
White blood cells will enter the simulation at pre-determined intervals (ex. every 15 seconds).

Players have the ability to select which white blood cell enters the simulation on the next interval.

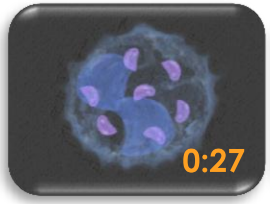
The available white blood cells to enter the simulation will be dependent on Cytokine levels, a per-unit “cool down”, and Dendritic cell activity.

RECRUITMENT

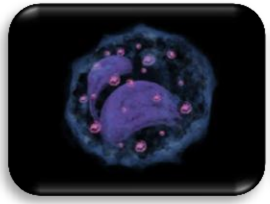
The interface for selecting the next white blood cell to enter the scene should show availability and requirements needed to recruit.



Neutrophil
READY!

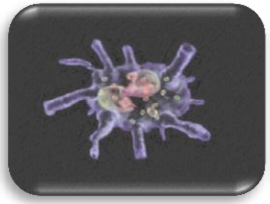


Basophil
WAITING

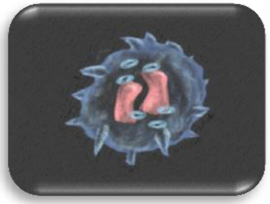


Eosinophil
READY!

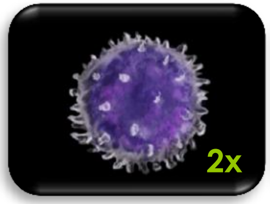
Macrophage
WAITING



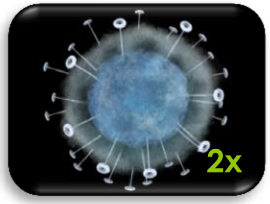
Dendritic
REQ 35 Cytokine



Natural Killer
REQ 50 Cytokine

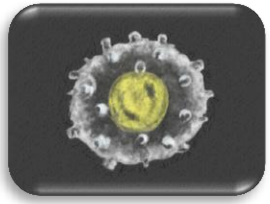


B-Cell
READY!



T-Helper
READY!

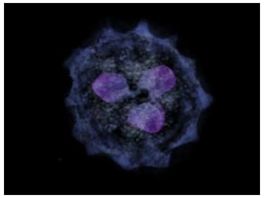
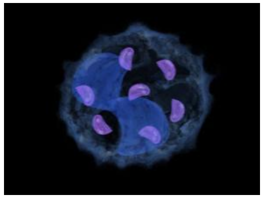
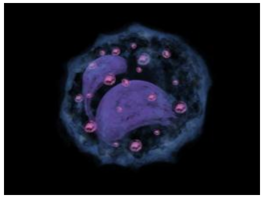
Use Dendritic cells to increase inventory of B-Cells and T-Helpers.



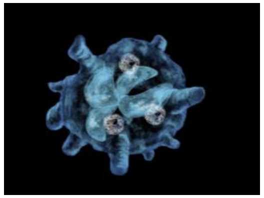
T-Killer
REQ 100 T-Cytokine

TYPICAL USE

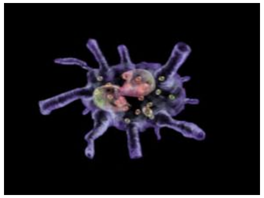
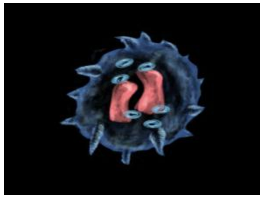
Start the fight with "phils" and build up cytokine levels.

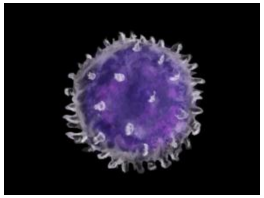
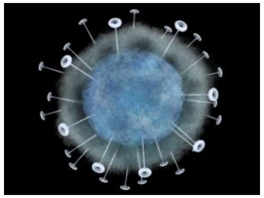
Bring in the Macrophages to kick some major ass and continue to raising cytokine levels.



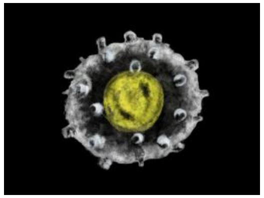
Call in the Dendritic to raise Lymph Node awareness. Call in Natural Killers if cells need a killin'.

Call in B-Cells and T-Helpers. These are big "hero" units that tear things up and signal for the big boy.

Call in the big T-Killer to wipe out infected or cancerous cells!



NANO-BOTS

Nano-bots are the primary mechanism for the player to influence the outcome of a simulation.

They are built specifically for one purpose much like a medicine or pharmaceutical drug.

Classes of nano-bots the player will have access to include: assassins, cleaners, repair, protection, trappers, enhancers, and amplifiers.

Nano-bots will learn and adapt to their job the more a player uses them. This is represented by leveling system with a skill tree for each class of nano-bot.

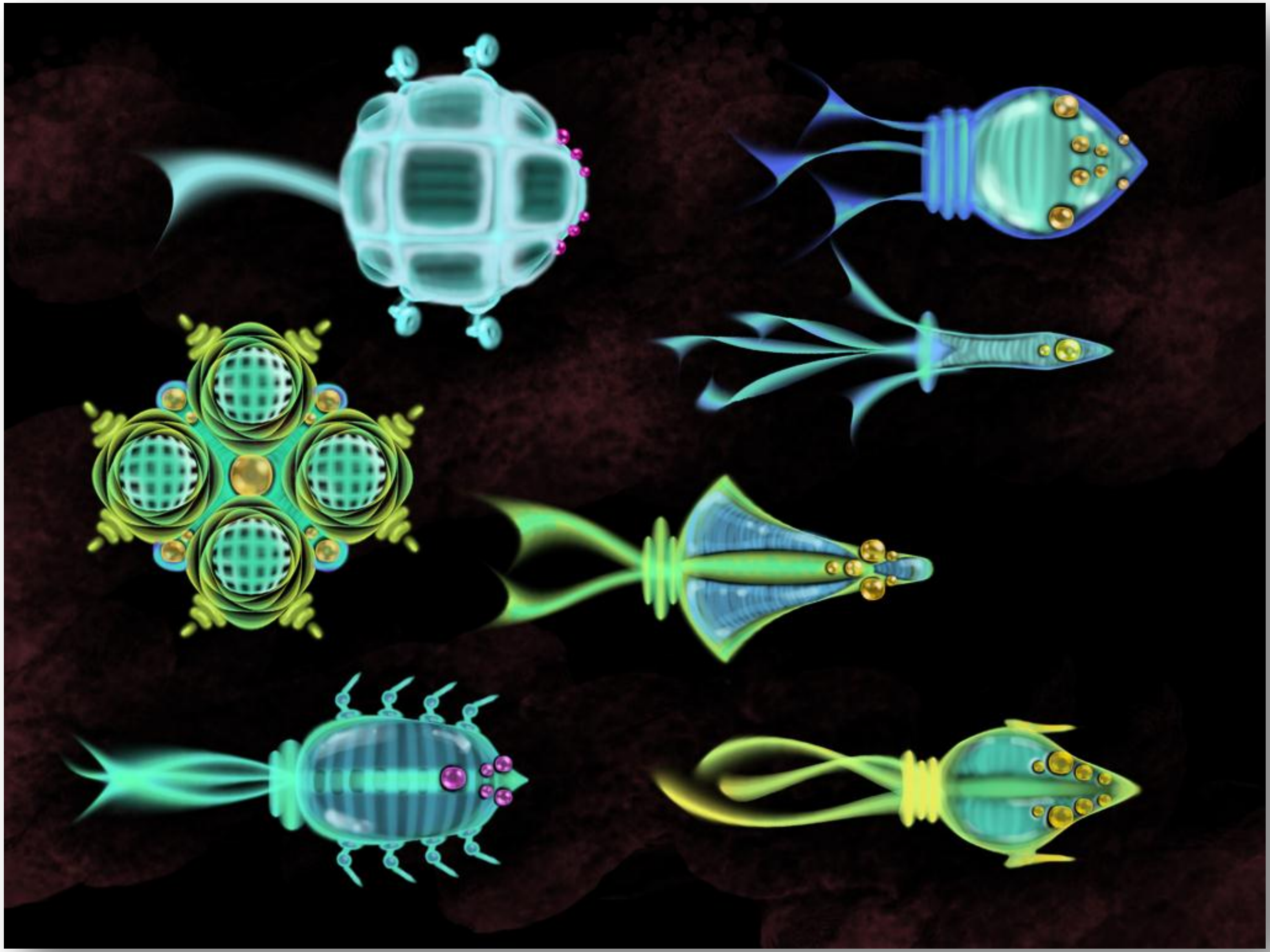
NANO-BOTS

Each nano-bot that has a fuel cell that is expended as they work. The nano-bot will return to a nano-factor to refuel when needed.

Nano-bots that run out of fuel in the field can be collected by a Scavenger or refueled by a Repair bot.

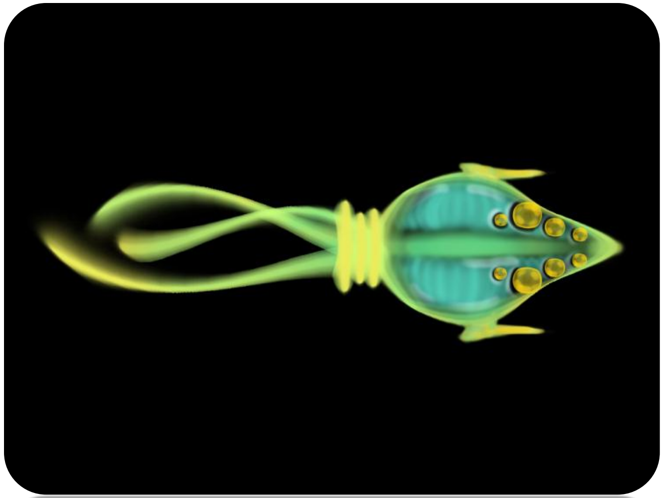
A player can retire a nano-bot that is no longer serving a purpose in the simulation. This will reduce the nano-rejection levels.

NANO-BOTS



Bionomicon has used deep sea biology research to inspire their nano-bot designs.

VIPER



Name: Viper

Purpose: Hunt and kill pathogens

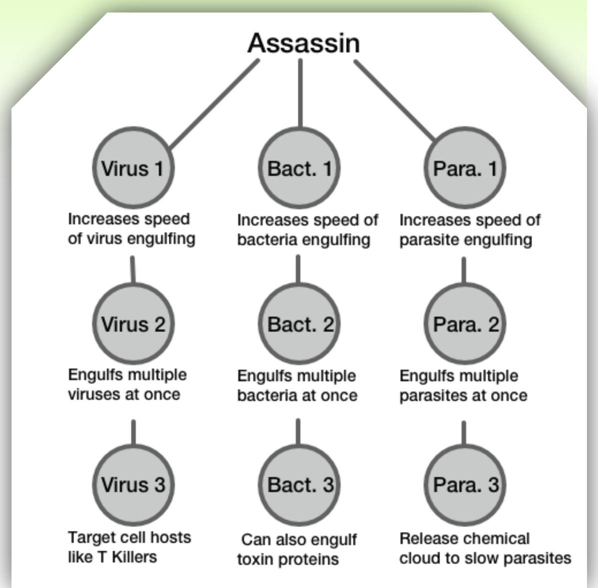
Size: Small

Speed: Fast

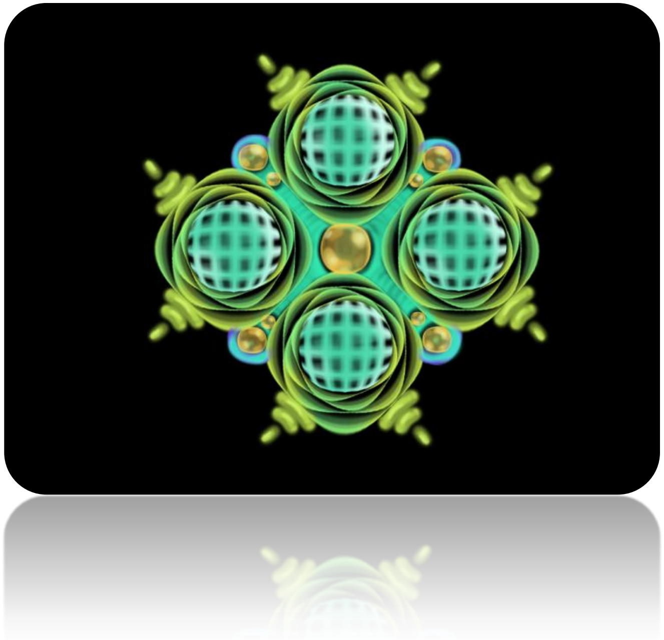
Genesis Time: Longest

Description

Vipers have an aerodynamic form with 3 flagellum for efficient maneuvering. They also have four grabber arms to attach to pathogens.



GULPER



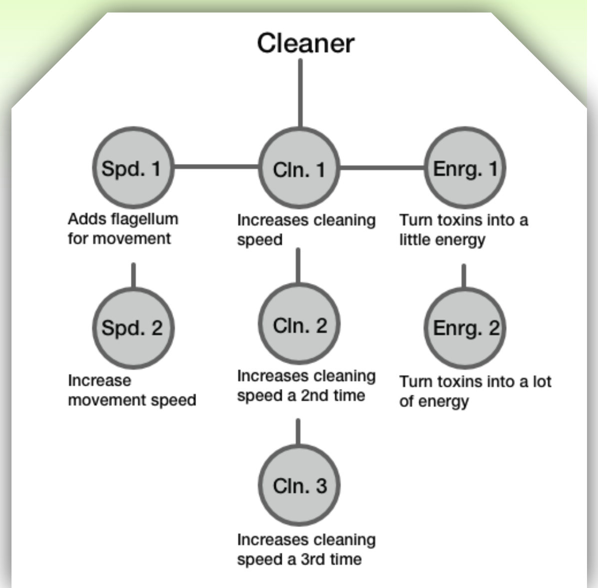
Name: Gulper

Purpose: Clean toxins produced by pathogens

Size: Large

Speed: Slow

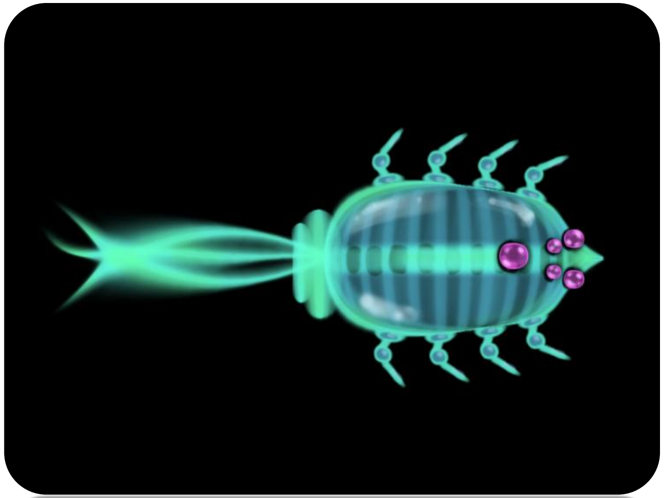
Genesis Time: Average



Description

Gulpers have large pouch like vacuoles that hold all the collected toxin proteins. These vacuoles really add to the size of the nano-bot. Small grabber arms are located around the vacuoles, and the bot uses them to efficiently grab toxin proteins and push them into the vacuoles.

AMPHIPOD



Name: Amphipod

Purpose: Repair damaged cells and refuel nano-bots.

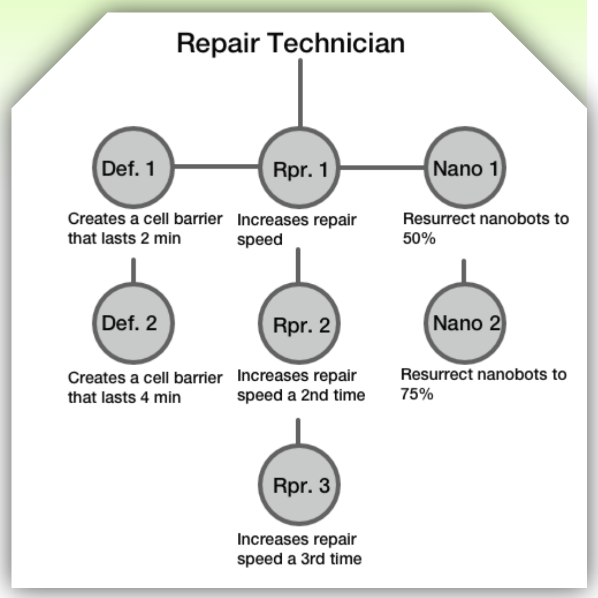
Size: Average

Speed: Average

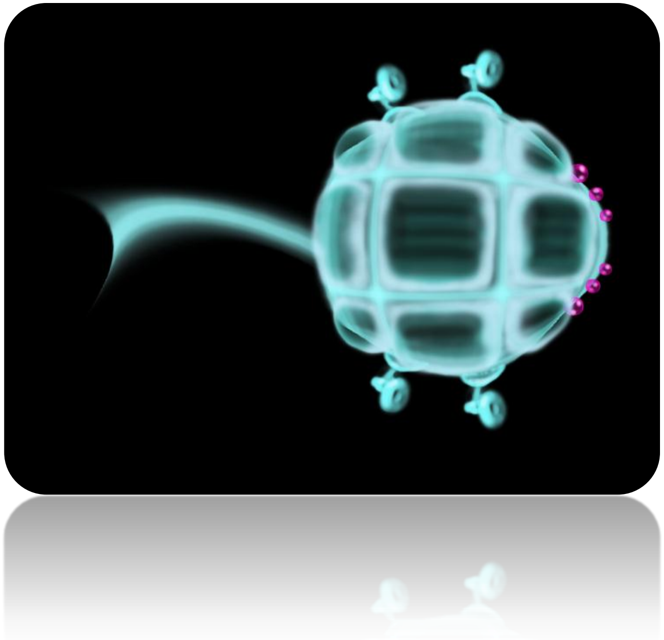
Genesis Time: Average

Description

The Amphipod has many grabbers that help the nano-bot latch onto cells, and it then releases chemicals that help the cell bind together again.



NAUTILUS



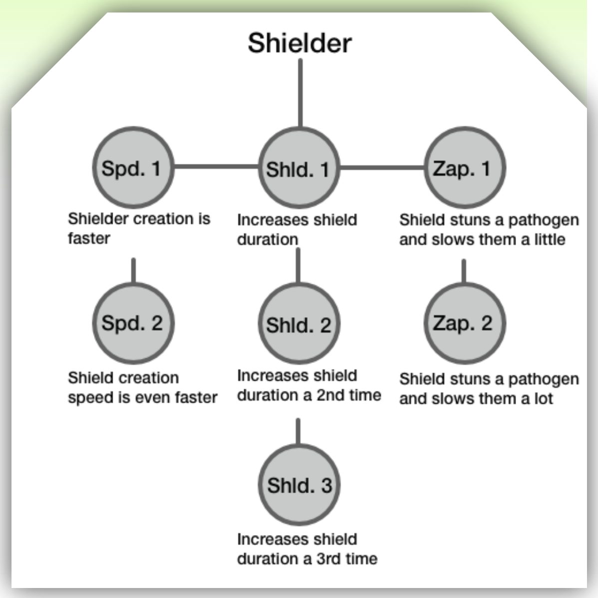
Name: Nautilus

Purpose: Create protective shield around cells

Size: Large

Speed: Slow

Genesis Time: Long

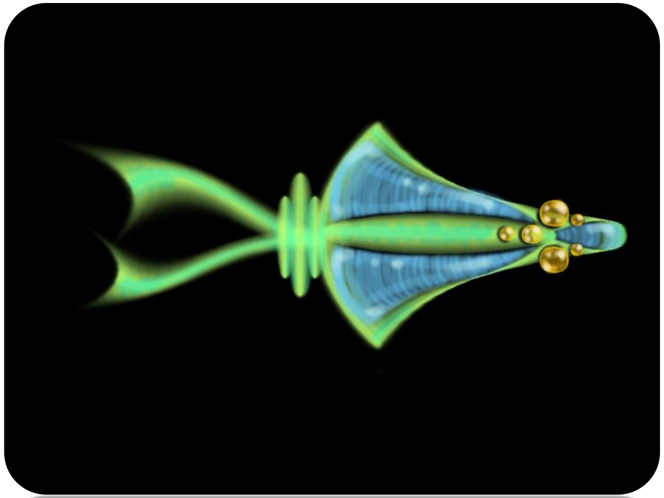


Description

Nautilus bots are bulky and slow. They hold a lot of proteins to build shields.

Nautilus bots should have a large bulky quality with a single flagellum that helps it navigate. It should have a "plate-like" quality, with large metal pieces that give a feeling of an armored unit.

ANGLER



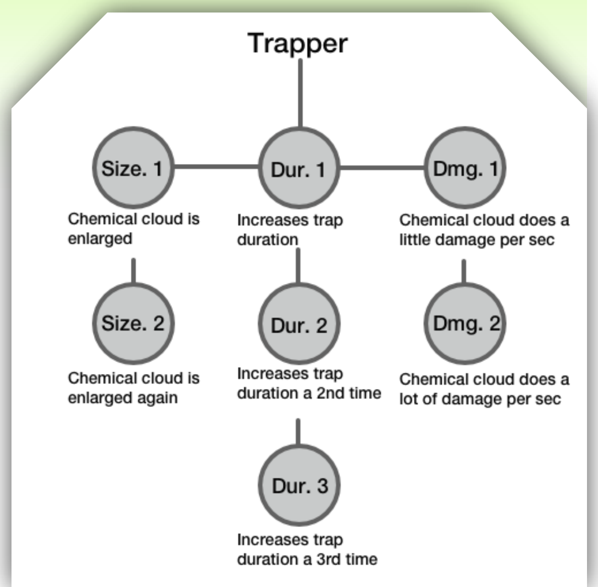
Name: Angler

Purpose: Release chemicals that slow down pathogens

Size: Average

Speed: Average

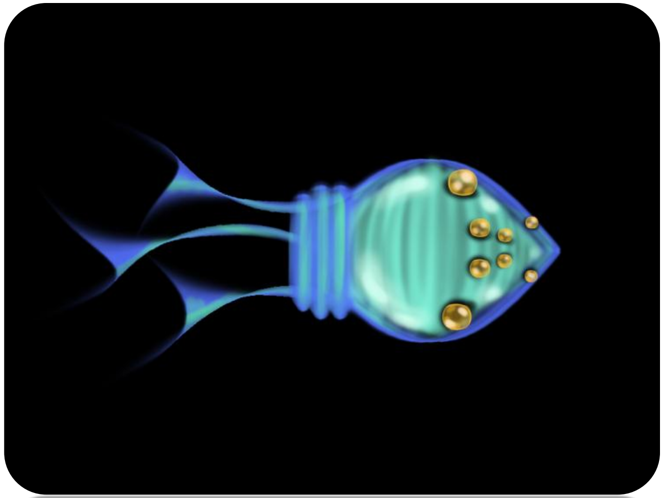
Genesis Time: Average



Description

Anglers are average sized, and they synthesize their own chemicals internally (hence the cool down time). They have two propulsion flagellum to help move efficiently, and they have lots of openings to disperse the chemicals.

CYTOSQUID



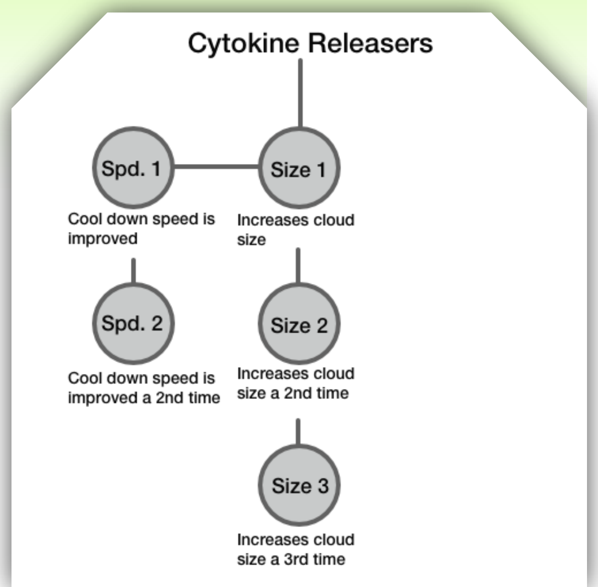
Name: Cytosquid

Purpose: Release cytokines

Size: Average

Speed: Fast

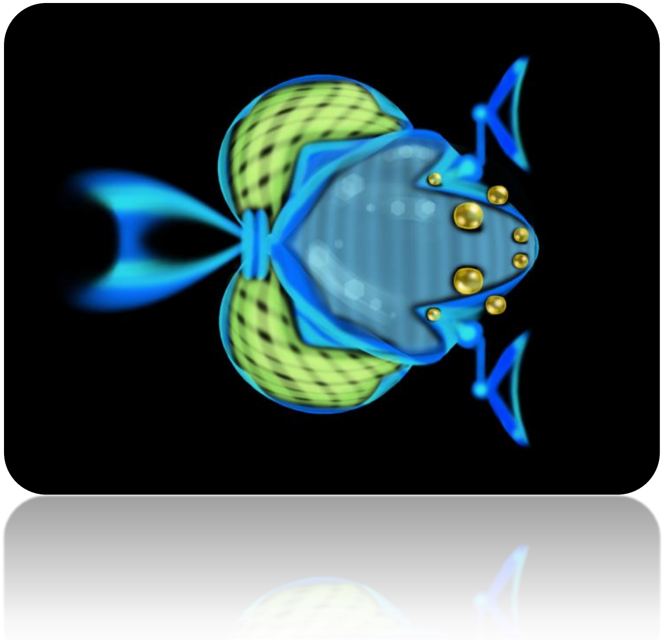
Genesis Time: Fast



Description

Cytosquids are round, porous, and small. They release a small cloud of cytokines then have a cool down timer before they can release more. They generally move close to white blood cells before releasing cytokines.

ISOPOD



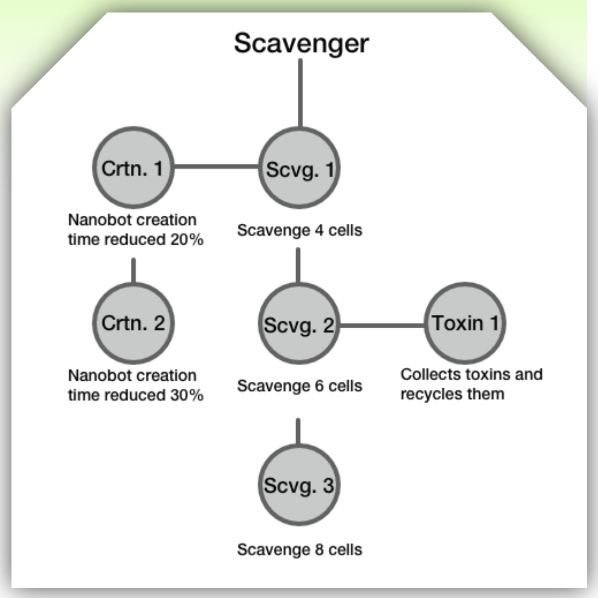
Name: Isopod

Purpose: Collect materials for nano-bot construction

Size: Average

Speed: Slow

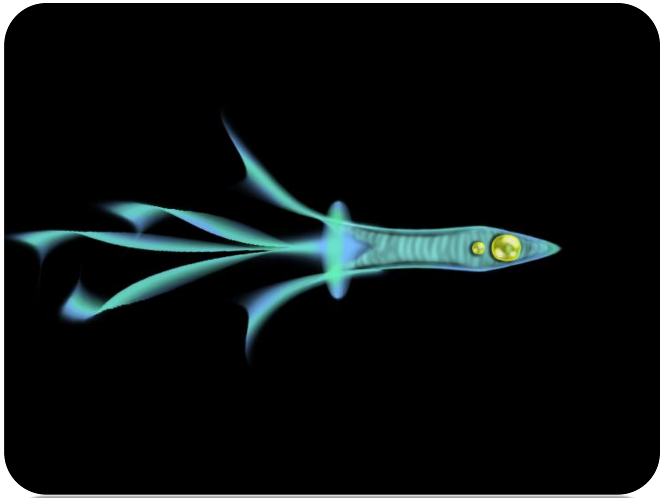
Genesis Time: Average



Description

These nano-bots hunt down dead cells or stranded nano-bots and bring them back to the factory to help with unit production.

CHIMAERA



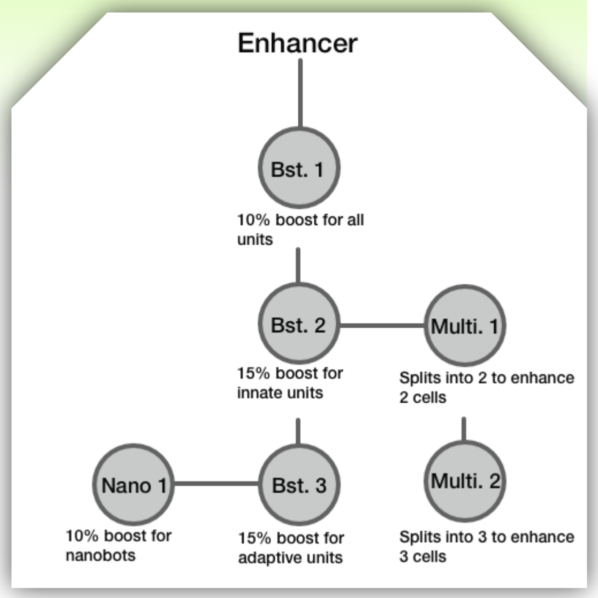
Name: Chimaera

Purpose: Augment the function of a white blood cell

Size: Small

Speed: Fast

Genesis Time: Super Fast

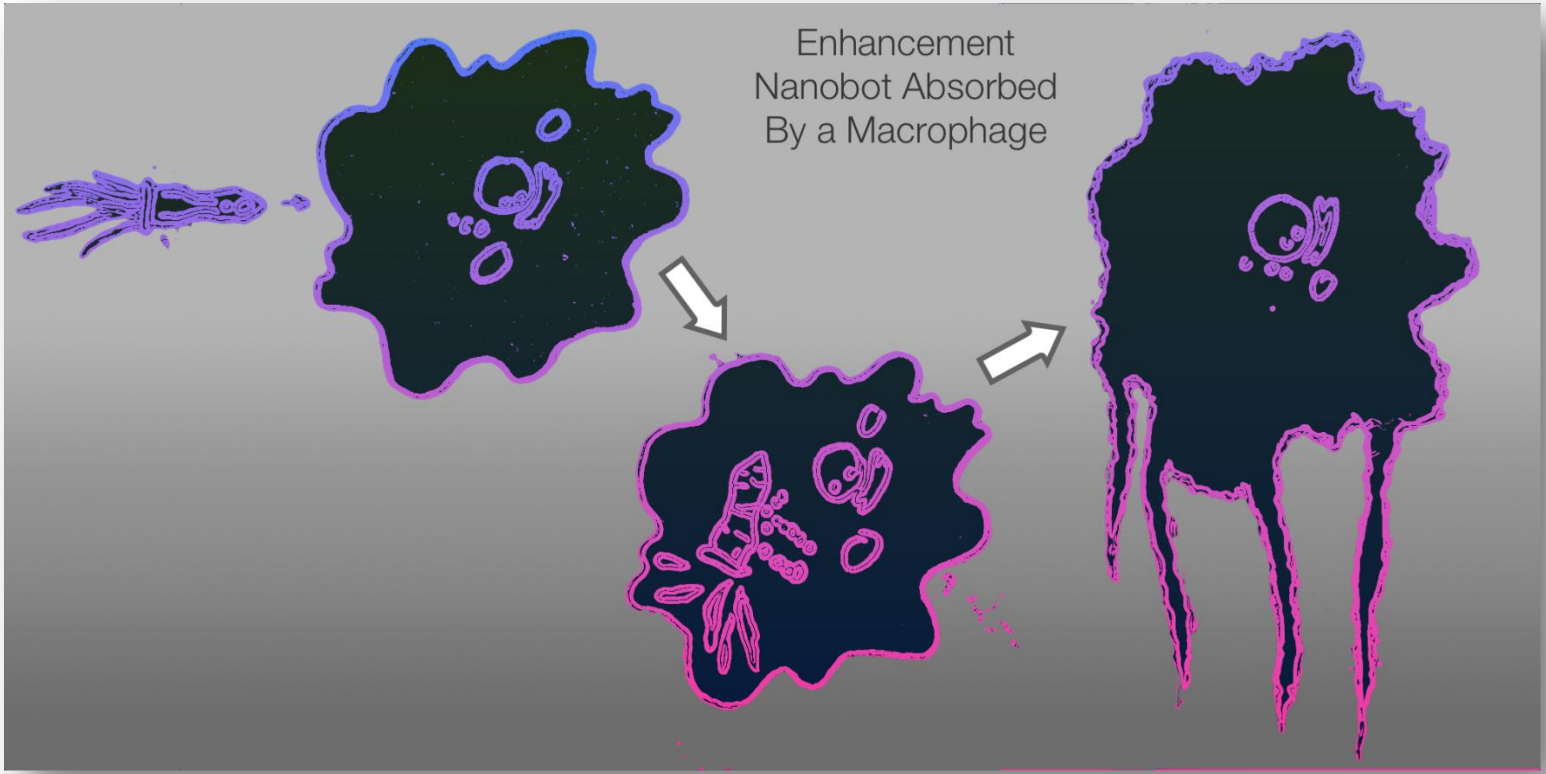


Description

These oddball nano-bots travel through the level and bind with white blood cells; augmenting movement speed, life span, engulfing capacity, and other unique stats.

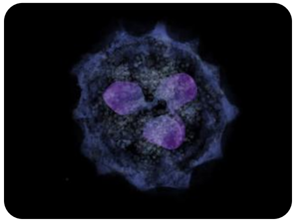
Enhancers are very small, smooth, and agile with a number of flagellum.

ENHANCER

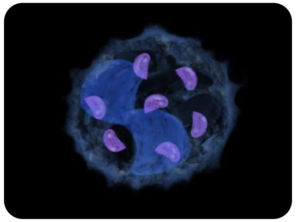


The Enhancer bot enters a cell much like a virus and hijacks the inner workings of the cell.

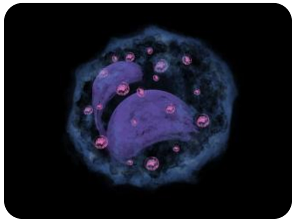
ENHANCER



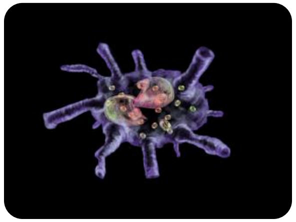
Neutrophil
10% speed boost
10% more cytokines
20% longer lifespan
Consume 2 pathogens



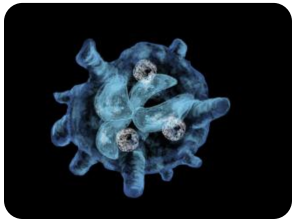
Basophil
10% speed boost
15% more histamines
20% longer lifespan
Consume 2 pathogens



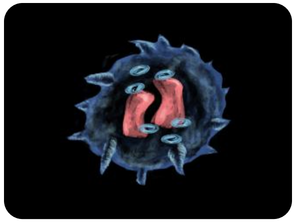
Eosinophil
10% speed boost
15% more toxins
20% longer lifespan
Consume 2 pathogens



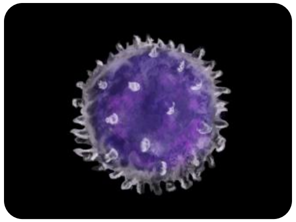
Dendritic Cell
20% speed boost
20% longer lifespan



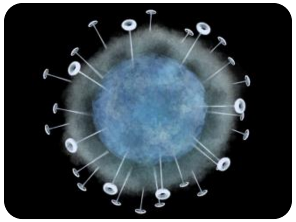
Macrophage
10% speed boost
10% more cytokines
20% longer lifespan
Consume 8 pathogens



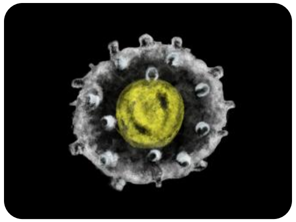
Natural Killer
10% speed boost
15% more toxins
20% longer lifespan



B Cell
10% speed boost
15% more antibodies
+1 memory cell



T Helper
20% speed boost
10% more cytokines
+1 memory cell



T Killer
20% speed boost

Each type of white blood cell is augmented in unique ways by the Enhancer bot.

NANO-FACTORIES

Nano-factories produce nano-bots during the simulation.

Players will place and construct nano-factories at key locations in a level.

Nano-factories are akin to towers in a Tower Defense game with the distinction of being able to produce any type of unit.

Players will queue up the production of nano-bots, much like a factory in a Real-Time Strategy game.

NANO-FACTORIES

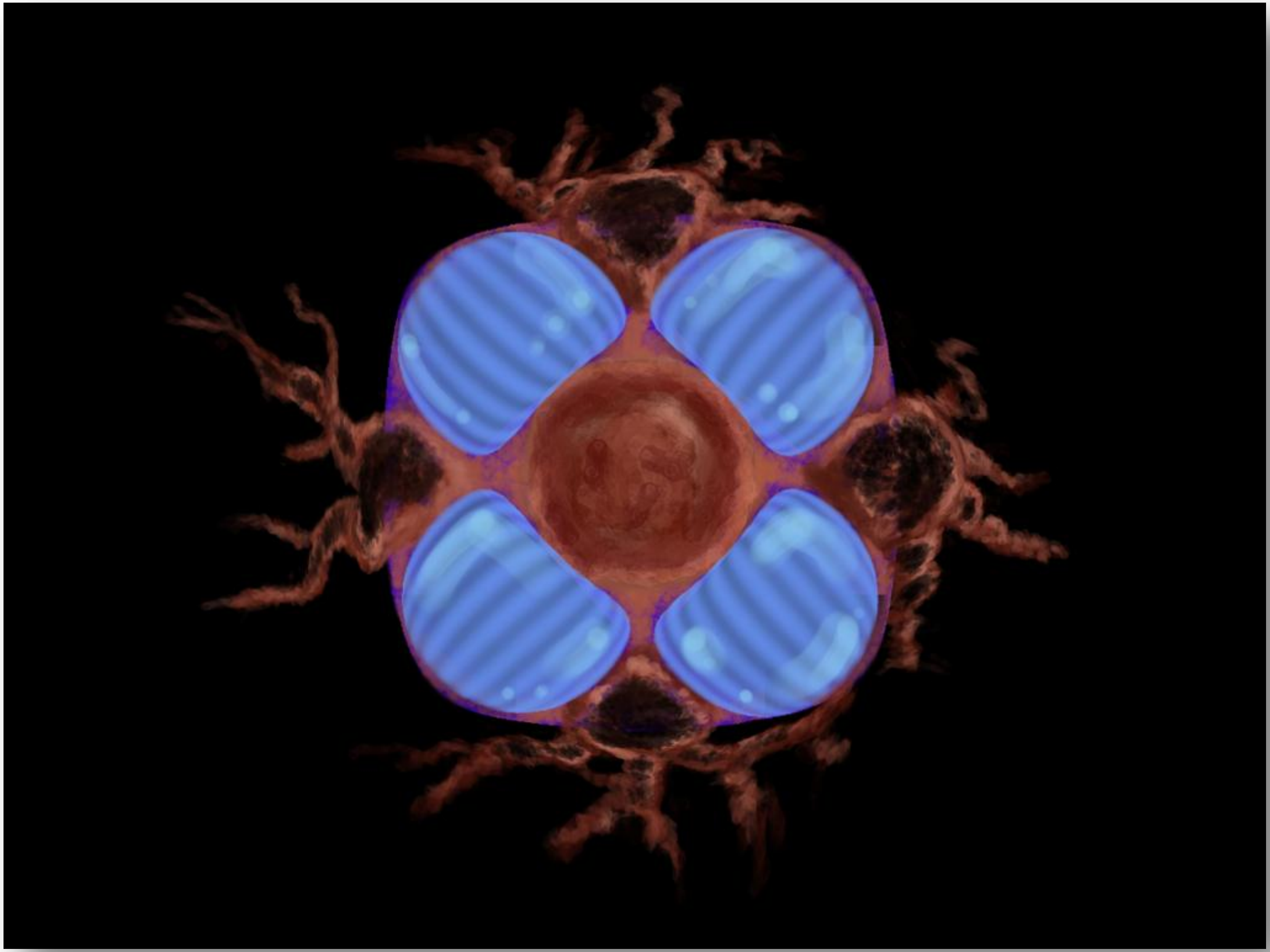
Nano-bots take a certain amount of time to produce. When construction is complete, the nano-bot is launched out of the factory and into the simulation.

There can only be a limited amount of nano-tech in an area before the immune system will attack it.

Nano-factories can be upgraded to provide nano-bot enhancements over the course of the simulation.

Factory upgrades improve the nano-bots they produce with enhancements like increased speed, better navigation, longer lifespan, and reduced genesis time.

NANO-FACTORIES



PATHOGENS

The simulator is programmed to model the following common classes of pathogen: Viruses, Bacteria, and Parasites.

Most pathogens target specific areas of the human body to wage war. Anthrax goes after the lungs, Malaria starts in the liver and ends up in the blood stream, etc.

The simulator will holographically generate an appropriate environment for the particular pathogen the player is fighting.

VIRUS

Viruses are the most common type of pathogen that the immune system encounters.

There are currently 170 known virus strains, represented by an H (1-17) and N (1-10) identifier.

A virus will penetrate a human cell, hijack it's internal replication engine, and produce more viruses.

Those new viruses spread to nearby cells and continue replicating.

BACTERIA

Bacteria are different than viruses in that they have fully functional cells.

This means that bacteria are able to survive and reproduce on their own.

Bacteria commonly form colonies and cause infections when they begin to reproduce within the human body.

PARASITE

A parasite is an organism that lives on or inside another organism (the host) and harms the host.

Parasites use human cells as a host to reproduce.

Parasitic offspring typically have a specific function like attacking red blood cells.

BIONOMICON

Scenarios

SCENARIOS

Simulated “scenarios” pit the player and their nano-bots against known and unknown pathogenic invaders.

Early in the game, these scenarios will include well-known diseases like Cancer, Diabetes, HIV, Malaria, and Anthrax.

As the game progresses, the player will take on fictional pathogens that might be encountered after the apocalypse.

SCENARIOS

We've started with level designs for some more prominent diseases. There is no end to the other diseases that levels could be built from including:

Smallpox, West Nile, Ebola, Hepatitis, Meningitis, Strep Throat, Measles, Polio, Typhoid Fever, Yellow Fever, Rabies, Diphtheria, Food Poisoning, Scarlet Fever, Tonsillitis, Tuberculosis, Ringworm, Mad Cow, Legionnaires' Disease, and the Black Death (Plague).

Fictional pathogens also provide limitless potential for additional content. The scenario building community will hopefully provide the majority of these.

MEMORY CELLS

B-Cells are stimulated by a T-Helper to replicate into Memory B-Cells that are coded to identify a specific pathogen in the future.

The game keeps track of the number of Memory B-Cells created in each scenario and persists that data.

When a scenario is replayed, the built up memory cells created for the specific pathogen influence the starting conditions for the scenario.

The scenario might start with a small inventory of B and T-Helper cells.

TUTORIALS

Tutorial scenarios will be used to get the player up to speed on the basics of the game play.

The basic actors like tissue cells, neutrophils, and macrophages will be introduced in the first tutorials.

A couple foundational nano-bots like the Assassin and Repair bots will be used in the final tutorials.

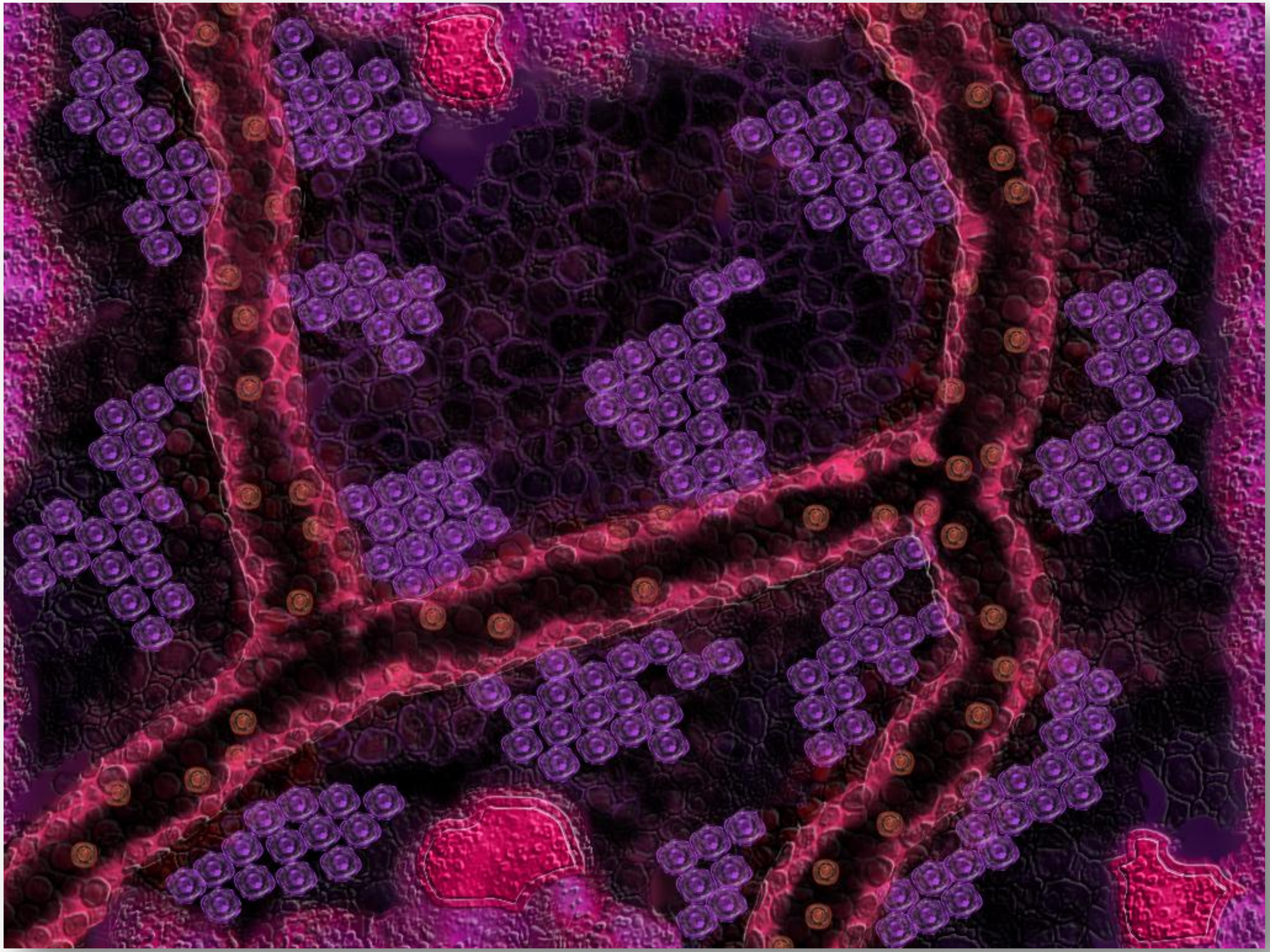
Teaching the player about the user interface will be a primary objective through most of the tutorials.

MALARIA

Malaria is caused by a parasite. The plasmodium parasite enters the blood stream of a human through the saliva of an anopheles mosquito. The parasite then makes its way through the blood stream to the liver, where it burrows into the liver cells.

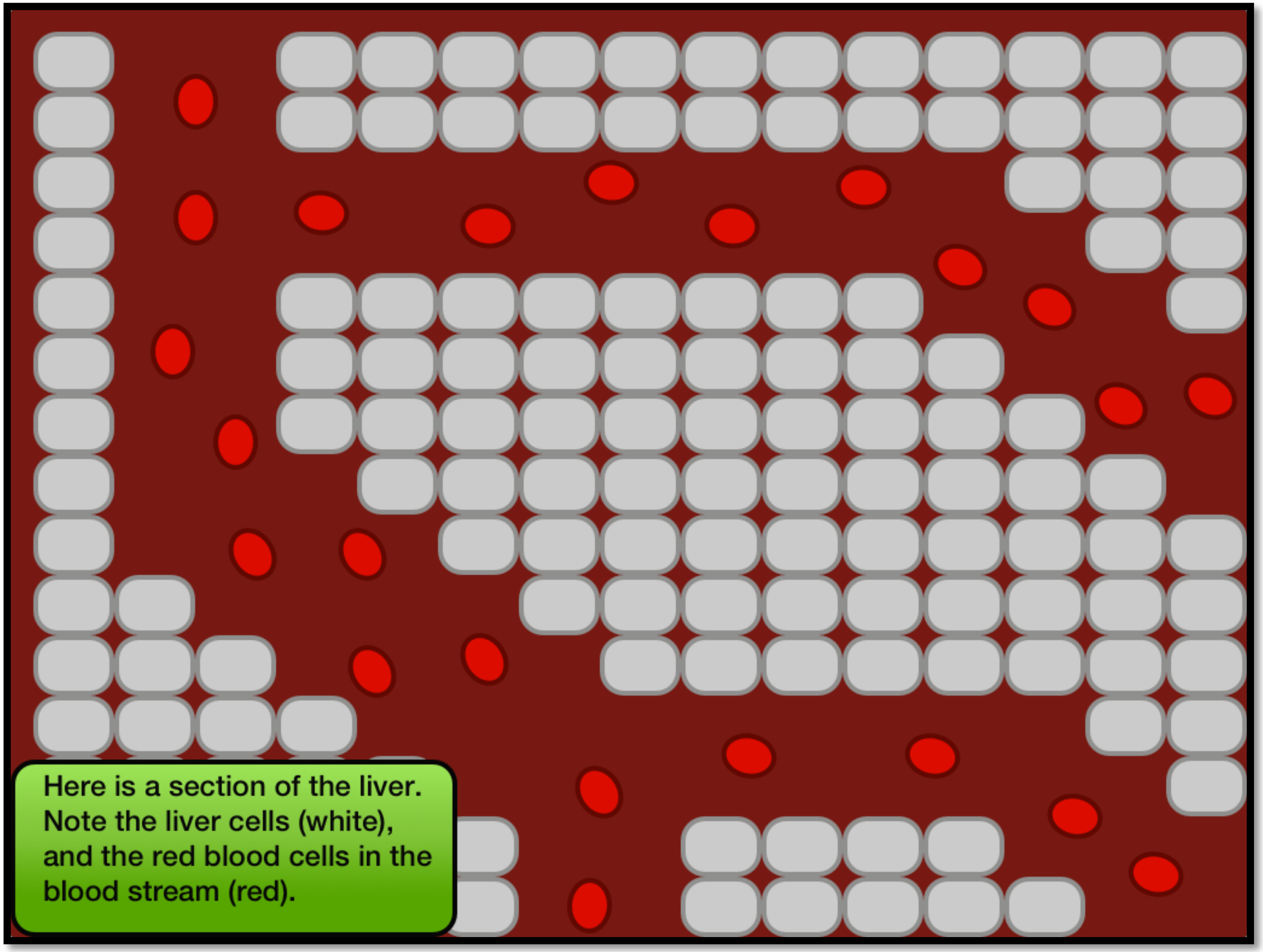
While in a liver cell, it reproduces. The merozoites then spread to the red blood cells and reproduce inside the red blood cells. The bloods cells that host the merozoites stick to the wall of the blood vessels until the new merozoites burst out, and the cycle continues.

MALARIA

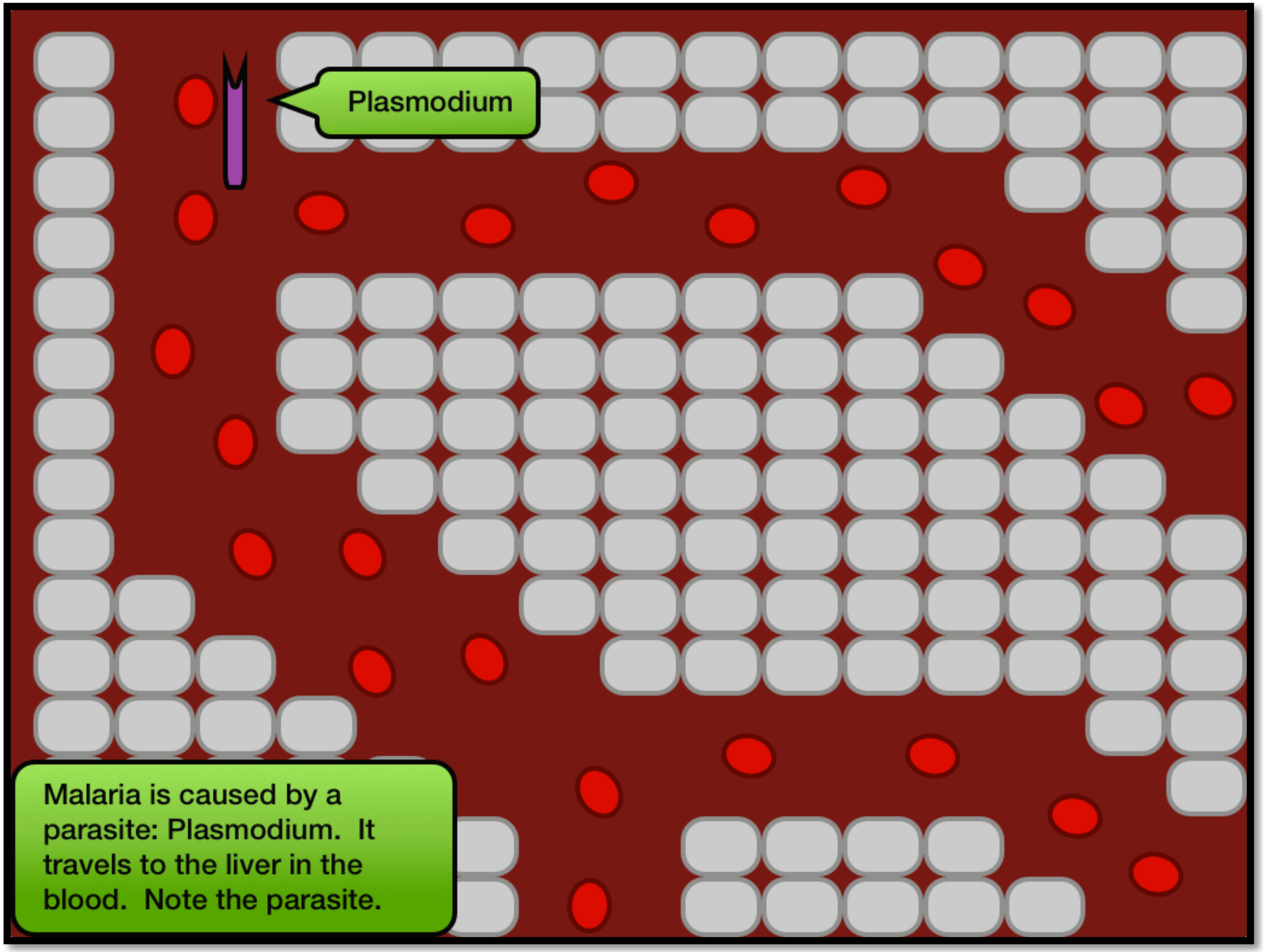


The battle to defeat Malaria takes place in the liver.

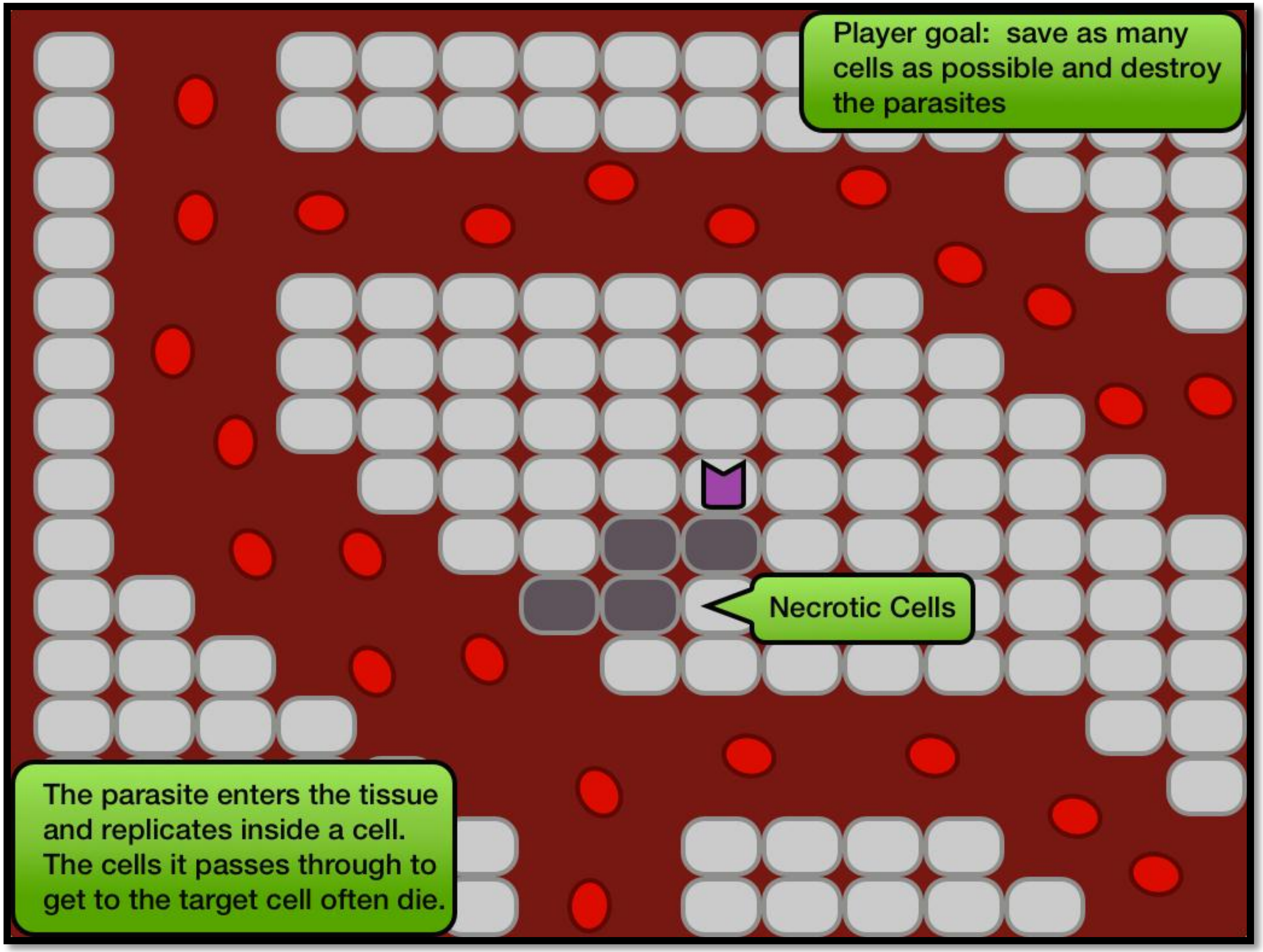
MALARIA


Here is a section of the liver.
Note the liver cells (white),
and the red blood cells in the
blood stream (red).

MALARIA

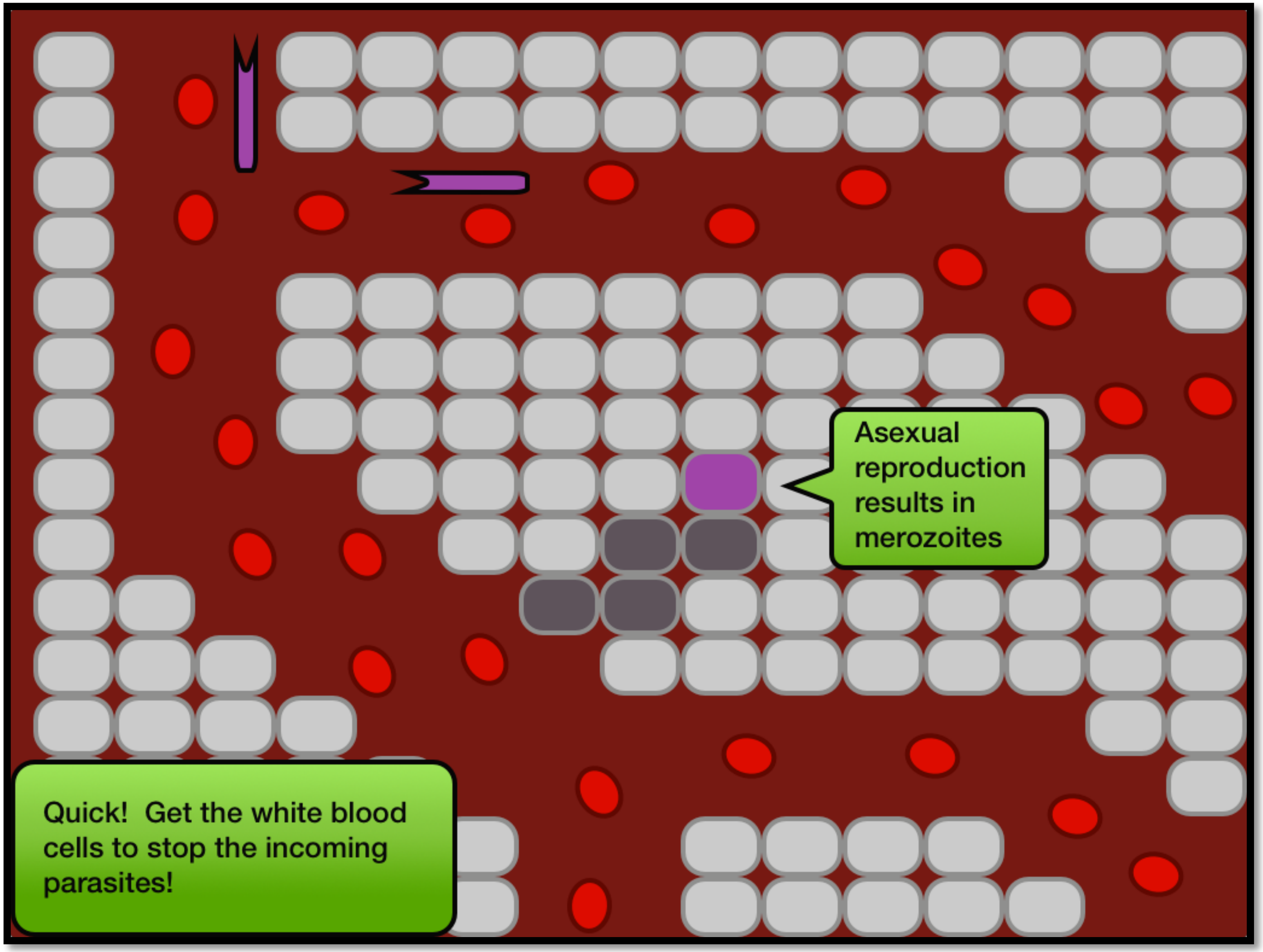

Plasmodium
Malaria is caused by a parasite: Plasmodium. It travels to the liver in the blood. Note the parasite.

MALARIA



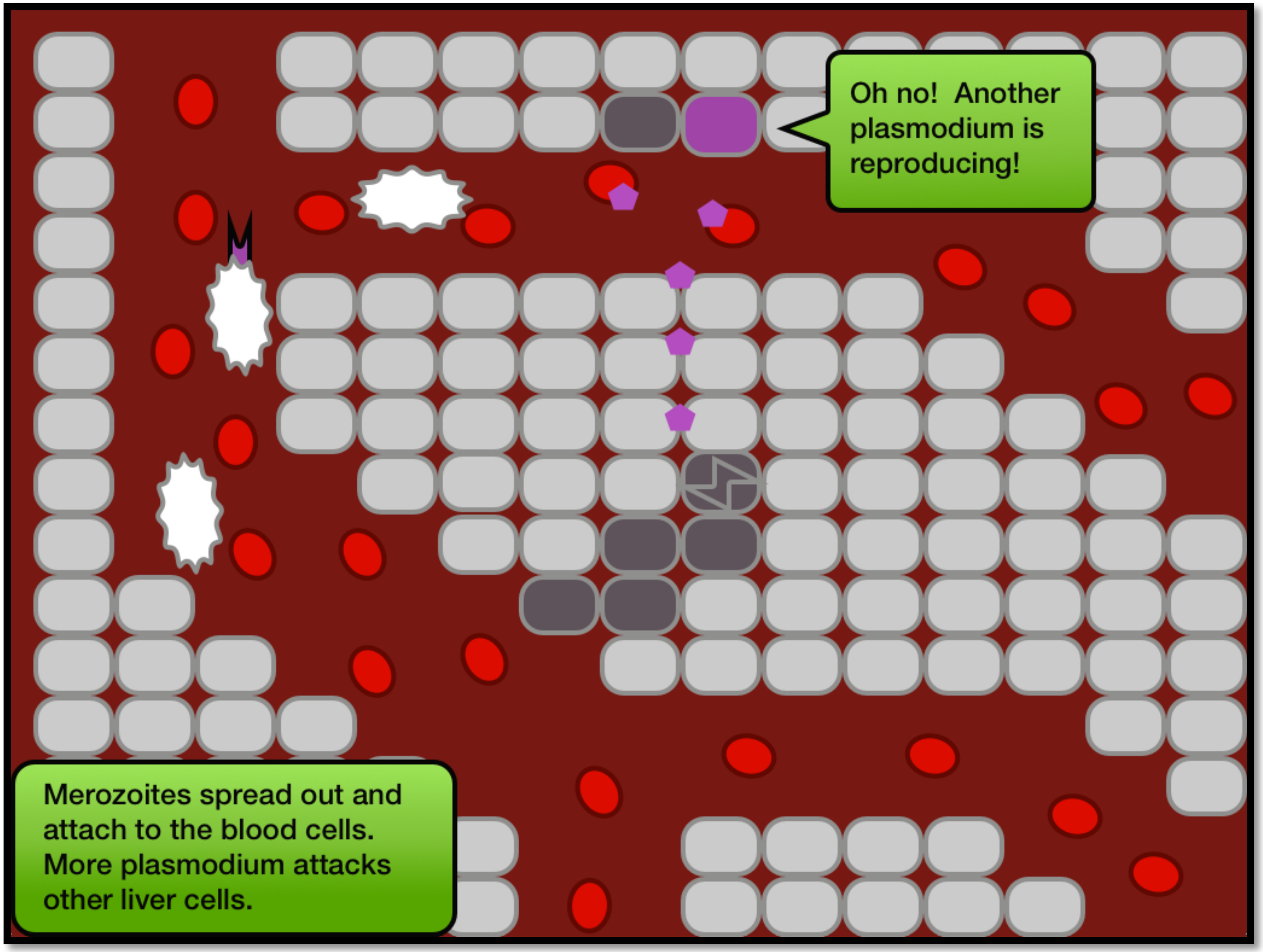
Player goal: save as many cells as possible and destroy the parasites
Necrotic Cells
The parasite enters the tissue and replicates inside a cell. The cells it passes through to get to the target cell often die.

MALARIA

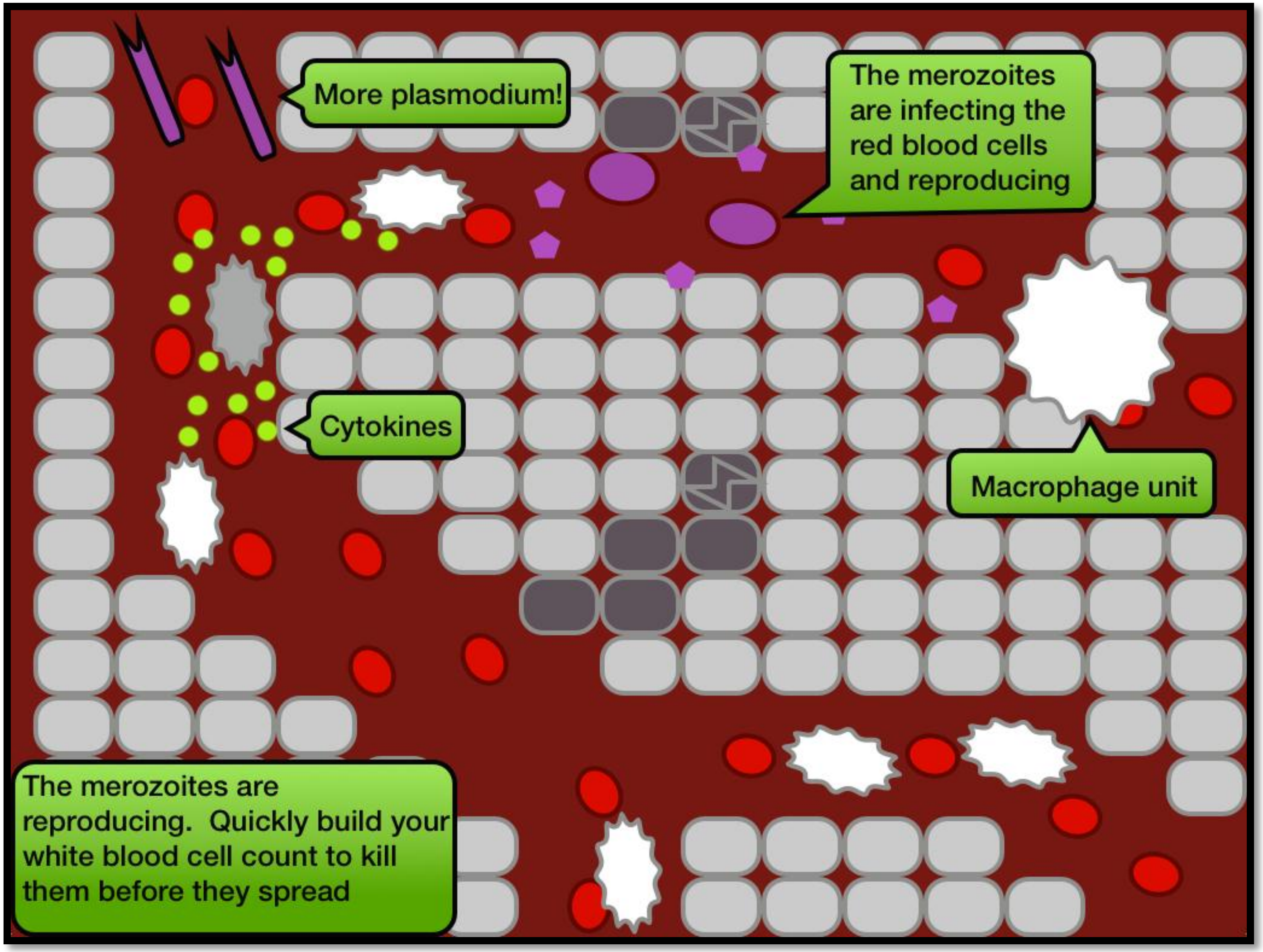


Asexual reproduction results in merozoites
Quick! Get the white blood cells to stop the incoming parasites!

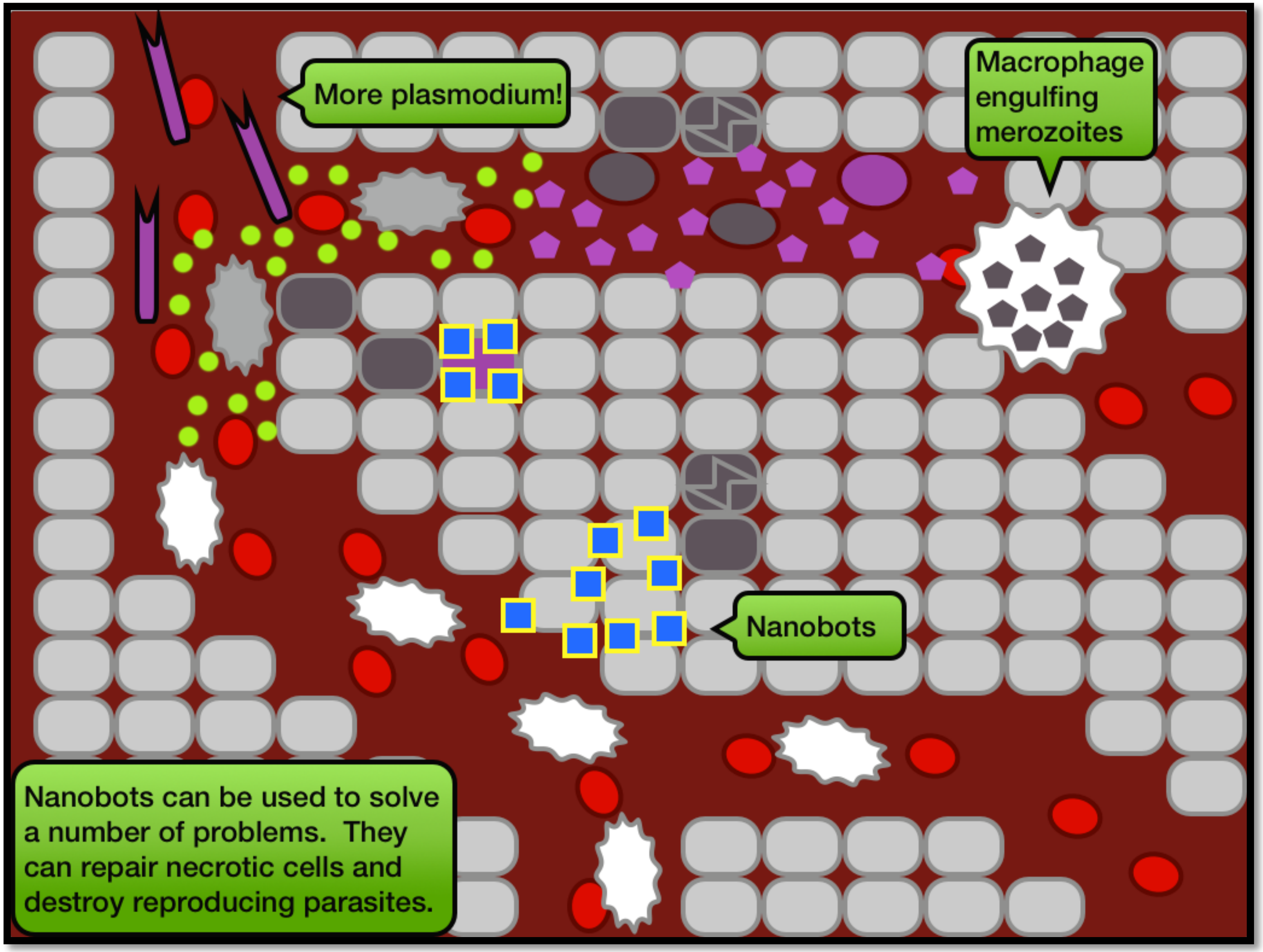
MALARIA


Oh no! Another plasmodium is reproducing!
Merozoites spread out and attach to the blood cells. More plasmodium attacks other liver cells.

MALARIA

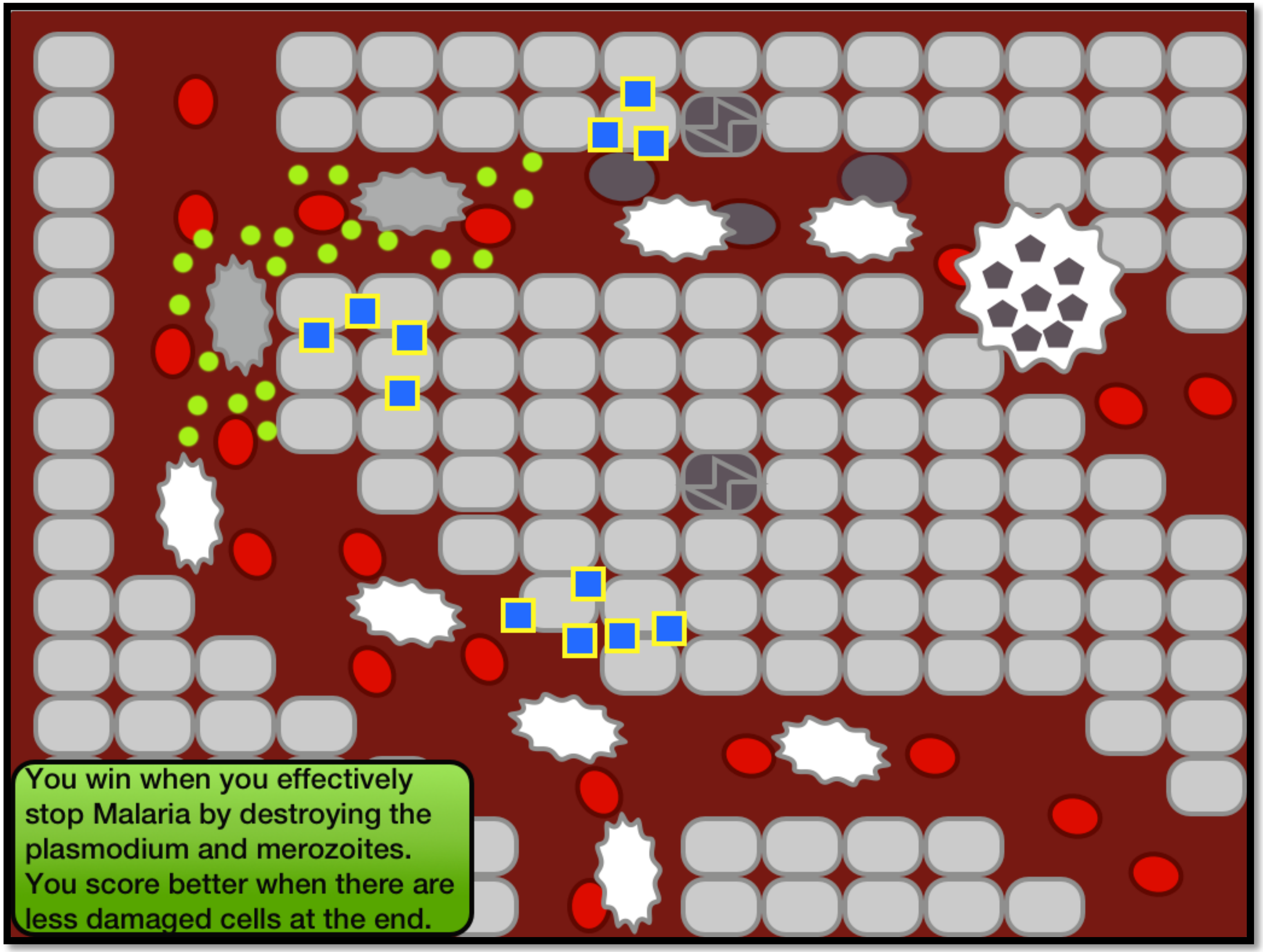

More plasmodium!
The merozoites are infecting the red blood cells and reproducing
Cytokines
Macrophage unit
The merozoites are reproducing. Quickly build your white blood cell count to kill them before they spread

MALARIA

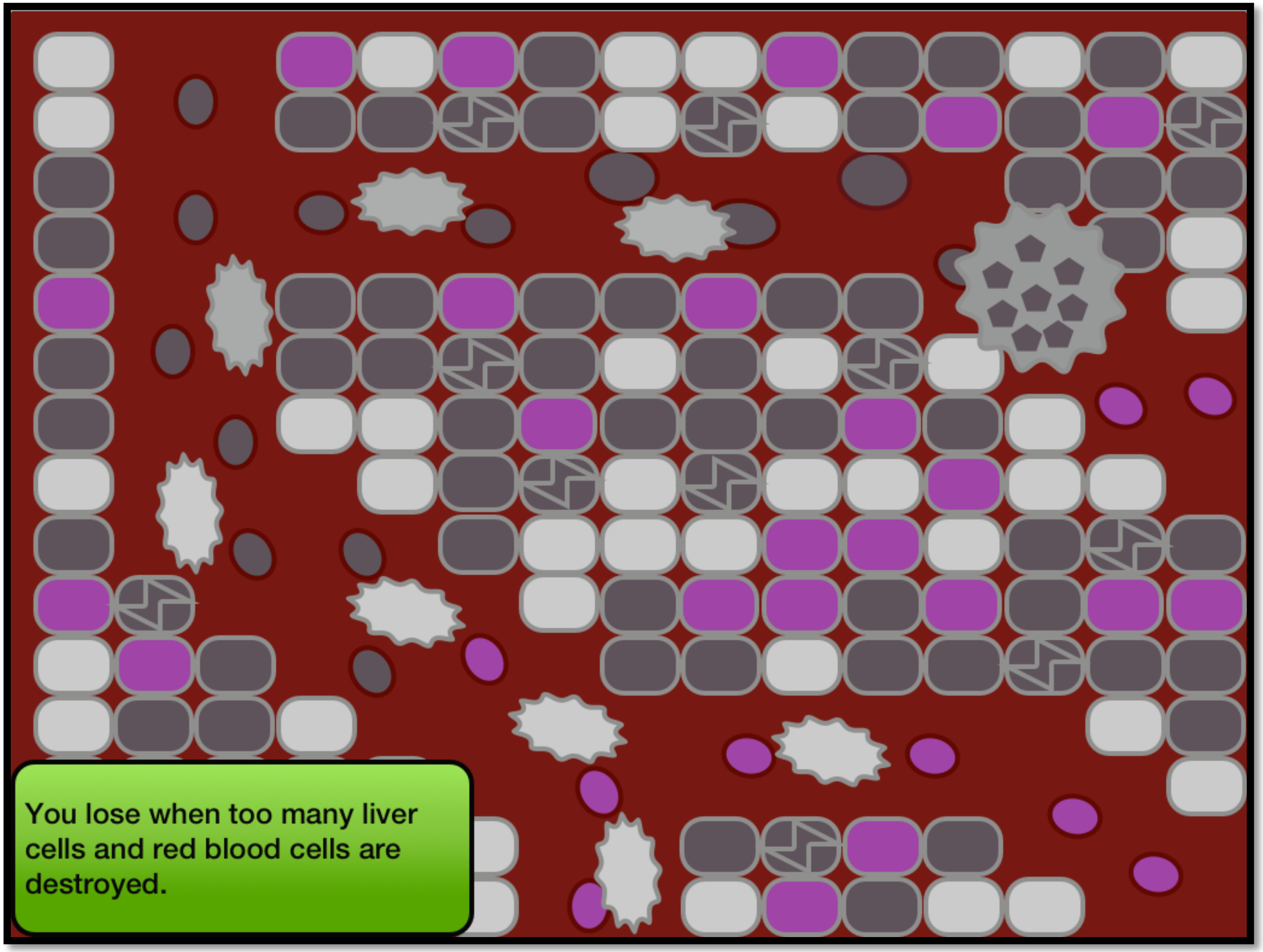


More plasmodium!
Macrophage engulfing merozoites
Nanobots
Nanobots can be used to solve a number of problems. They can repair necrotic cells and destroy reproducing parasites.

MALARIA


You win when you effectively stop Malaria by destroying the plasmodium and merozoites. You score better when there are less damaged cells at the end.

MALARIA


You lose when too many liver cells and red blood cells are destroyed.

ANTHRAX

Anthrax is caused by Bacillus anthracis. The bacteria can exist dormant in its endospore stage for many years, hidden in the ground. It can then be inhaled or ingested, or create an infection through a cut in the surface of the skin.

When a human inhales the spores they travel to their lungs and rest in the alveolar sacs. Macrophages are then alerted to the presence of the spores and leave the blood stream to engulf the endospores.

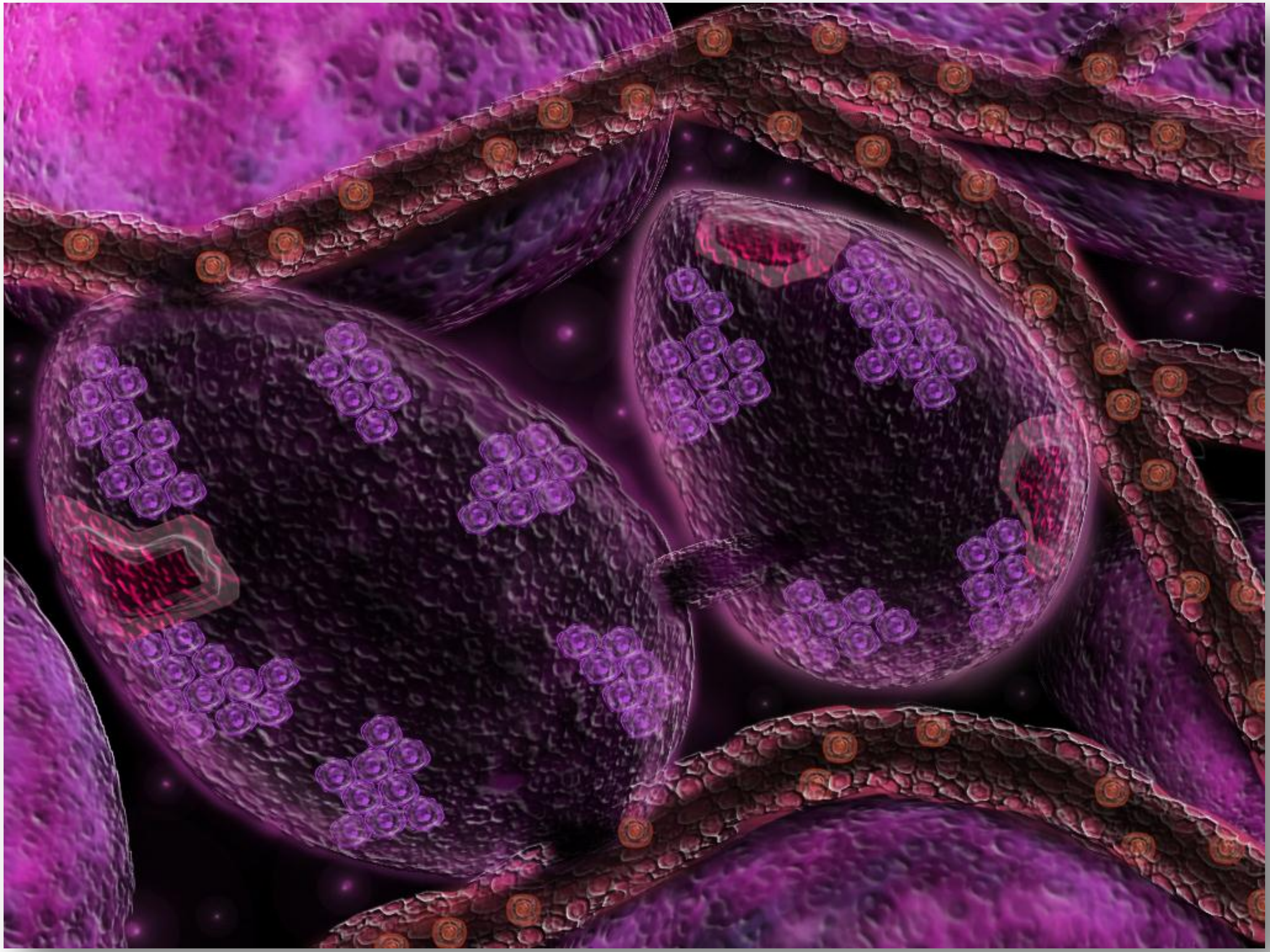
ANTHRAX

Usually a pathogen is destroyed when they are engulfed by macrophages, but in this instance the endospores germinate inside the macrophage and exit as vegetative Bacillus.

The Bacillus anthracis then spreads and multiplies by invading cells. They also release exotoxins that attach to the cells and kill them.

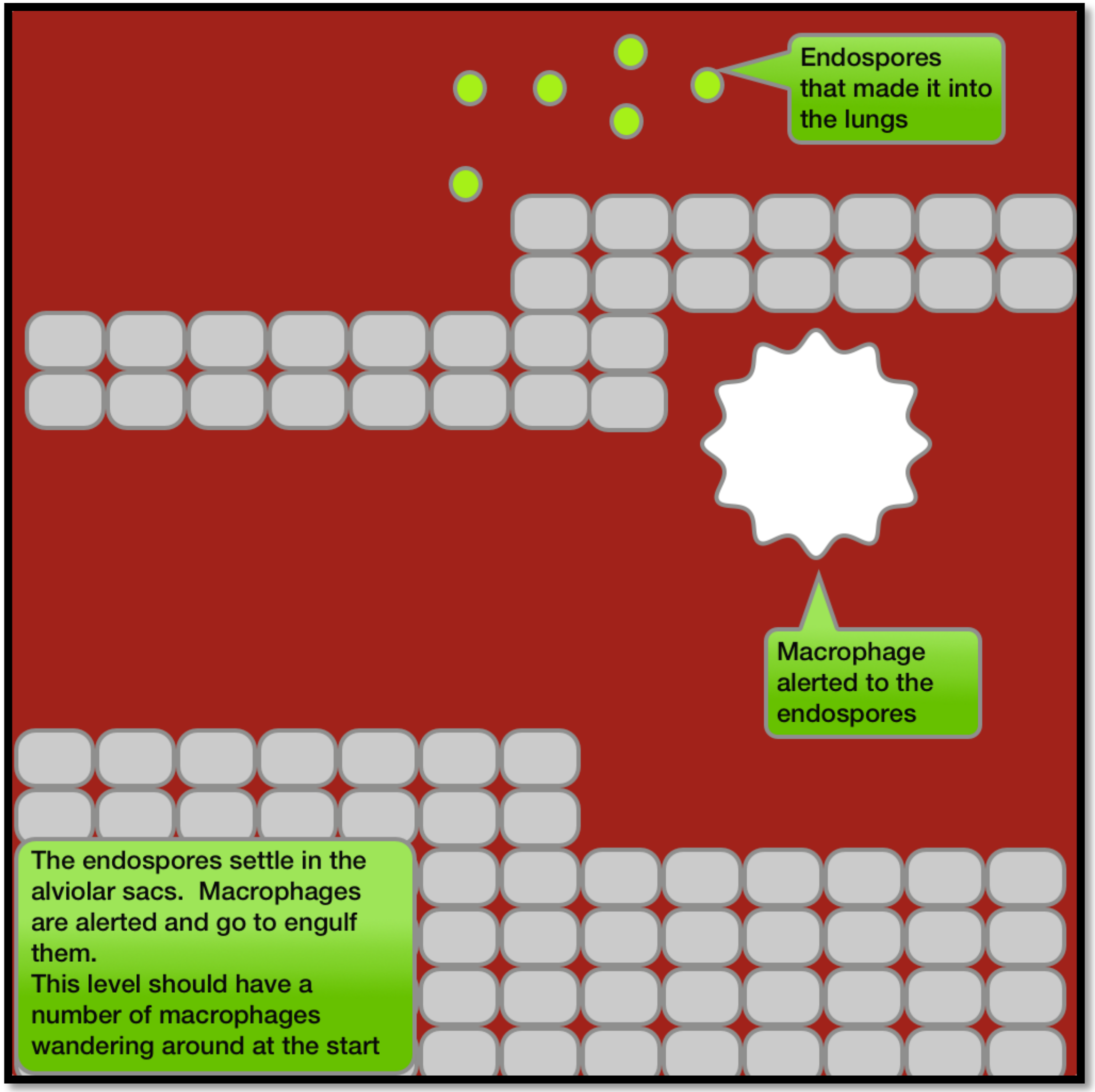
These exotoxins travel through the blood stream and cause blood poisoning on their way to the cells they destroy.

ANTHRAX



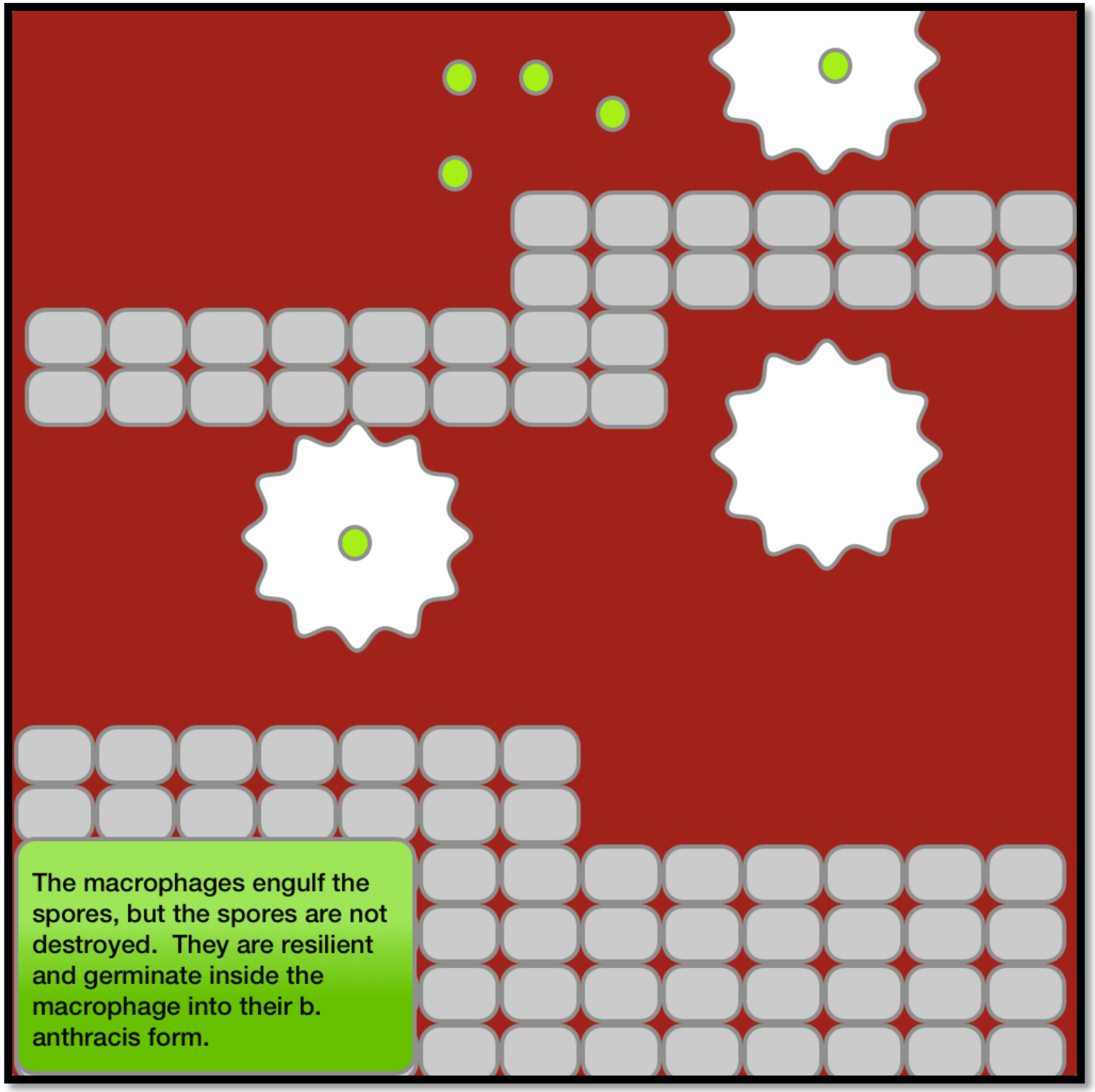
The battle to defeat Anthrax takes place in the lymph nodes.

ANTHRAX

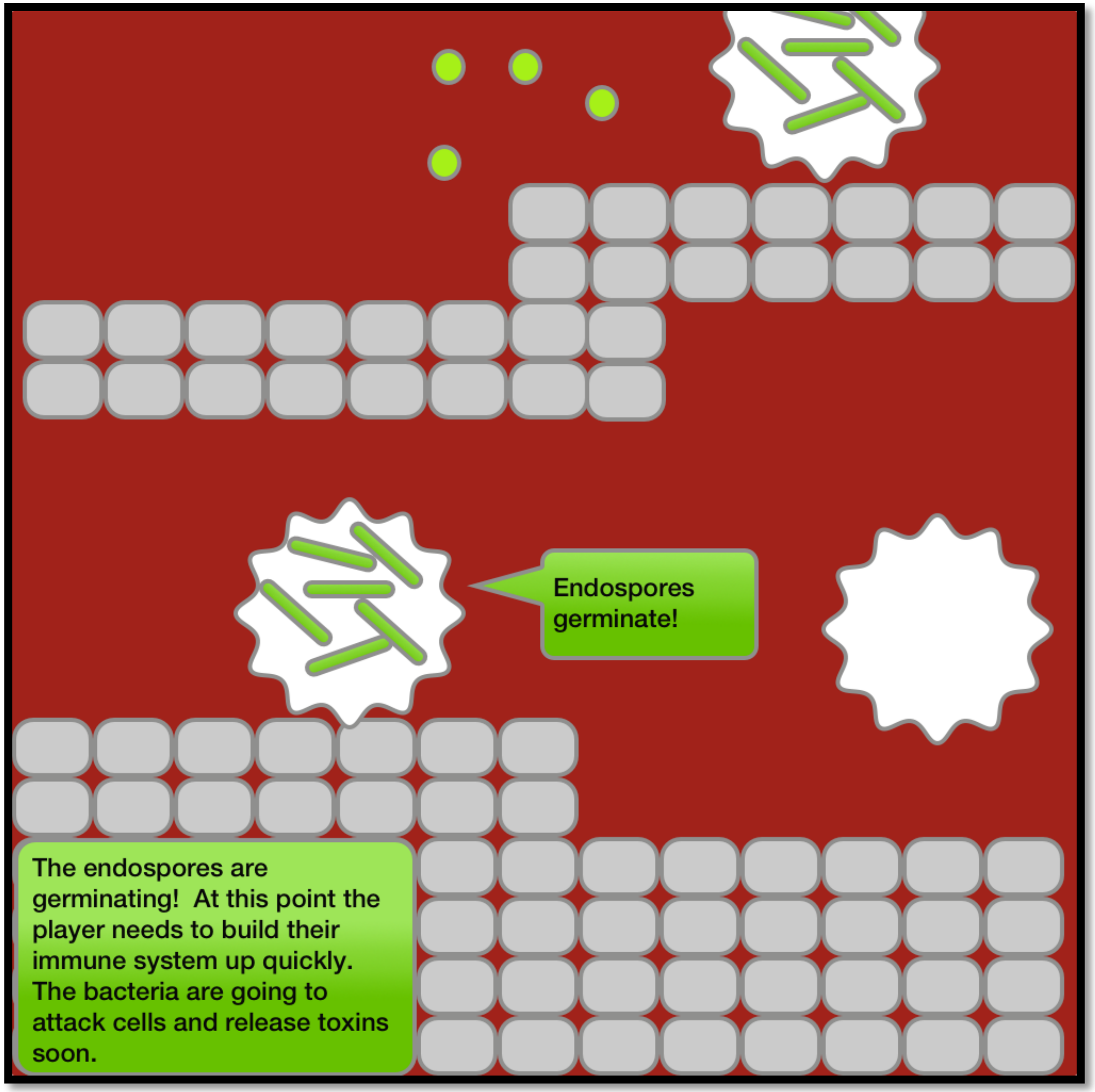


Endospores that made it into the lungs
Macrophage alerted to the endospores
The endospores settle in the alviolar sacs. Macrophages are alerted and go to engulf them.
This level should have a number of macrophages wandering around at the start

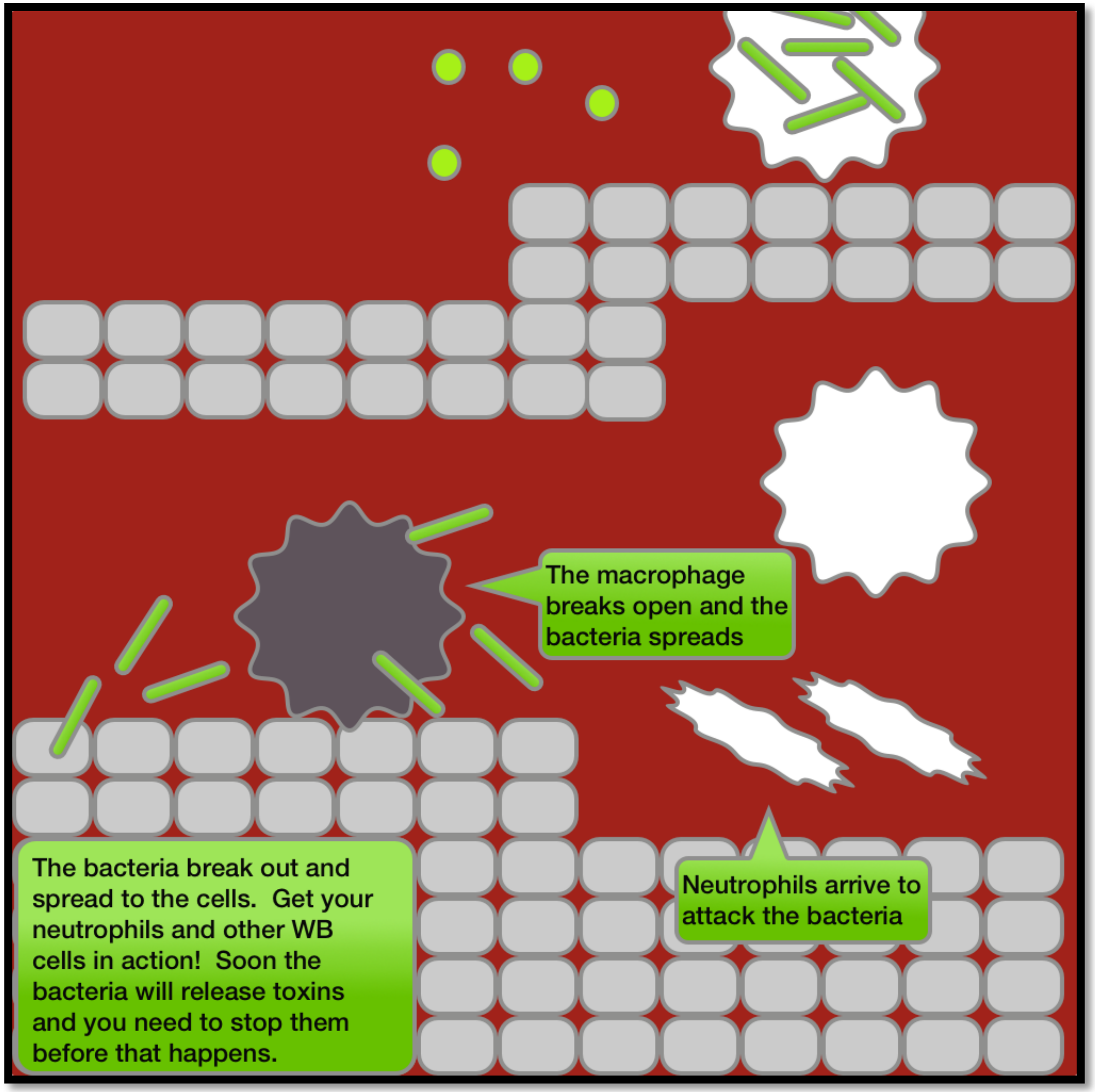
ANTHRAX


The macrophages engulf the spores, but the spores are not destroyed. They are resilient and germinate inside the macrophage into their b. anthracis form.

ANTHRAX

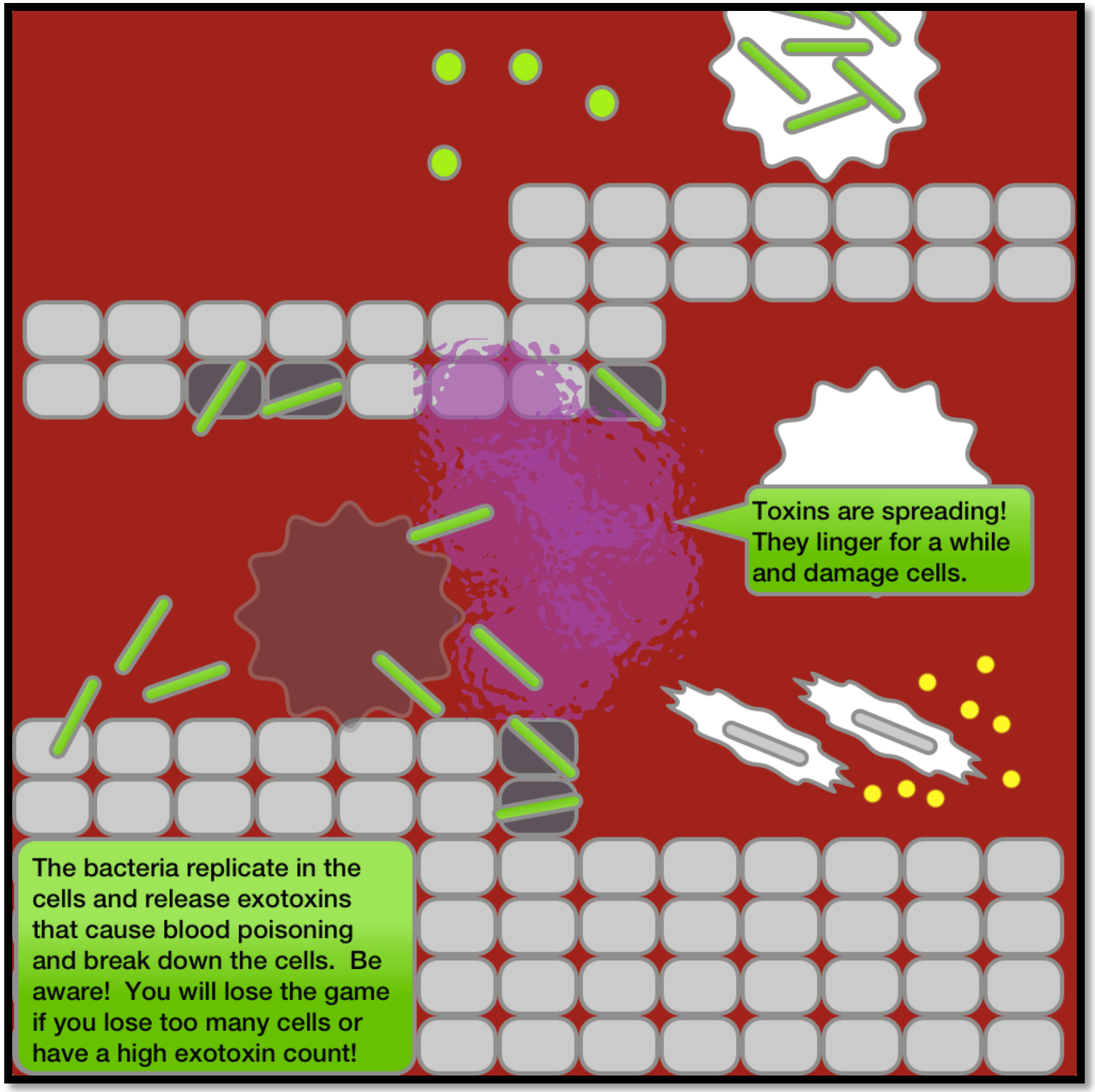

Endospores germinate!
The endospores are germinating! At this point the player needs to build their immune system up quickly. The bacteria are going to attack cells and release toxins soon.

ANTHRAX



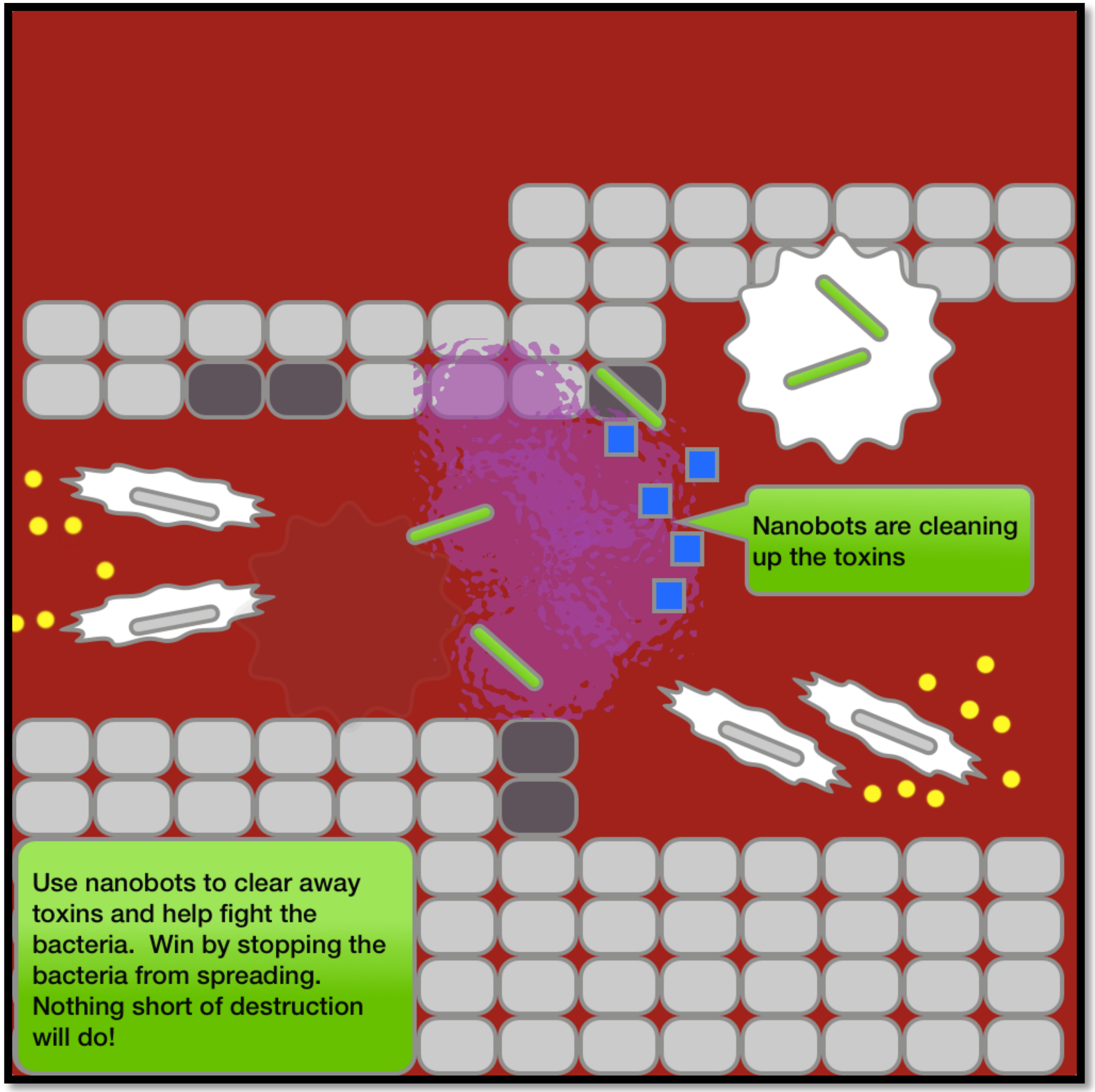
The macrophage breaks open and the bacteria spreads
Neutrophils arrive to attack the bacteria
The bacteria break out and spread to the cells. Get your neutrophils and other WB cells in action! Soon the bacteria will release toxins and you need to stop them before that happens.

ANTHRAX



Toxins are spreading! They linger for a while and damage cells.
The bacteria replicate in the cells and release exotoxins that cause blood poisoning and break down the cells. Be aware! You will lose the game if you lose too many cells or have a high exotoxin count!

ANTHRAX



Nanobots are cleaning up the toxins
Use nanobots to clear away toxins and help fight the bacteria. Win by stopping the bacteria from spreading. Nothing short of destruction will do!

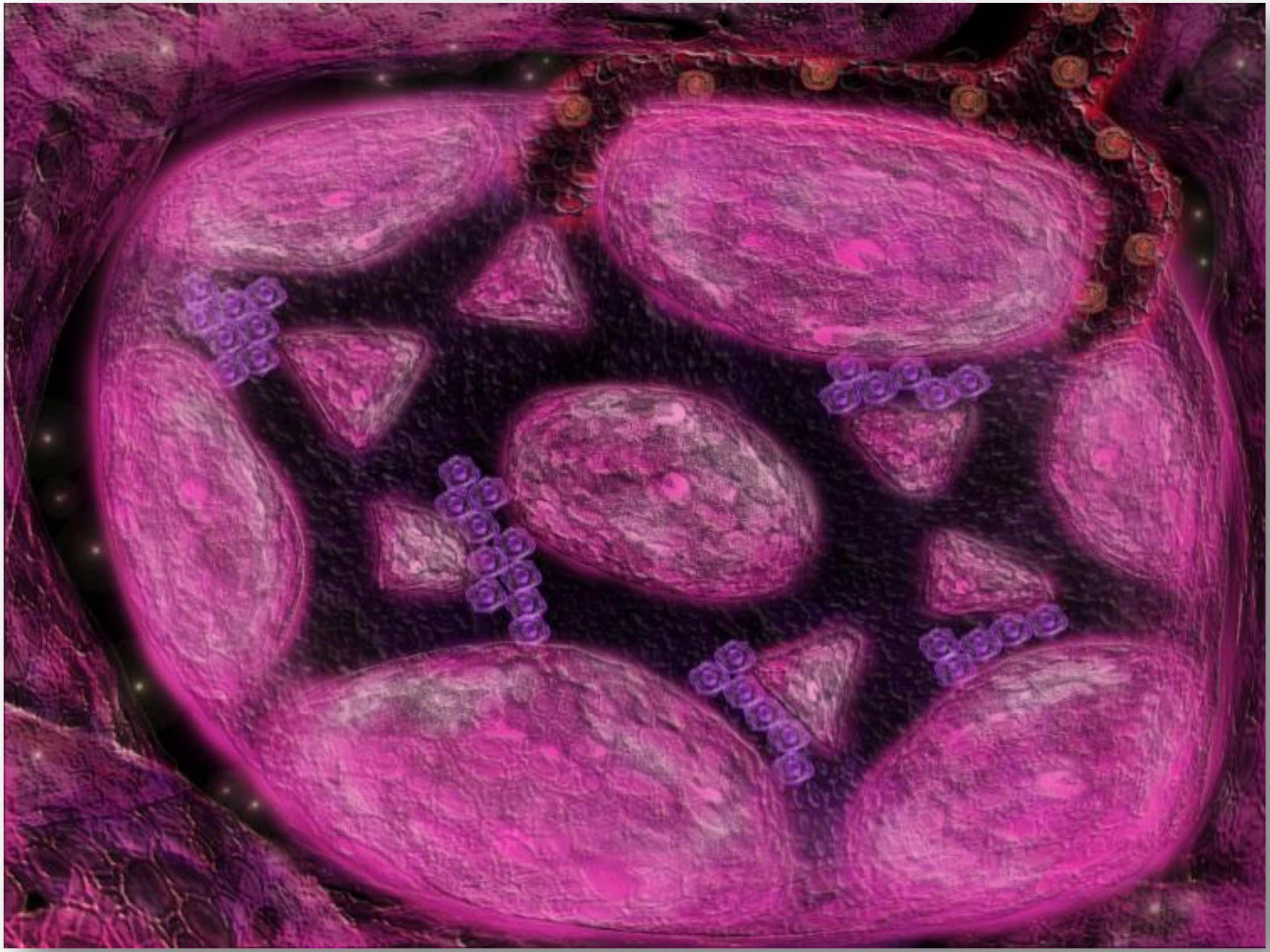
CANCER

A natural immune response can fight cancer. Natural Killer cells, B-Cells and T-Killers all work well against the cancer cells.

T-Killer cells can be targeted to kill reproducing cells and B -Cells can create antibodies that effectively mark replicating cells.

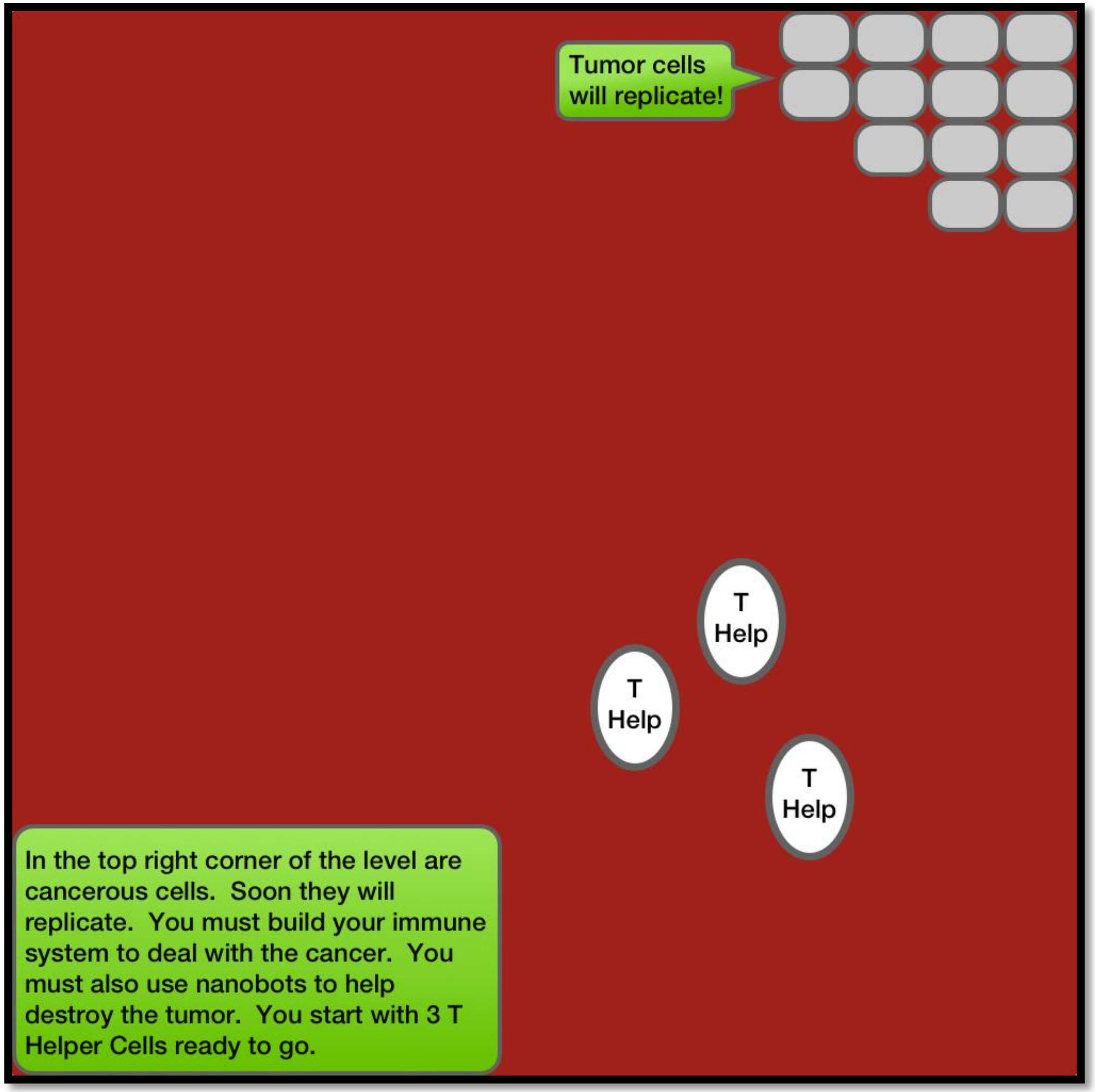
However, the immune system is usually unable to totally crush cancer. Often times chemotherapy is a solution, but in our simulation we have the capacity to include fictional nano-bots to help destroy the cancerous cells.

CANCER

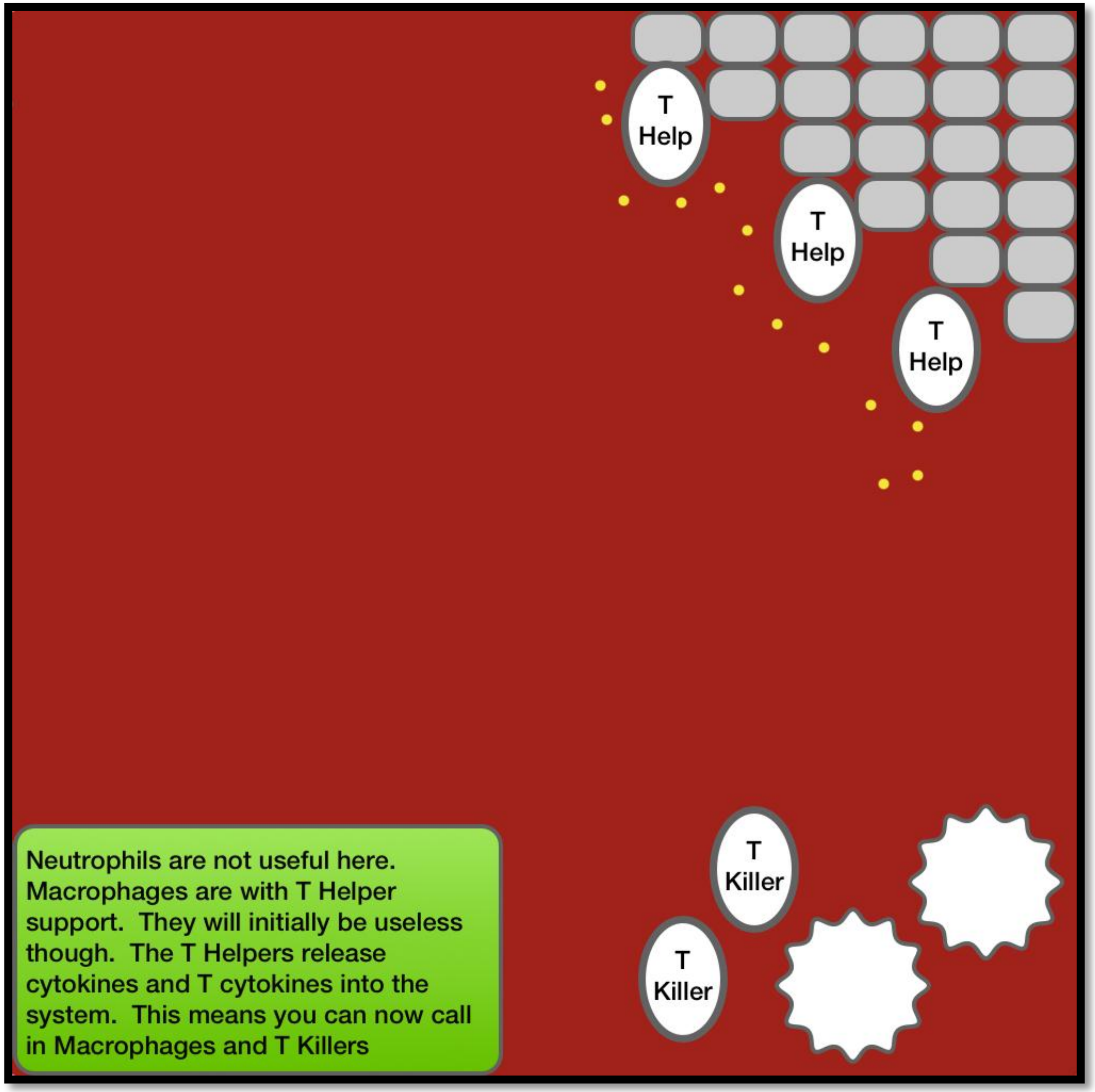


The battle to defeat Cancer takes place in the parathyroid.

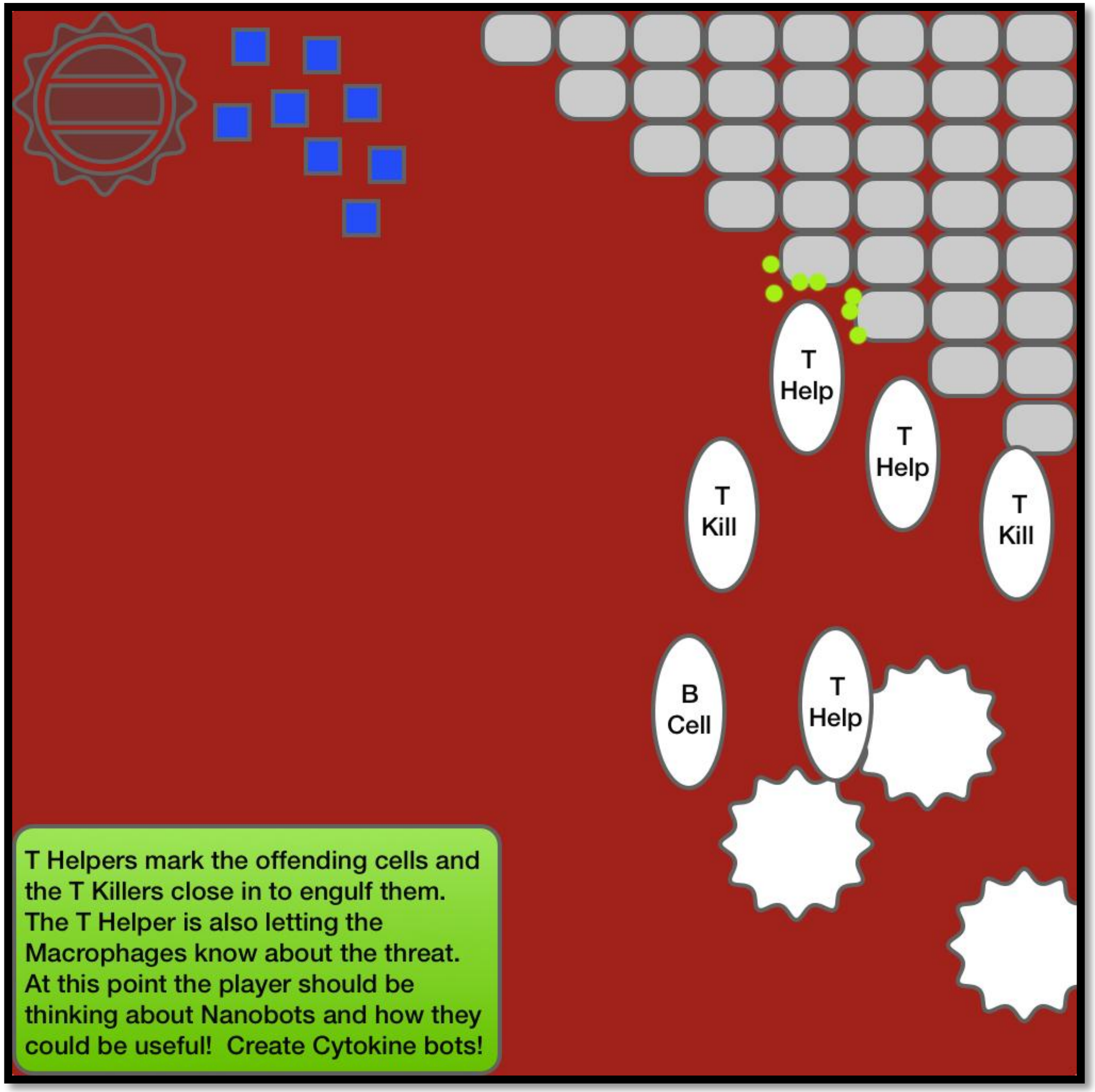
CANCER


Tumor cells will replicate!
T Help
T Help
T Help
In the top right corner of the level are cancerous cells. Soon they will replicate. You must build your immune system to deal with the cancer. You must also use nanobots to help destroy the tumor. You start with 3 T Helper Cells ready to go.

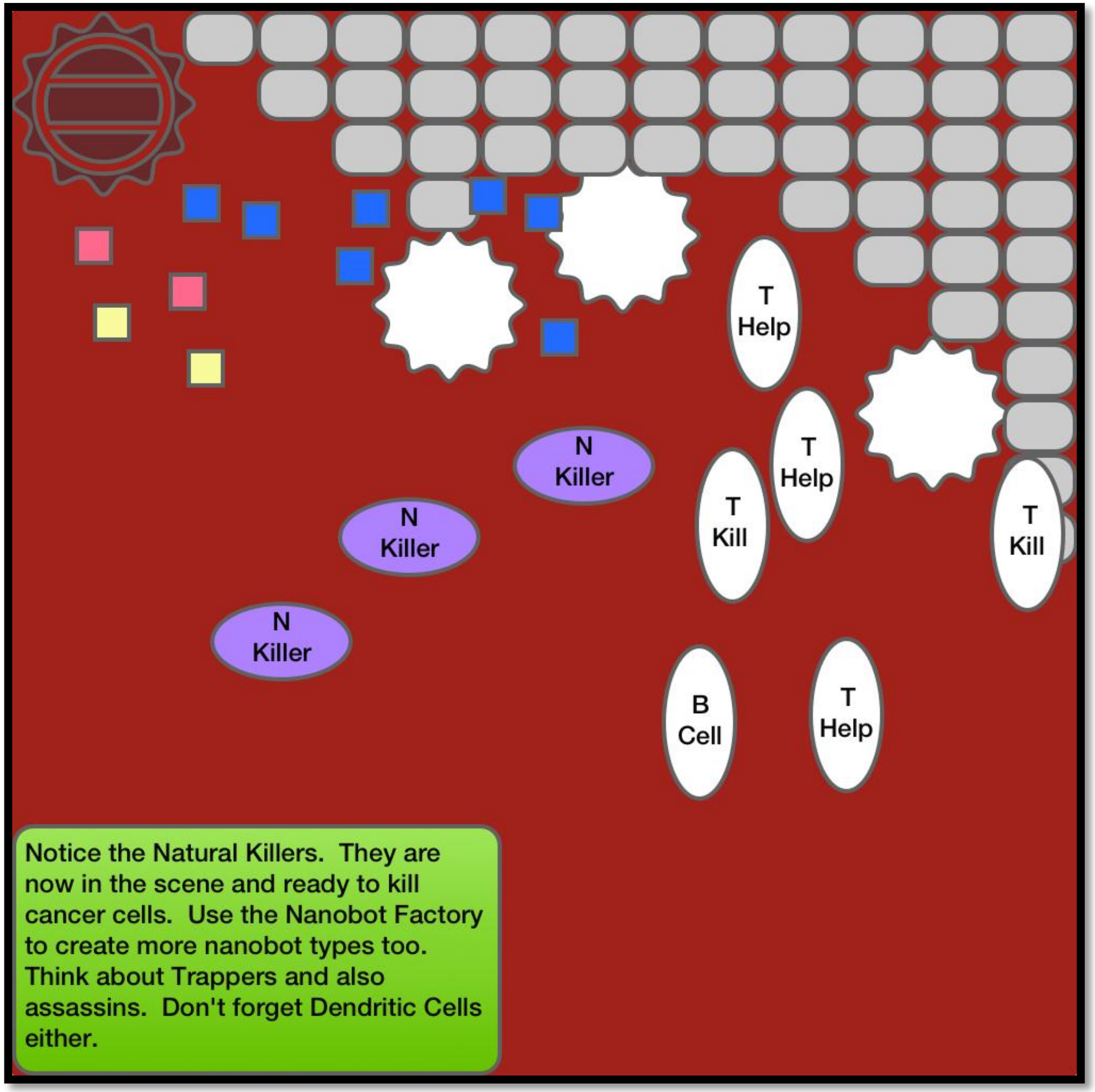
CANCER


T Help
T Help
T Help
Neutrophils are not useful here. Macrophages are with T Helper support. They will initially be useless though. The T Helpers release cytokines and T cytokines into the system. This means you can now call in Macrophages and T Killers
T Killer
T Killer

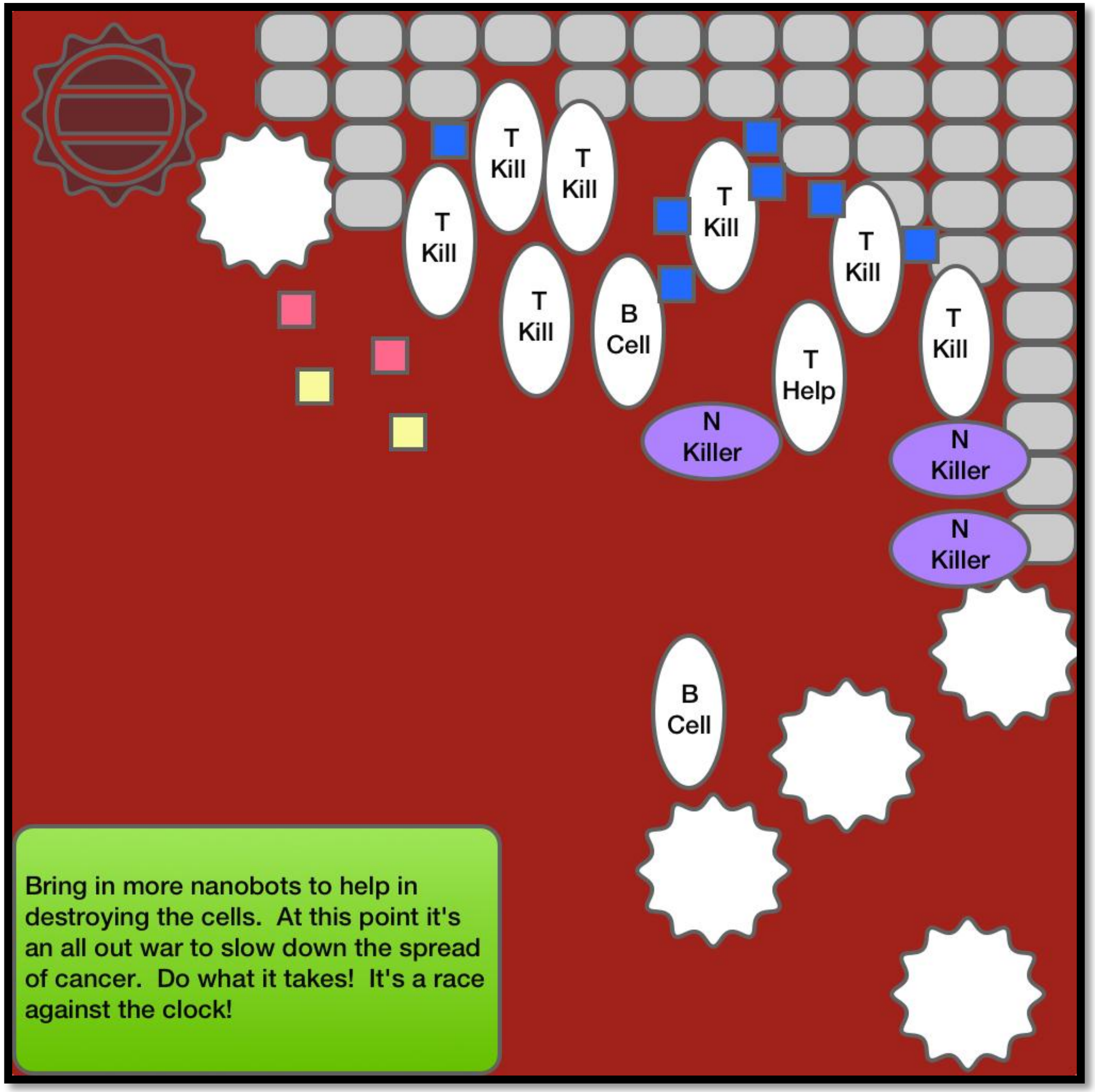
CANCER


T
Help
T
Help
T
Kill
T
Kill
B
Cell
T
Help
T Helpers mark the offending cells and the T Killers close in to engulf them. The T Helper is also letting the Macrophages know about the threat. At this point the player should be thinking about Nanobots and how they could be useful! Create Cytokine bots!

CANCER


T
Help
T
Help
T
Kill
T
Kill
N
Killer
N
Killer
N
Killer
B
Cell
T
Help
Notice the Natural Killers. They are now in the scene and ready to kill cancer cells. Use the Nanobot Factory to create more nanobot types too. Think about Trappers and also assassins. Don't forget Dendritic Cells either.

CANCER


T Kill
T Kill
T Kill
T Kill
T Kill
T Kill
T Kill
B Cell
T Help
N Killer
N Killer
N Killer
B Cell
Bring in more nanobots to help in destroying the cells. At this point it's an all out war to slow down the spread of cancer. Do what it takes! It's a race against the clock!

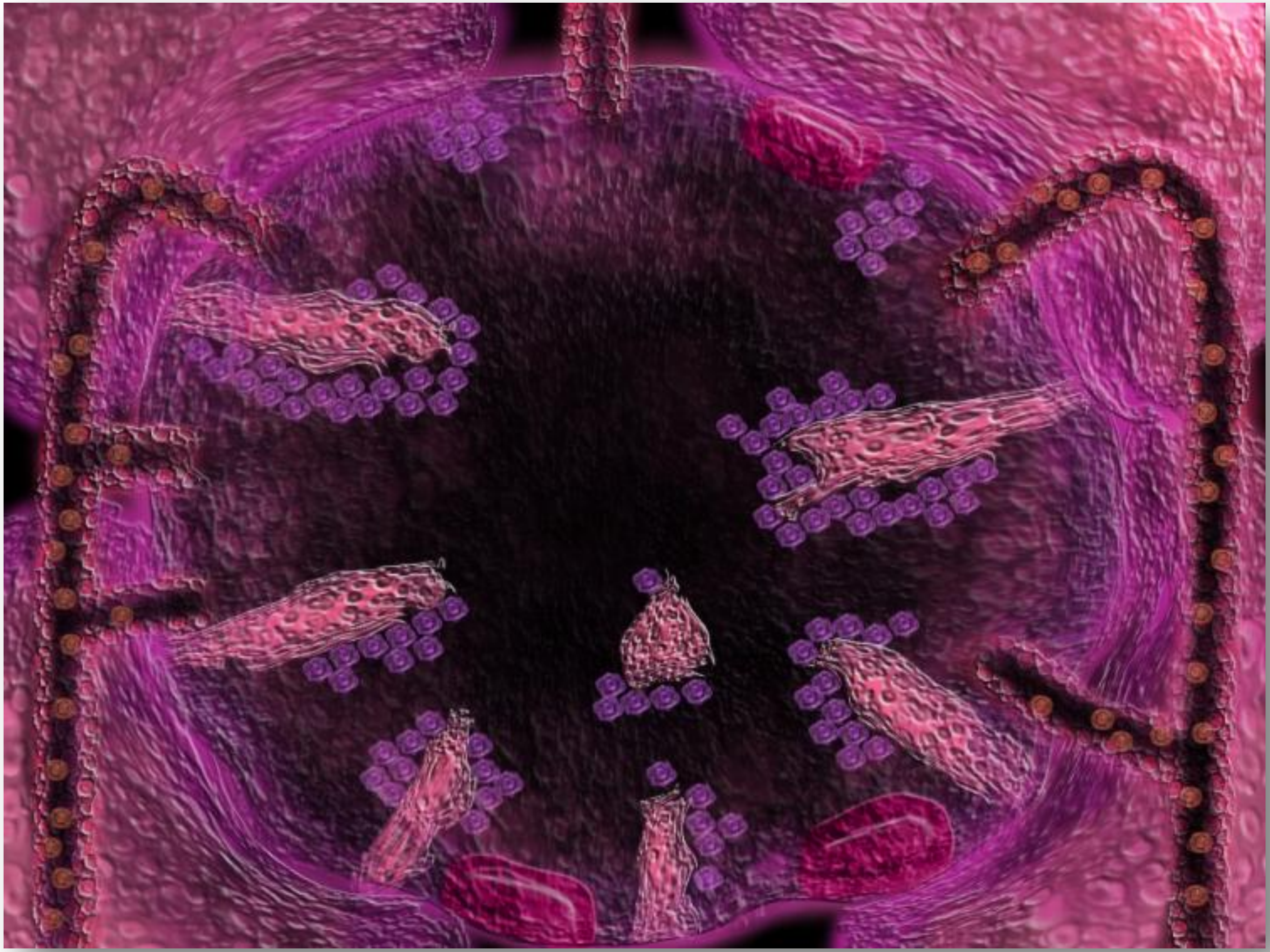
HIV

The human immunodeficiency virus (HIV) is a slowly replicating retrovirus that causes the acquired immunodeficiency syndrome (AIDS), a condition which progressive failure of the immune system allows life-threatening infections and cancers to thrive.

HIV infects vital cells in the human immune system such as helper T-Helpers, Macrophages, and Dendritic cells.

When T-Helper cell counts decline below a critical level, cell-mediated immunity is lost, and the body becomes progressively more susceptible to opportunistic infections.

HIV

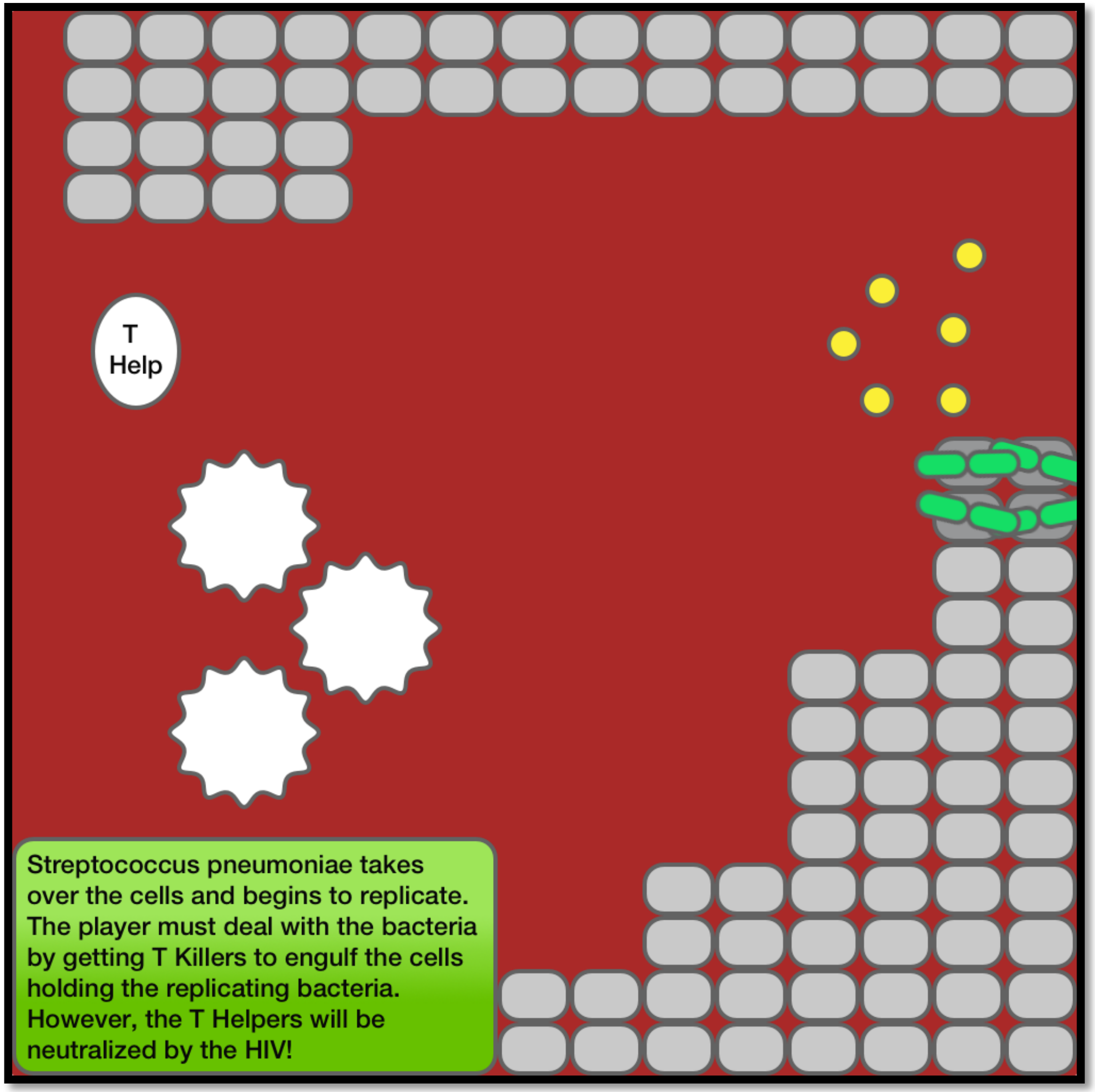


The battle to defeat HIV and Pneumonia takes place in the Alveolar Sacs.

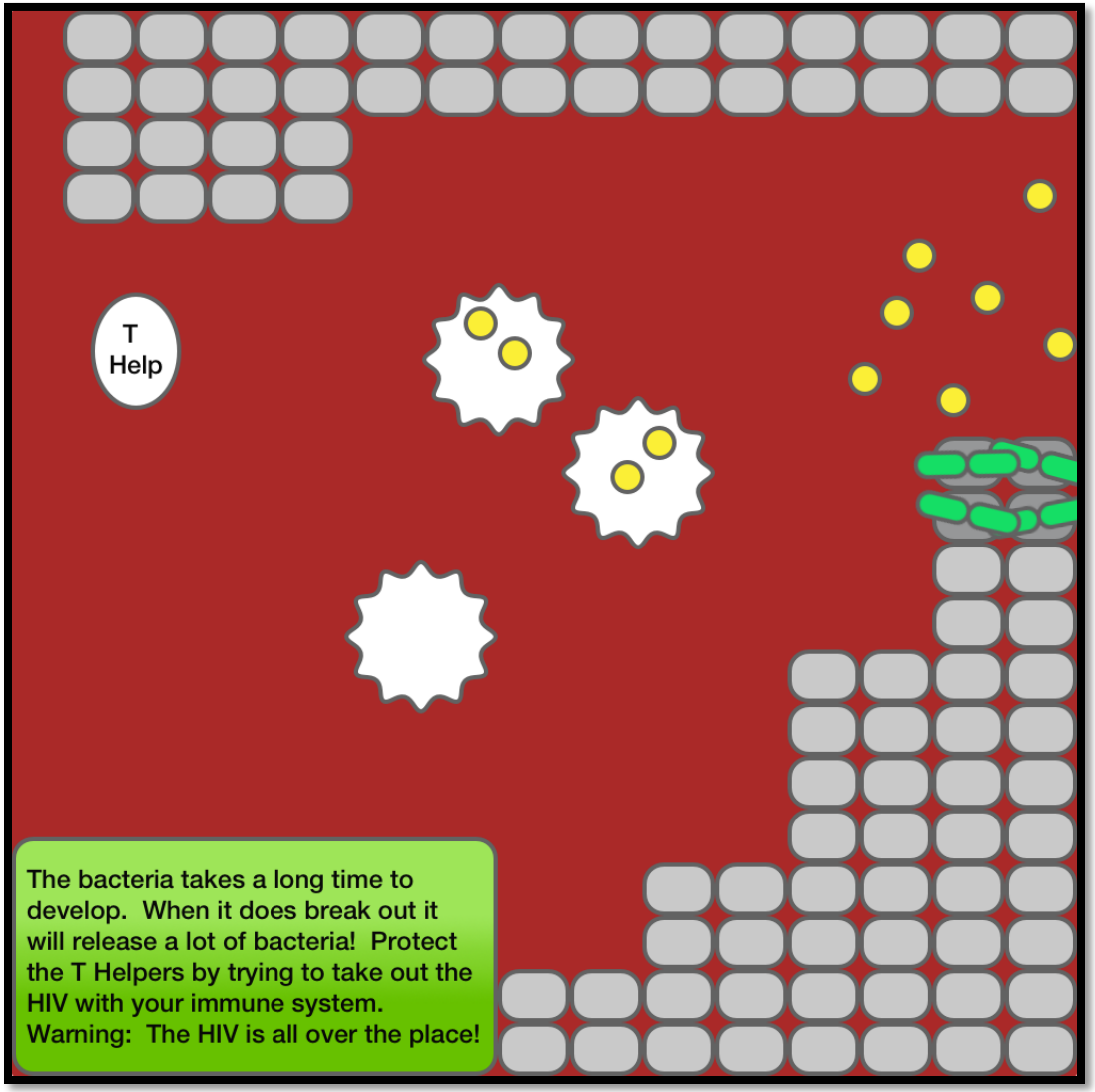
HIV


HIV
bacteria
Here are a number of cells in the alveolar sacs. They are at risk from the streptococcus pneumoniae. Furthermore, HIV is rampant in this system. The player must get their innate and adaptive immune systems up and running!

HIV


T
Help
Streptococcus pneumoniae takes over the cells and begins to replicate. The player must deal with the bacteria by getting T Killers to engulf the cells holding the replicating bacteria. However, the T Helpers will be neutralized by the HIV!

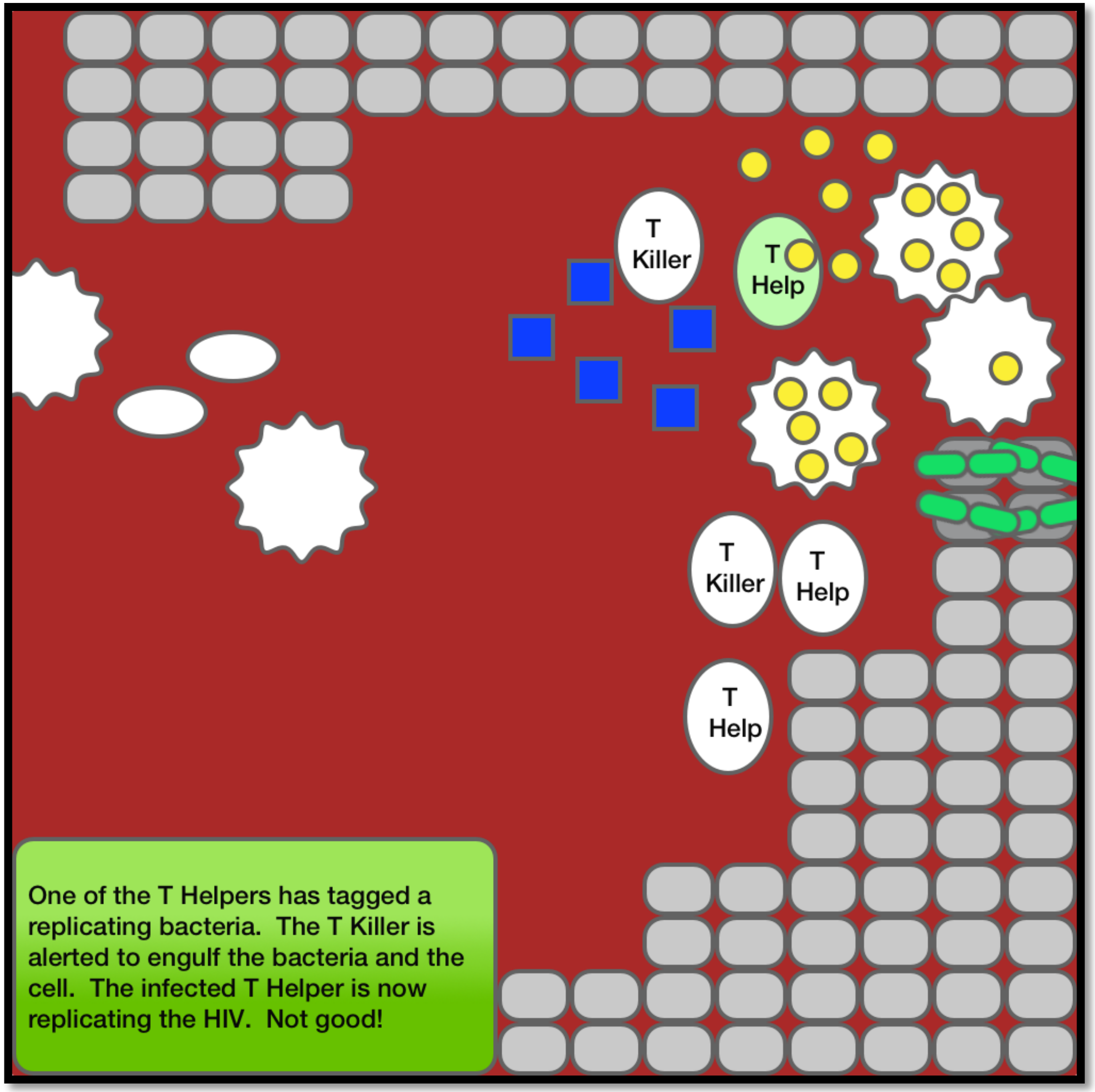
HIV


T
Help
The bacteria takes a long time to develop. When it does break out it will release a lot of bacteria! Protect the T Helpers by trying to take out the HIV with your immune system. Warning: The HIV is all over the place!

HIV


T
Help
T
Help
T
Killer
T
Killer
T
Help
The bacteria is still replicating. Consider using nanobots that are good at killing viruses to protect the T Helpers. One T Helper is close to tagging the replicating cells! Make sure it succeeds! One T Helper has been infected. Not good!

HIV

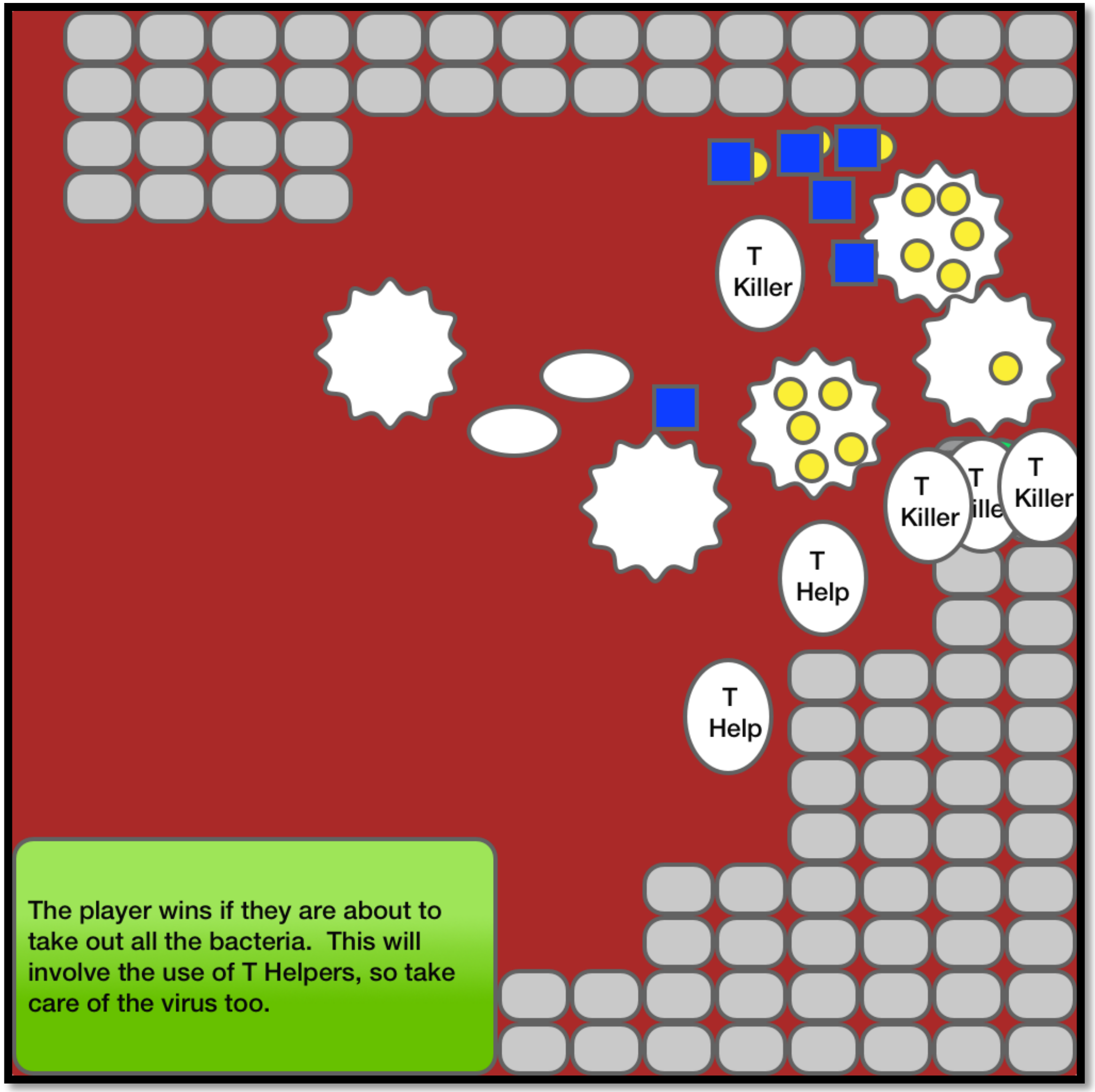


One of the T Helpers has tagged a replicating bacteria. The T Killer is alerted to engulf the bacteria and the cell. The infected T Helper is now replicating the HIV. Not good!

HIV

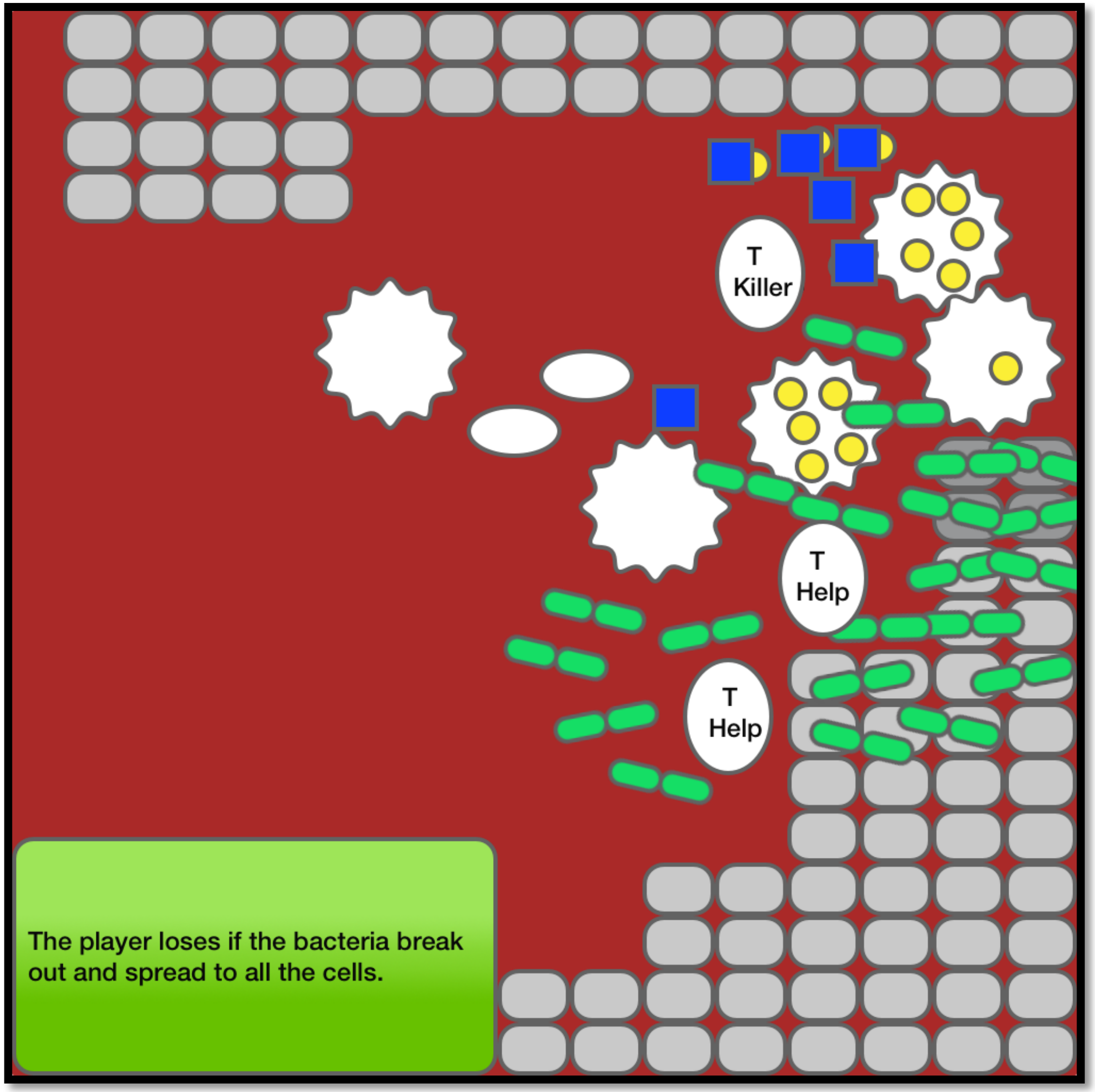

T
Killer
T
Killer
T
Help
T
Help
T Killers will take out the T Helpers that have been infected with HIV. However, the HIV replicates quickly! Make sure to have lots of T Killers ready! Notice the T Killer taking out the bacteria inside the cells. Tag them all and take them out.

HIV



T Killer
T Killer
T Killer
T Killer
T Help
T Help
The player wins if they are about to take out all the bacteria. This will involve the use of T Helpers, so take care of the virus too.

HIV

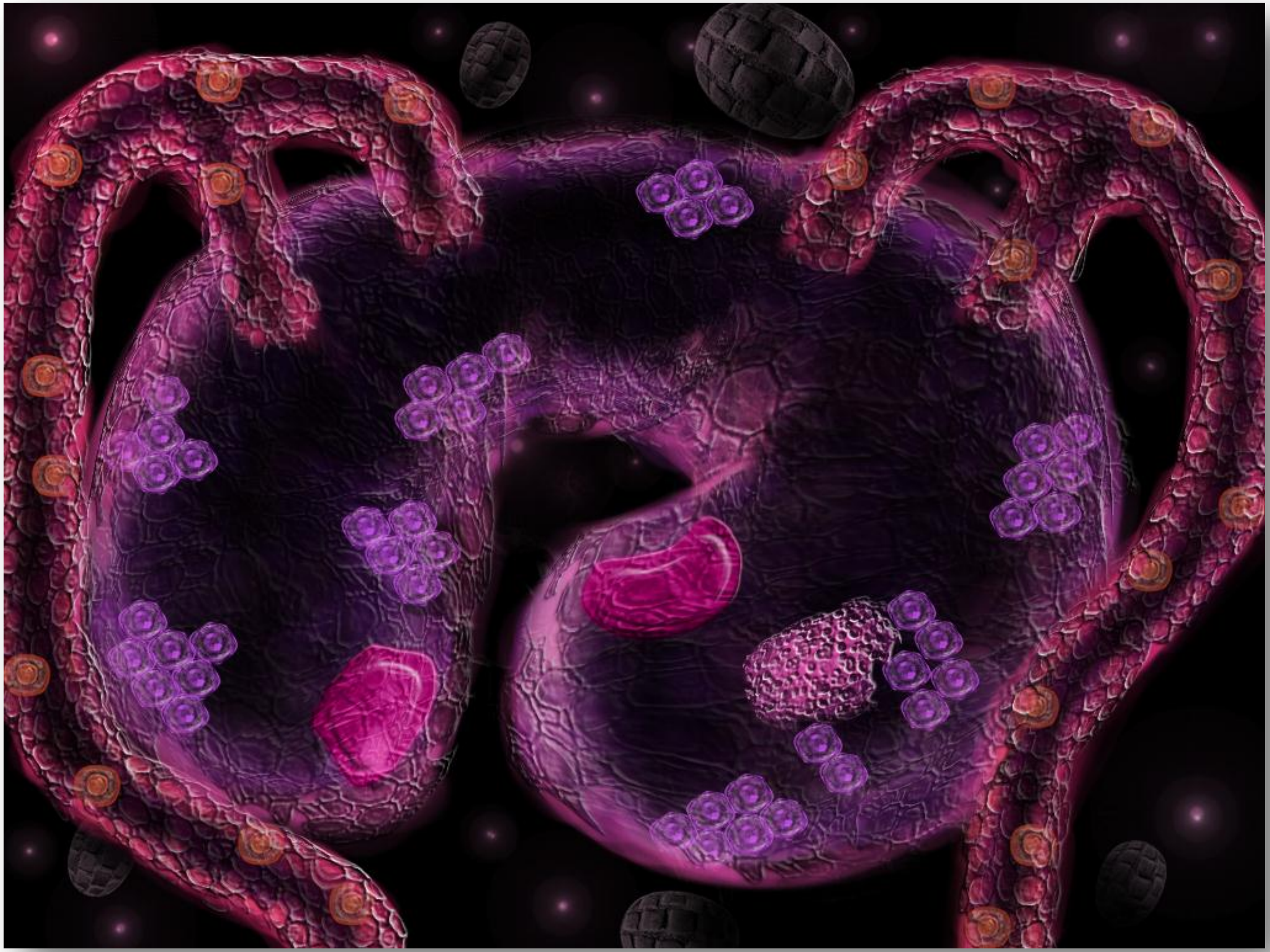

T
Killer
T
Help
T
Help
The player loses if the bacteria break out and spread to all the cells.

DIABETES

Type 1 Diabetes is a form of diabetes mellitus that results from the autoimmune destruction of the insulin-producing beta cells in the pancreas.

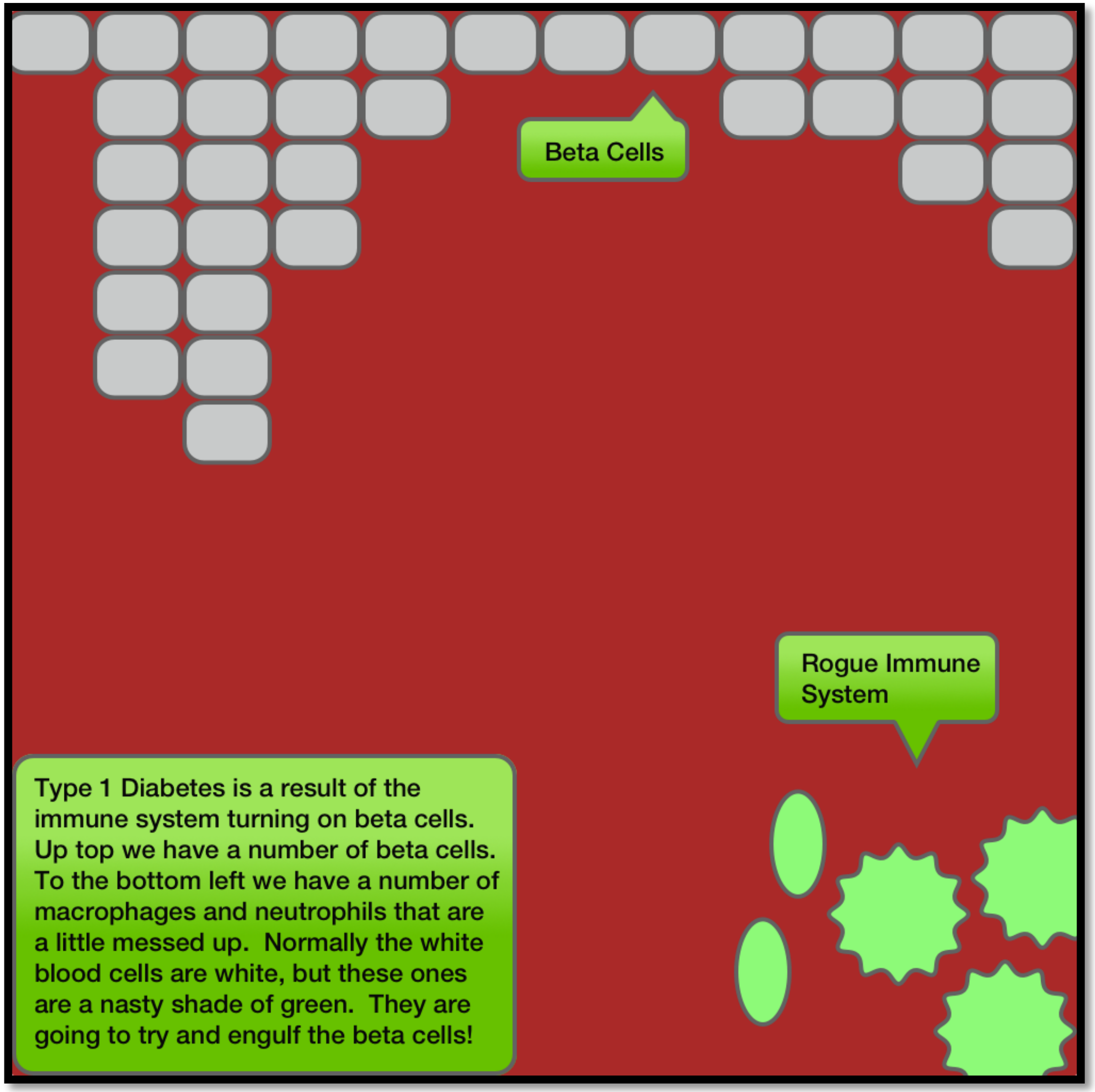
The subsequent lack of insulin leads to increased blood and urine glucose. Untreated, type 1 diabetes is ultimately fatal.

DIABETES

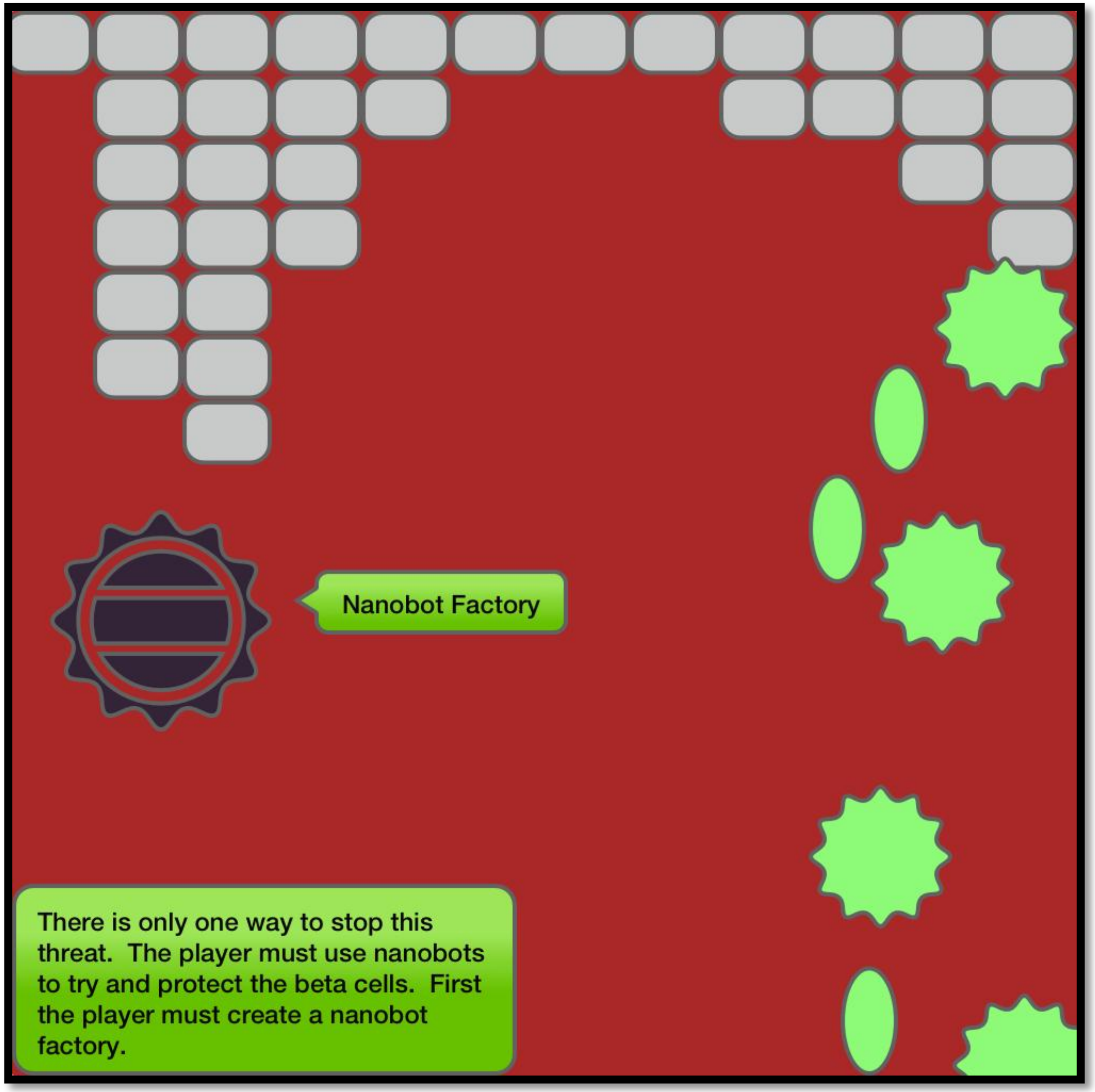


The battle to defeat Diabetes takes place in the pancreas.

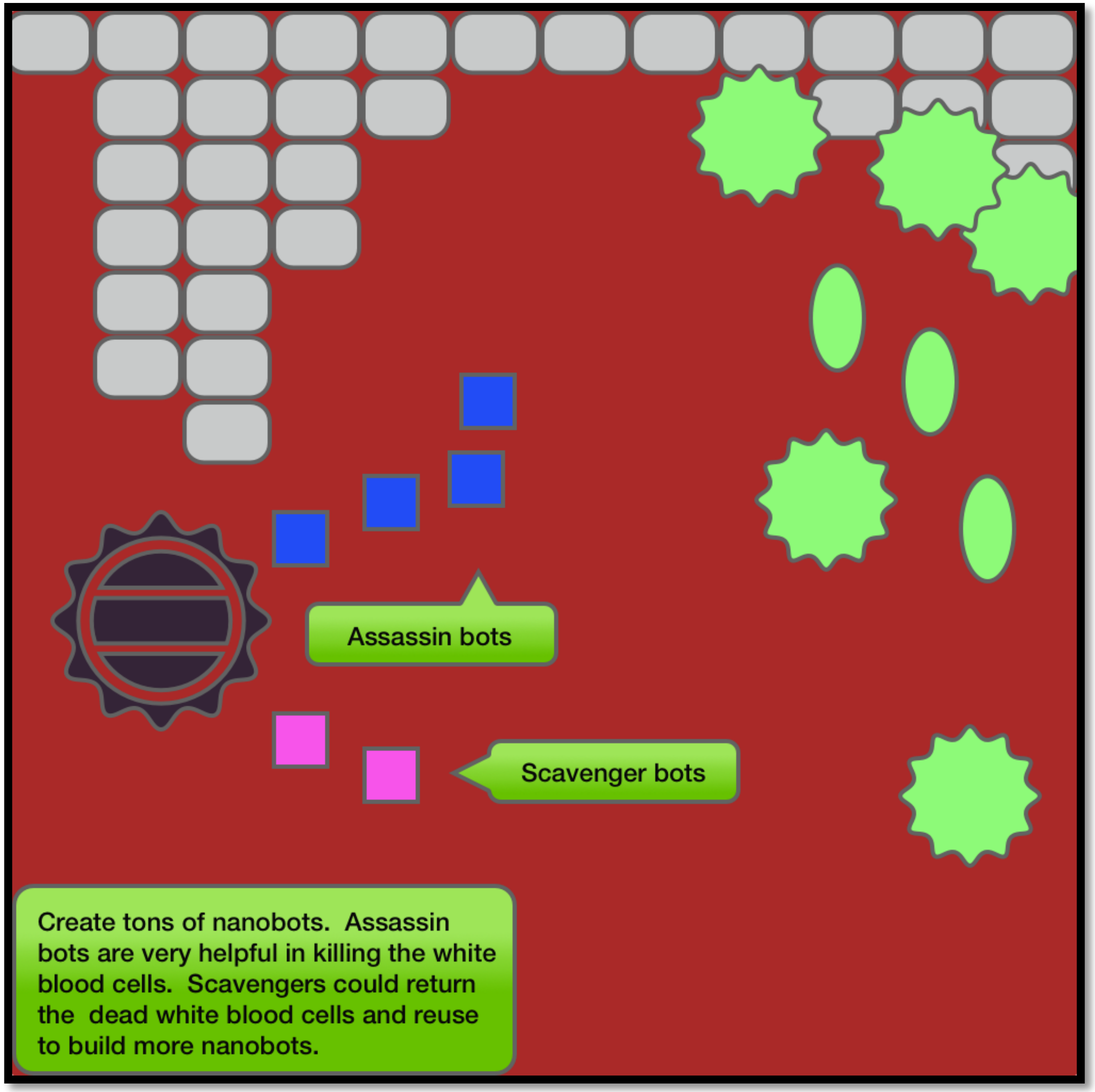
DIABETES


Beta Cells
Rogue Immune System
Type 1 Diabetes is a result of the immune system turning on beta cells. Up top we have a number of beta cells. To the bottom left we have a number of macrophages and neutrophils that are a little messed up. Normally the white blood cells are white, but these ones are a nasty shade of green. They are going to try and engulf the beta cells!

DIABETES

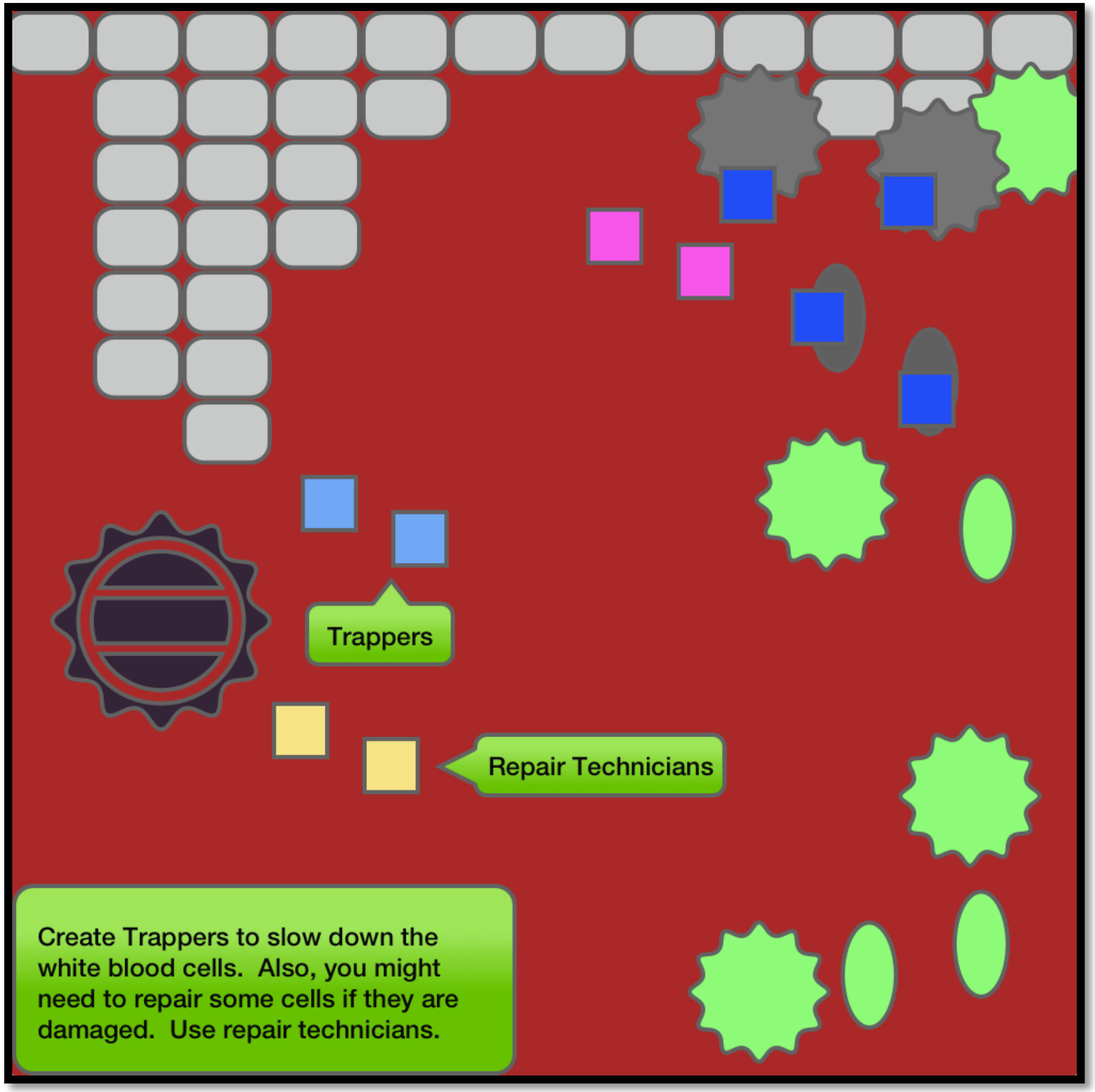

Nanobot Factory
There is only one way to stop this threat. The player must use nanobots to try and protect the beta cells. First the player must create a nanobot factory.

DIABETES



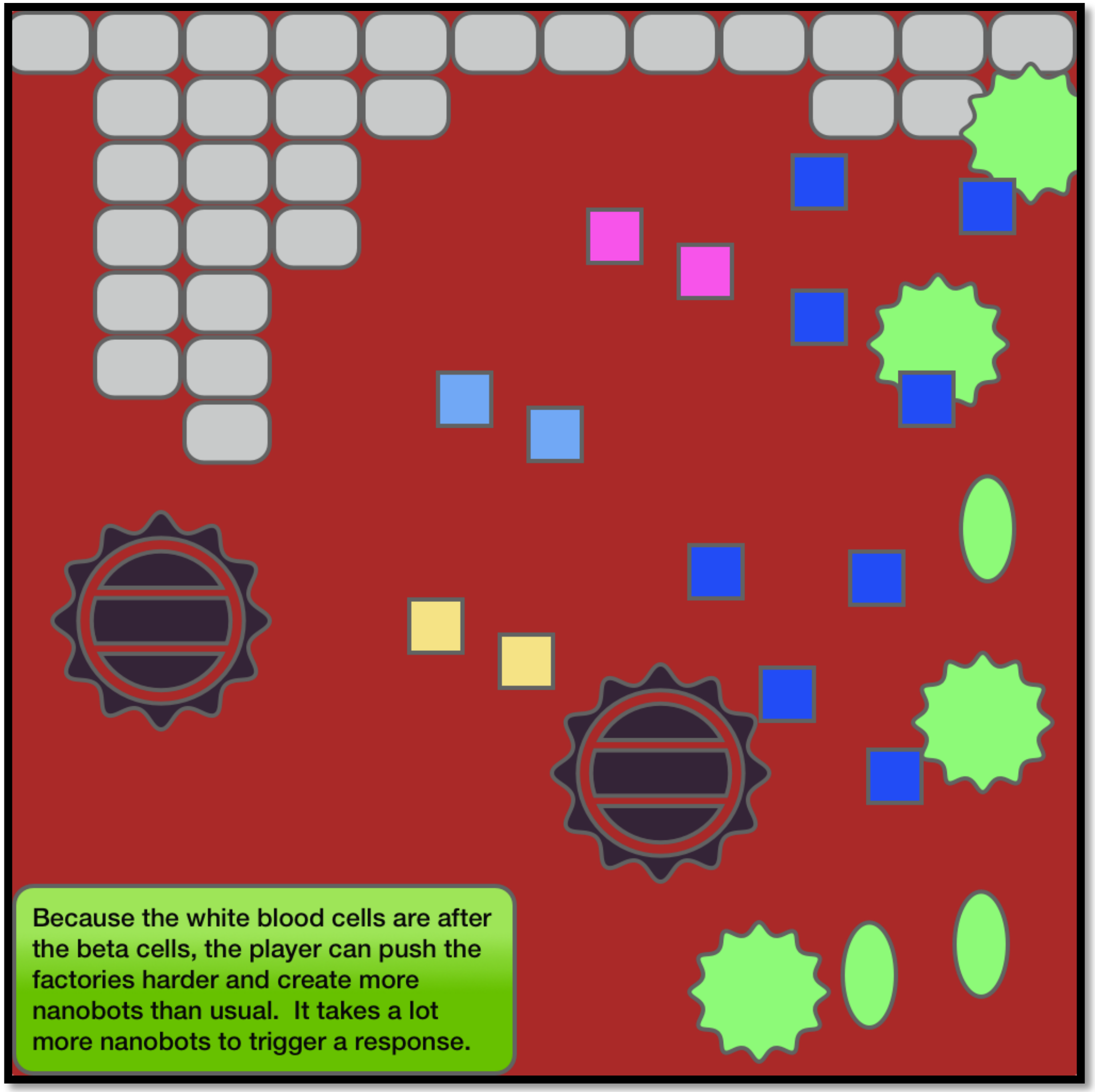
Assassin bots
Scavenger bots
Create tons of nanobots. Assassin bots are very helpful in killing the white blood cells. Scavengers could return the dead white blood cells and reuse to build more nanobots.

DIABETES



Trappers
Repair Technicians
Create Trappers to slow down the white blood cells. Also, you might need to repair some cells if they are damaged. Use repair technicians.

DIABETES


Because the white blood cells are after the beta cells, the player can push the factories harder and create more nanobots than usual. It takes a lot more nanobots to trigger a response.

DIABETES


One strategy could be to turn the white blood cells on the nanobots. The cells would engulf the bots and die after eating a few of them. You win if you protect at least a number of the beta cells from the immune system. You lose if a certian amount of beta cells are destroyed after all the waves.

PATHOGEN X

Pathogen X is an extraterrestrial parasite unlike any faced.

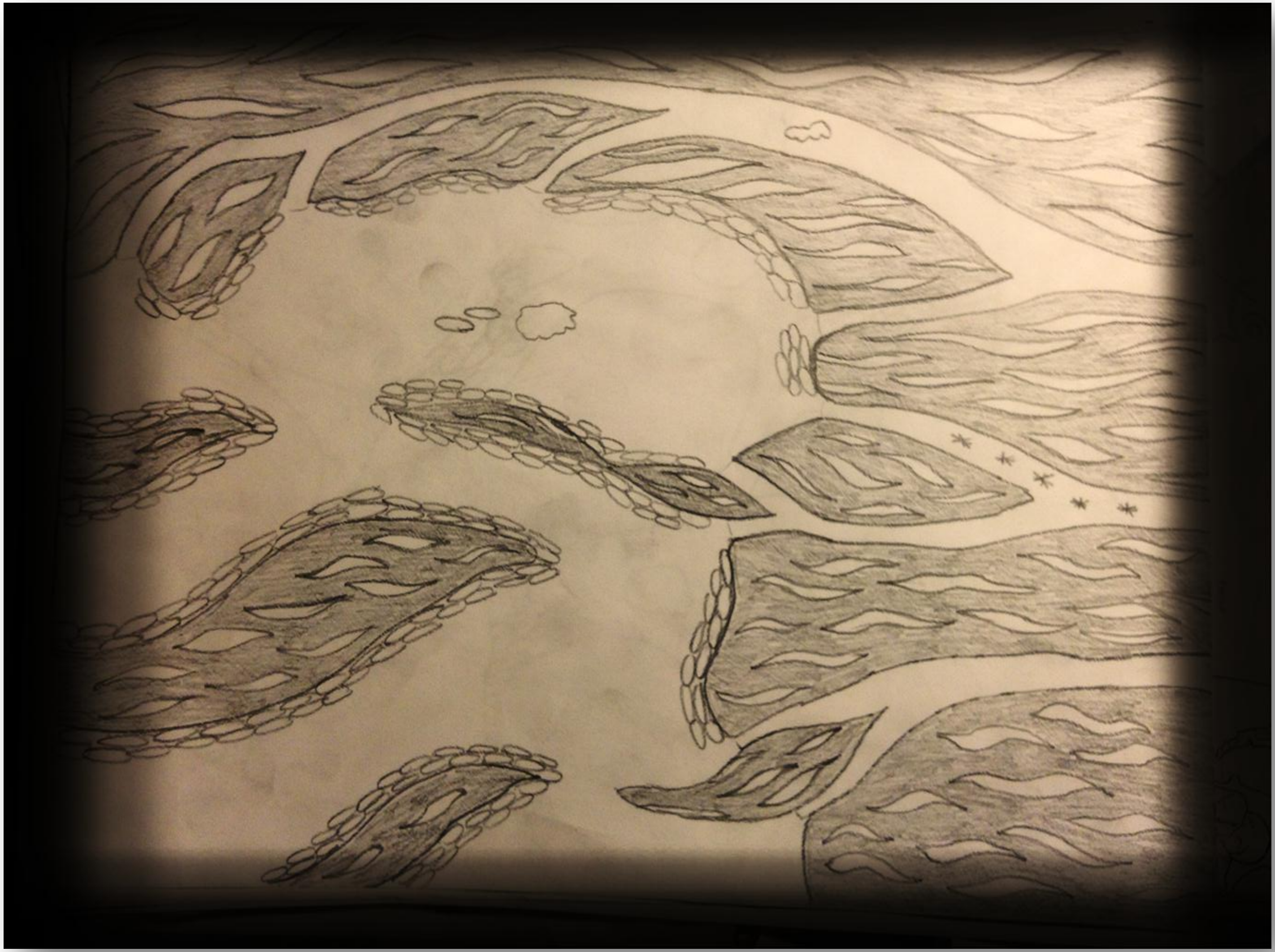
It travels to the heart and begins to reproduce in cells and break them apart for nutrition to fuel reproduction.

The player should have to use all of their skills they have learned to efficiently deal with this threat.

This parasite has arms (like a macrophage) that extend to engulf cytokines and antibodies.

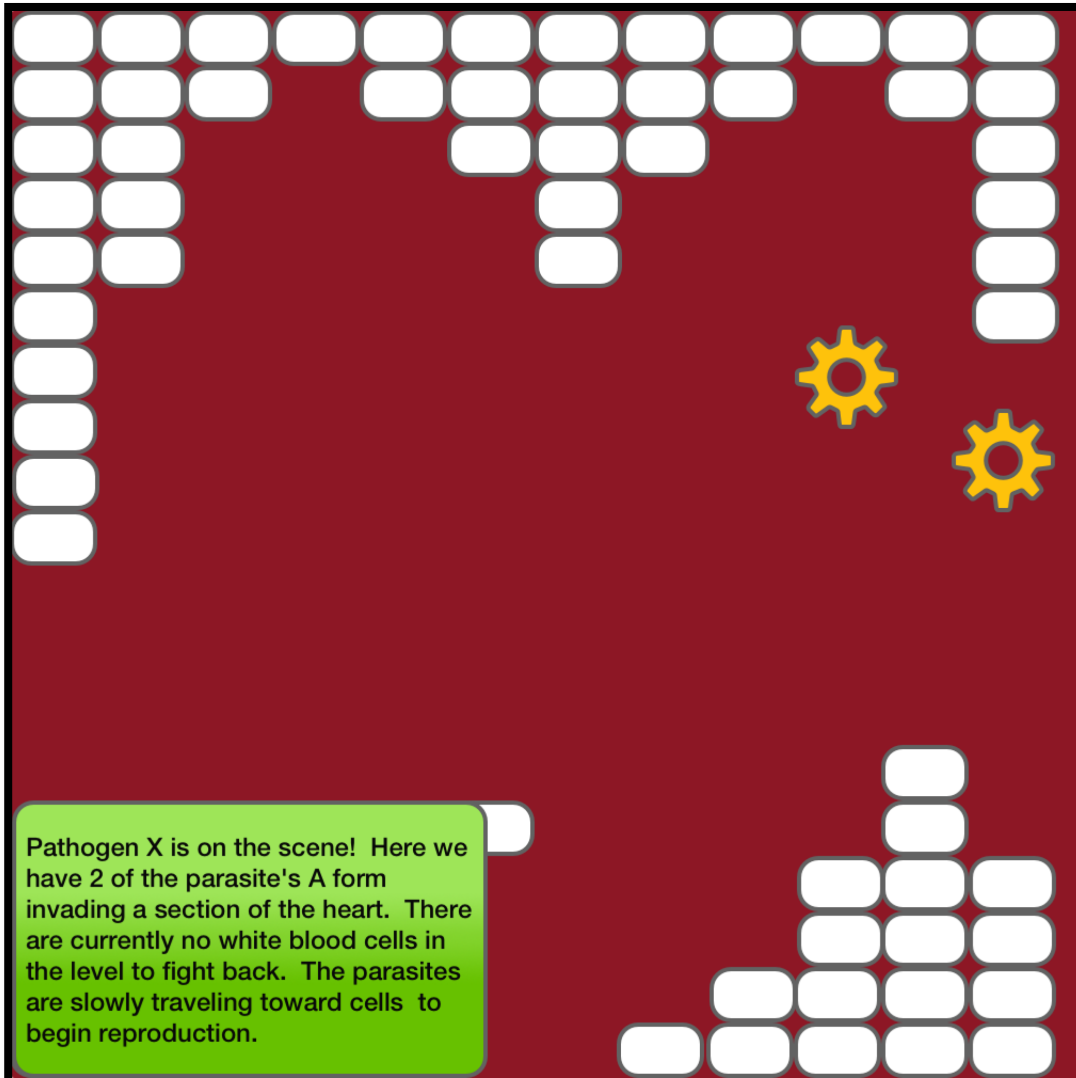
The parasite reproduces a variant that collects materials from tissue cells in order to fuel reproduction.

PATHOGEN X

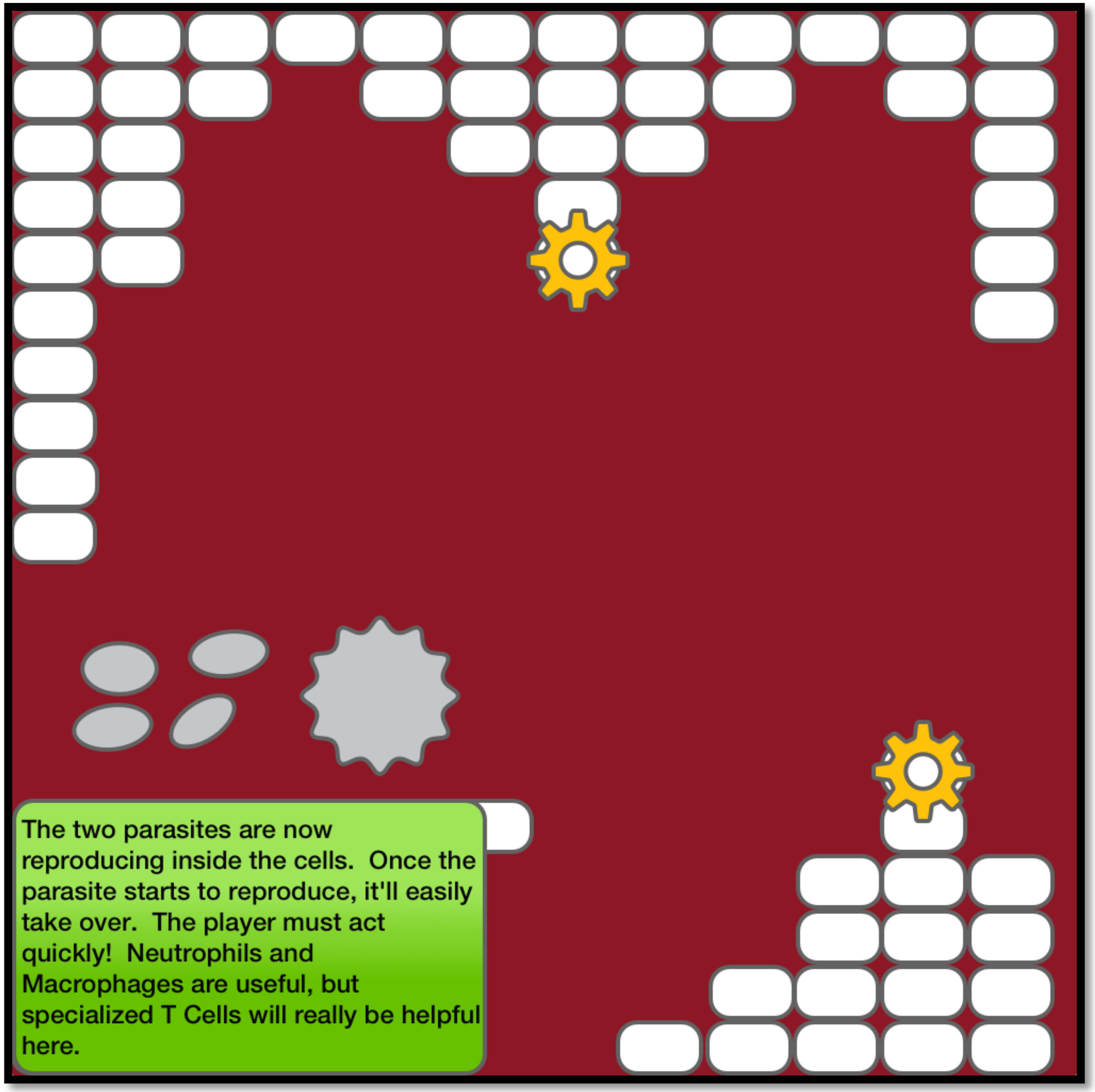


The battle to defeat Pathogen X takes place in the heart.

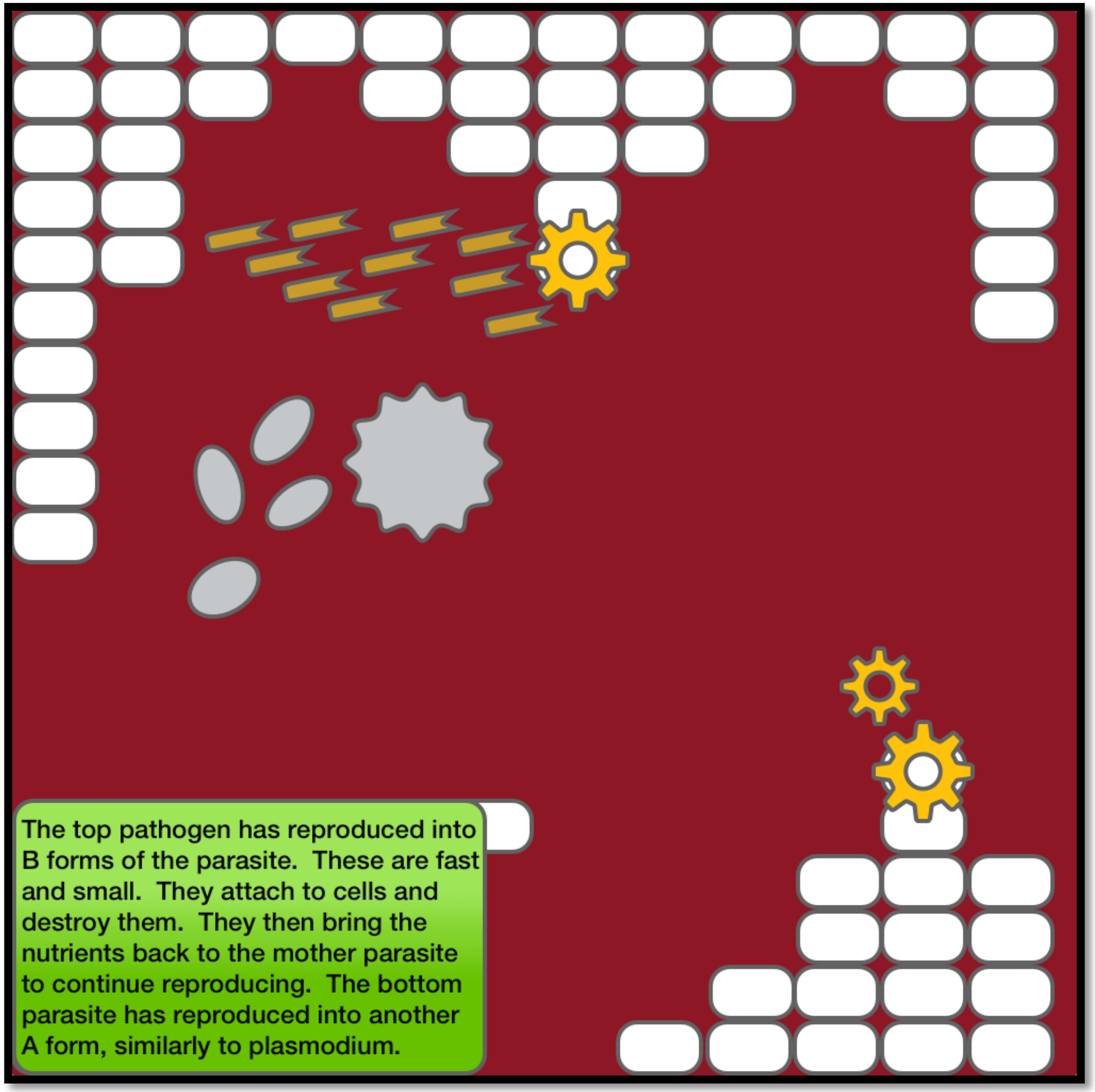
PATHOGEN X


Pathogen X is on the scene! Here we have 2 of the parasite's A form invading a section of the heart. There are currently no white blood cells in the level to fight back. The parasites are slowly traveling toward cells to begin reproduction.

PATHOGEN X

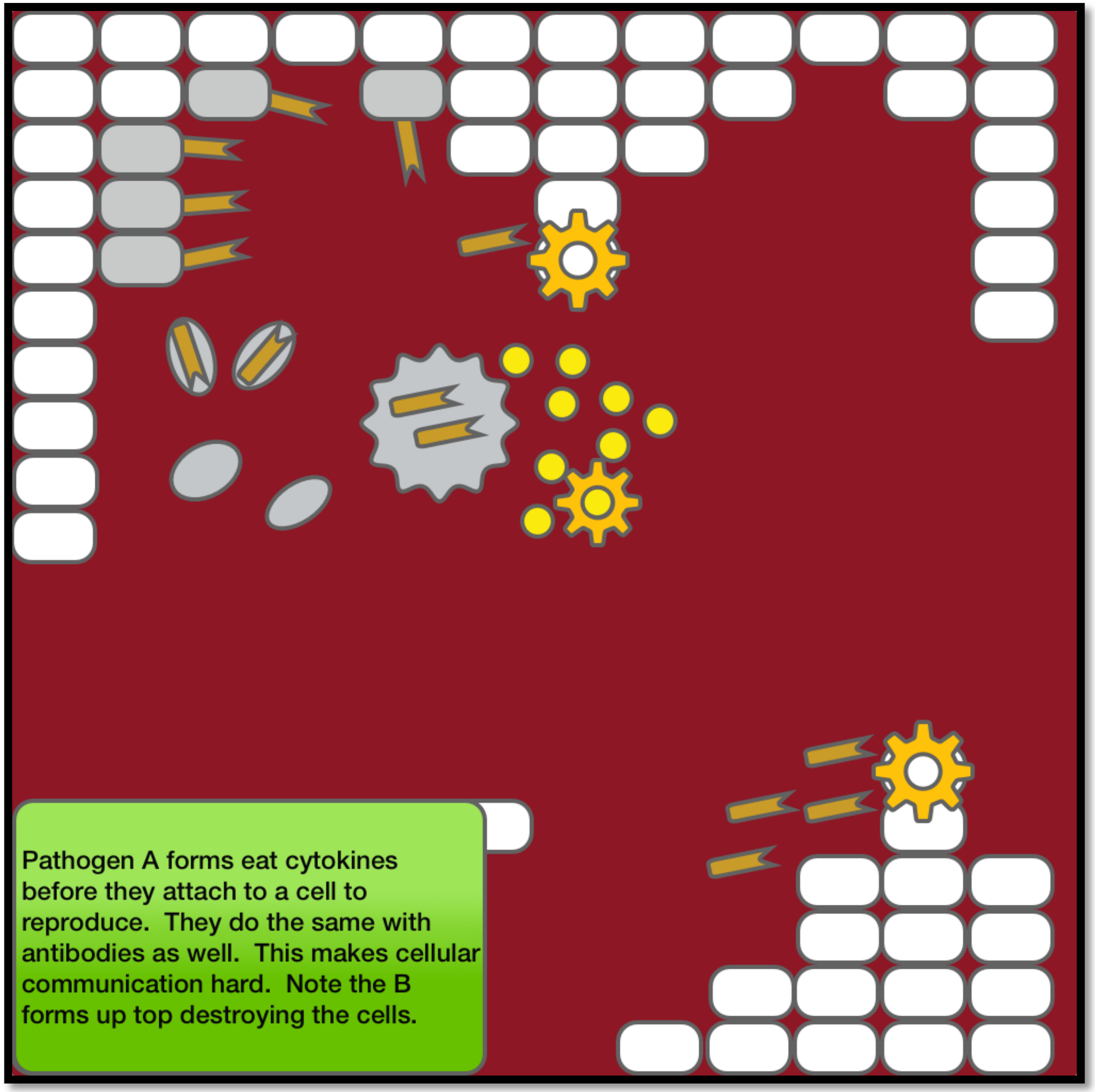

The two parasites are now reproducing inside the cells. Once the parasite starts to reproduce, it'll easily take over. The player must act quickly! Neutrophils and Macrophages are useful, but specialized T Cells will really be helpful here.

PATHOGEN X

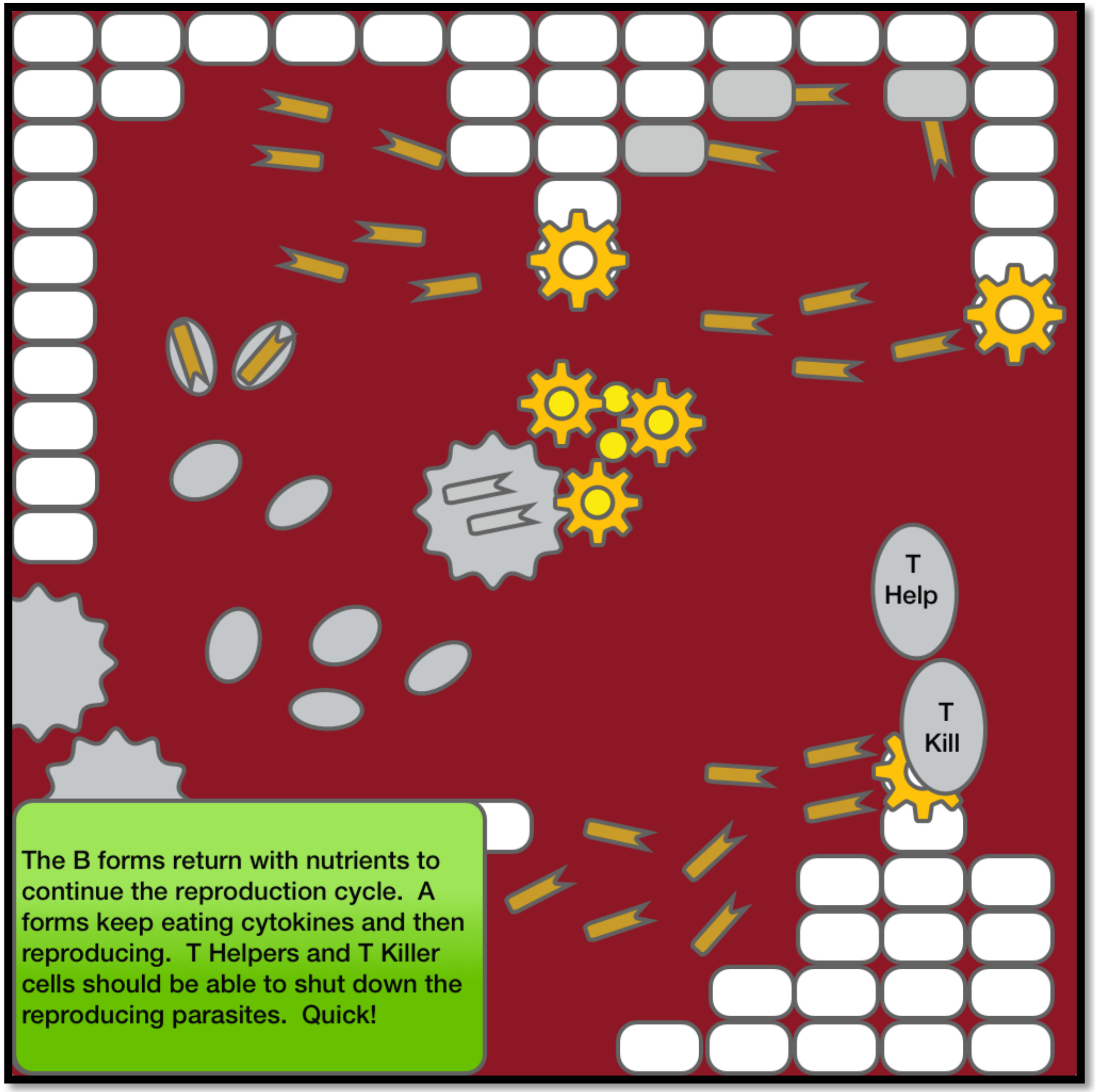


The top pathogen has reproduced into B forms of the parasite. These are fast and small. They attach to cells and destroy them. They then bring the nutrients back to the mother parasite to continue reproducing. The bottom parasite has reproduced into another A form, similarly to plasmodium.

PATHOGEN X

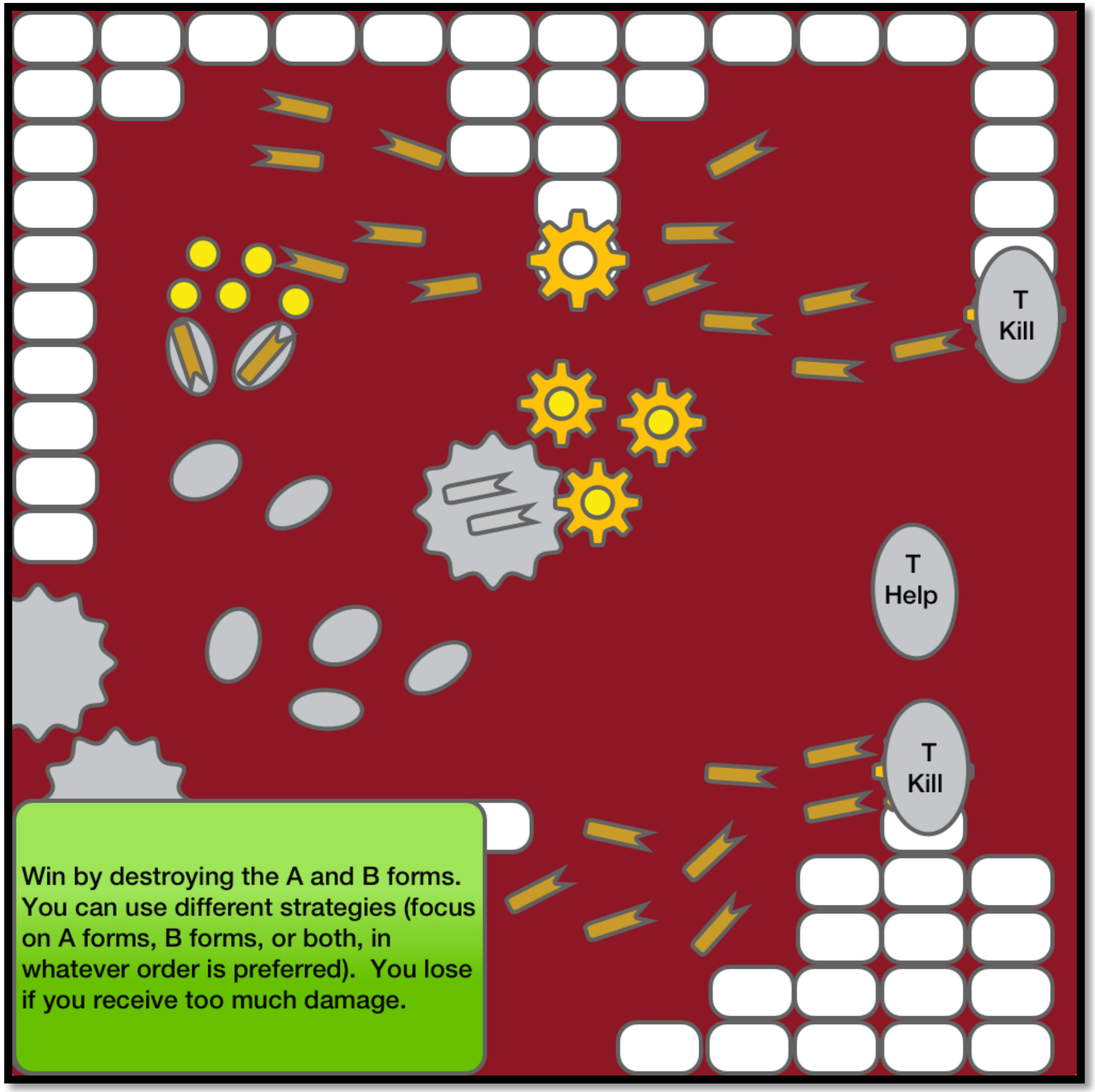

Pathogen A forms eat cytokines before they attach to a cell to reproduce. They do the same with antibodies as well. This makes cellular communication hard. Note the B forms up top destroying the cells.

PATHOGEN X



The B forms return with nutrients to continue the reproduction cycle. A forms keep eating cytokines and then reproducing. T Helpers and T Killer cells should be able to shut down the reproducing parasites. Quick!

PATHOGEN X



Win by destroying the A and B forms. You can use different strategies (focus on A forms, B forms, or both, in whatever order is preferred). You lose if you receive too much damage.

BIONOMICON

Modding

LEVEL EDITOR

The level editor allows the player to pick a specific location in the body and build unique experiences from existing game content.

The player can choose from a number of pieces that connect to create unique levels.

The player can choose the flow direction of a blood vessel and also where that vessel connects to other areas in the level.

The player can choose where the tissue cells should be placed that need to be protected.

LEVEL EDITOR

The player can choose the number and speed of enemy waves as well as the intervals for white blood cell arrival.

The player can choose the locations for nano-factories.

Finally, the player can set the starting conditions for the scenario. This might include initial actors in the scene, starting cytokine levels, etc.

Players should be able to share their creations with the community using something like Steam Workshop.

PATHOGEN EDITOR

The pathogen editor allows players to create custom enemy types and share them with the community.

The pathogen class (virus, parasite, or bacteria) determines the general behavior. Modifiers provide the player with a layer of behavioral customization.

The shape, size, color, and décor (like cilia or flagellum) of a pathogen body can also be customized.

Special abilities like releasing a cloud of toxins on death can be attached to particular classes of pathogen.

SERIOUSLY

Can we make it now?